

# Automorphisms of the Surcomplex Numbers

Real forms, Hahn symmetries,  
valuation, rigidity, and finite symmetry descent

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## Abstract

The phrase “automorphism of the surcomplex numbers” has several inequivalent meanings. For the algebraically closed field  $\mathbb{K} = \mathbf{No}(i)$ , the automorphisms fixing  $\mathbf{No}$  pointwise are just identity and conjugation. Those preserving  $\mathbf{No}$  setwise form  $\text{Aut}(\mathbf{No}) \times C_2$ , but this is a proper subgroup of the unrestricted field automorphisms. We prove this distinction explicitly, using Hahn-series phase twists that fix every ordinary complex number and every valuation value while moving the surreal real axis.

We develop the coefficient, character, value-group, and leading-term-kernel layers of the valued automorphism group; construct automorphisms moving ordinary real constants by infinitesimals; and separate strong summation from fine continuity. A further contrast is that nonidentity surcomplex field automorphisms can be everywhere fine-differentiable with derivative zero. We examine class-sized homogeneity, fixed fields, involutions, inequivalent real forms, norm and circle preservation, simplicity, the omega-map, and exponential structure. The repository’s existing exponential faithfulness argument is incorporated as prior work, not presented as a new discovery. That repository report classifies the automorphisms preserving the logarithmic modulus,  $\text{Aut}_L(\mathbb{K}) \simeq \text{Aut}_{\text{exp}}(\mathbf{No}) \times C_2$ ; the present report studies plain field automorphisms, of which those form a proper subgroup of the real-axis stabilizer. The final sections identify precise formalization targets and distinguish proved statements, imported theorems, and unresolved classification problems.

A second source (Sections 16–30) conjugates finite groups of valued Hahn-field automorphisms to their coefficient action by a norm section, and classifies the involutions of  $\mathbf{No}(i)$  preserving the valuation and  $\mathbf{Oz}[i]$ : exactly  $2^c$  conjugacy types, indexed by real closed fields  $F$  of transcendence degree  $c$ , with fixed field  $F((t^{\mathbf{No}}))$ , a copy of  $\mathbf{No}$  exactly when  $F \cong \mathbb{R}$ . A third source (Sections 31–40) re-derives this independently, adds an explicit strong inverse of the conjugator by formal inversion and a descent for strong finite actions in every characteristic, and shows that the  $2^c$  fixed rings have the same finite quotients and the same solvable integer equations.

**Scope and status.** An AI-assisted research draft, not refereed. A mathematical research exposition, not a claim of complete classification or of historical priority. Written proofs are supplied for the main reductions and constructions. No new Lean verification is supplied here. Its labels have no dedicated implementation mappings; existing generic Lean results cover ordered displacement, exponential rigidity, logarithmic-modulus classification and the valuation kernel (Section 1.5). The original repository comparison is pinned to the revision recorded in Section 13; the relations to the collection as it now stands are in Section 1.5 and Appendix C. The word “automorphism” is always qualified by the structure it preserves (Section 1.4). Every label carries the prefix `saut`.

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# 1 Which structure is being preserved?

Let  $\mathbf{No}$  be Conway's real closed ordered field of surreal numbers and put

$$\mathbb{K} = \mathbf{No}(i) = \{x + iy : x, y \in \mathbf{No}\}, \quad i^2 = -1.$$

This is the surcomplex field. Its algebraic closedness follows from real closedness of  $\mathbf{No}$ . Write

$$c(x + iy) = x - iy, \quad N(x + iy) = x^2 + y^2, \quad |x + iy| = \sqrt{x^2 + y^2}.$$

The square root is the nonnegative surreal square root. In particular, the modulus is *surreal-valued*, not an ordinary real-valued norm. The underlying surreal construction and its normal forms are classical [1, 2].

The elementary equation  $\alpha(i)^2 = -1$  forces any field automorphism  $\alpha$  of  $\mathbb{K}$  to send  $i$  to  $i$  or  $-i$ . It does *not* force  $\alpha(\mathbf{No}) = \mathbf{No}$ . This is the first logical distinction on which the subject turns. Once  $\mathbf{No}$  is preserved, order is automatic on that subfield; without the distinguished subfield, the pure algebraically closed field has no intrinsic real axis.

Three groups should therefore never be conflated:

$$\text{Aut}(\mathbb{K}/\mathbf{No}) \subsetneq G_{\mathbf{No}} := \{\alpha : \alpha(\mathbf{No}) = \mathbf{No}\} \subsetneq \text{Aut}(\mathbb{K}). \quad (1.1)$$

The left-hand group fixes  $\mathbf{No}$  *pointwise*. The middle group preserves it *setwise*. The right-hand group has no such condition. We will prove

$$\text{Aut}(\mathbb{K}/\mathbf{No}) = \{\text{id}, c\}, \quad G_{\mathbf{No}} \simeq \text{Aut}(\mathbf{No}) \times C_2,$$

and exhibit witnesses to both strict inclusions.

Other reasonable structures produce other answers. One may mark ordinary  $\mathbb{C}$ , the natural valuation ring, every individual valuation value, the Conway monomials, strong Hahn summation, conjugation, modulus, simplicity, a derivation, or a real exponential. Naming one does not silently name all the others. For example, preserving every ordinary complex number and every valuation value still permits a rotation of some real monomials into imaginary ones.

## 1.1 Literature and the repository

Kuhlmann–Serra developed decompositions of valued Hahn-field automorphisms [3]; their work on Hahn groups supplies a second layer of structure [4]. Kaplan–Krapp–Serra apply these ideas to  $\mathbf{No}$ , including strong summation and additional exponential structure [5]. We use these as the structural background, while proving the complex-coefficient reductions and explicit examples below directly.

The supplied repository already contains a manuscript proving faithful value-group action for surreal exponential automorphisms, with a logarithmic-modulus application to  $\mathbf{No}(i)$  [10]. That argument is reproduced with attribution in Section 11. Its result is important: one must not simply repeat the corresponding question in an earlier source as unanswered. Section 15 separates that mathematical issue from publication status and priority.

## 1.2 Foundations: a collection of class maps

We work in a class theory such as NBG with global choice. An individual surreal number is a set, but  $\mathbf{No}$ ,  $\mathbb{K}$ , and the graphs of their automorphisms are proper classes. The notation  $\text{Aut}(\mathbb{K})$  denotes a *definable collection of class maps*; it is not itself a class-object in NBG. A group isomorphism formula means a unique factorization

statement about individual automorphisms, with the stated composition law. This convention avoids forming a nonexistent set or class of all class functions. Likewise, a finite subgroup means a faithful action of an ordinary finite group  $G$ , encoded by one class relation in  $G \times \mathbb{K} \times \mathbb{K}$  whose slices are field automorphisms. No set of proper-class graphs is formed; the fixed field is defined by quantifying over the finite index set  $G$ . For a collection of all class automorphisms, a displayed fixed-field identity is instead read pointwise: it asserts that an element is fixed by every individual map exactly when it belongs to the named field. It does not invoke class comprehension with a quantifier over class maps.

Global choice is used for the nonconstructive extension arguments in Section 9. The explicit Hahn maps need no class-sized transcendence basis: their values are defined on the set-sized support of one input at a time. Every infinite sum in this article has a set-sized index family. No proper-class support is admitted.

For a set-based treatment, most local algebra and all explicit formulas can instead be read in a Hahn field  $\mathbb{C}((t^\Gamma))$  for a set-sized divisible ordered abelian group  $\Gamma$ . A self-map requires the chosen parameters to preserve that workspace. Global homogeneity and the topology of the full surreal field should not be transferred to an arbitrary fixed workspace.

### 1.3 Notation

Throughout the full-field discussion,

$$\Gamma = (\mathbf{No}, +, <), \quad t^g := \omega^{-g}.$$

Thus  $t = \omega^{-1}$  is positive infinitesimal. The sign in this convention is essential. In  $t$ -notation, supports increase and are well-ordered; in  $\omega$ -notation, normal-form exponents decrease.

| Symbol                            | Meaning                                                         |
|-----------------------------------|-----------------------------------------------------------------|
| $\text{Aut}(\mathbb{K}/E)$        | Field automorphisms fixing the indicated subfield $E$ pointwise |
| $G_{\mathbf{No}}$                 | Setwise stabilizer of the distinguished surreal real axis       |
| $G_v$                             | Automorphisms stabilizing the natural valuation ring            |
| $v, \text{lc}$                    | Minimum support exponent and its nonzero coefficient            |
| $\mathcal{U}$                     | Automorphisms fixing the leading term of every nonzero input    |
| $\text{Aut}_{\text{ord}}(\Gamma)$ | Ordered <i>additive-group</i> automorphisms of $\Gamma$         |
| $\text{st}$                       | Residue/standard-part map on the finite subring                 |
| $c$                               | The chosen conjugation $x + iy \mapsto x - iy$                  |

### 1.4 Conventions: one meaning per word

This subsection and the next were written when the manuscript joined the collection (Appendix C). They fix vocabulary and state relations. The only mathematics they add is the elementary observation (1.2) below, which is marked as an observation of the edit.

**Kinds of automorphism.** An unqualified “automorphism” of  $\mathbb{K}$  or of  $\mathbf{No}$ , and the notation  $\text{Aut}(\mathbb{K})$ ,  $\text{Aut}(\mathbf{No})$ , always mean a *plain field automorphism*: a bijective ring map, with no further condition. Every other kind is named by the structure it preserves.

- A *valued automorphism* is a field automorphism  $\alpha$  with  $\alpha(\mathcal{O}) = \mathcal{O}$ , that is, an element of the group  $G_v$  of Section 6. Equivalently  $v(\alpha z) = \tau_\alpha(vz)$  for an ordered automorphism  $\tau_\alpha$  of the value group. A valued automorphism need not fix any value. Every field automorphism of  $\mathbf{No}$  is valued (Lemma 2.1 and Section 6); a field automorphism of  $\mathbb{K}$  need not be (Proposition 9.3).

- A valued automorphism is *value-fixing* when  $\tau_\alpha = \text{id}$ , that is,  $v(\alpha z) = v(z)$  for every  $z \neq 0$ . This is the meaning of “exactly value-preserving” and of “values fixed” in Theorem 4.2 and Appendix A.
- A *1-automorphism* is an element of  $\mathcal{U}$  in (5.1) (or of  $\mathcal{U}_{\mathbb{R}}$  for **No**):  $v(\alpha z - z) > v(z)$  for every  $z \neq 0$ , which is the same as fixing every leading term. It is value-fixing, and it fixes  $i$ . The phase twists  $P_\theta$ ,  $\theta \neq 0$ , are value-fixing but are not 1-automorphisms. This leading-term notion is the one of Kaplan–Krapp–Serra [5] that the collection’s rigidity report checked against the pinned version (its Corollary 5.2 uses only that a 1-automorphism fixes every value).
- An *exponential automorphism* is a field automorphism  $\sigma$  of **No** with  $\sigma(\exp x) = \exp(\sigma x)$  for all  $x \in \mathbf{No}$ ; they form  $\text{Aut}_{\text{exp}}(\mathbf{No})$ . The notion is defined for **No** only: no surcomplex exponential is used in this report. By (11.2), the identity is the only exponential automorphism that is value-fixing, and a fortiori the only exponential 1-automorphism.
- An *L-automorphism* is a field automorphism  $\alpha$  of  $\mathbb{K}$  with  $\alpha(L(z)) = L(\alpha z)$  for every  $z \neq 0$ , where  $L(z) = \log |z|$  (11.3). They form  $\text{Aut}_L(\mathbb{K})$ , the notation of the rigidity report; the source wrote  $\text{Aut}(\mathbb{K}, L)$ .
- *Strong* and *strongly  $\mathbb{C}$ -linear* are the support conditions of Section 3. They are not continuity conditions (Section 7).

The inclusions that hold are

$$\mathcal{U} \subseteq \{\text{value-fixing}\} \subseteq G_v \subseteq \text{Aut}(\mathbb{K}), \quad \{\text{id}, c\} = \text{Aut}(\mathbb{K}/\mathbf{No}) \subseteq \text{Aut}_L(\mathbb{K}) \subseteq G_{\mathbf{No}} \subseteq G_v.$$

The first chain is immediate from the definitions. In the second, the middle inclusion is the range argument of Theorem 11.4 and the last is Proposition 7.1. *Observation of the edit.* Combining Theorem 2.2 with Theorem 11.4,

$$\text{Aut}_L(\mathbb{K}) = \{\alpha \in G_{\mathbf{No}} : \alpha|_{\mathbf{No}} \in \text{Aut}_{\text{exp}}(\mathbf{No})\} \subsetneq G_{\mathbf{No}}. \quad (1.2)$$

The equality is Theorem 11.4 read through the factorization (2.3). Strictness is witnessed by  $S_q$  for any rational  $q > 0$ ,  $q \neq 1$ : it lies in  $G_{\mathbf{No}}$  (Section 4), and its restriction to **No** induces the value-group dilation  $g \mapsto qg$ , which no exponential automorphism induces (Proposition 11.3). Whether the first inclusion  $\{\text{id}, c\} \subseteq \text{Aut}_L(\mathbb{K})$  is strict is the question whether  $\text{Aut}_{\text{exp}}(\mathbf{No})$  is nontrivial; neither this report nor the rigidity report decides it.

**Rigidity.** In this report the word labels three specific statements: exact norm rigidity (Theorem 2.3), the simplicity statement of Section 11, and the faithful value-group action of exponential automorphisms (Theorem 11.2); the last is faithfulness, not triviality, exactly as in the rigidity report. None of them is the all-scale polynomial rigidity `f:thm-rigidity of surcomplex/analysis` or the scalar rigidity of `surreal/gamma-functions`.

**Phase.** The phases of the twists  $P_\theta$  are the coefficient multipliers  $e^{i\theta\ell(g)}$  of (4.6). They are not an argument of  $z$ , not the finite-angle `cis` or the global phase of `surcomplex/trigonometry`, and not the phase of the differential-equations report; `NOTATION.md` lists those meanings.

**Real form.** A real form is a real closed subfield  $F \subset \mathbb{K}$  with  $\mathbb{K} = F(i)$  (Section 10). In `ent:prop:real-form of surcomplex/entire-functions-at-arbitrary-rank` and in `analytic:subsec:fixed-real of surcomplex/analytic-geometry` the phrase names the real-coefficient version of a theorem, an unrelated use.

**Symbols.** Here  $\Gamma = (\mathbf{No}, +, <)$  is a proper class; this departs from the recommendation of NOTATION.md that  $\Gamma$  be a set-sized group, and the set-sized reading is the one described in the previous subsection. The conventions  $t^g = \omega^{-g}$ ,  $v = \min \text{supp}$  and  $v(0) = \infty$  are the collection's.  $\mathbb{K} = \mathbf{No}(i)$  is the field that NOTATION.md writes  $\text{SC} = \mathbf{No}[i]$  and the analysis report writes  $\mathbb{K}$ ;  $K_1$ ,  $K_2$  and  $F(i)$  in Sections 2 and 9 are generic. Six symbols of the source were renamed so that each has one meaning.

- The source wrote  $E_{\rho, \tau}$  for the coefficient-and-exponent lift and  $E_d$  for the Taylor motion of Theorem 5.1. Here they are  $M_{\rho, \tau} = M_{\rho, \tau, 1}$  and  $\Phi_d$ , so that  $E$  never denotes a lift. Otherwise  $E$  denotes an ordered exponential (Theorem 11.2), as in the rigidity report, except that  $\text{Aut}(\mathbb{K}/E)$  in the notation table and the proofs of Proposition 9.3 and Theorem 10.1 use  $E$  locally for a set-sized subfield of  $\mathbb{K}$ . (Corrected in batch 32: this sentence previously said that  $E$  denotes only an ordered exponential.)
- The derivation of Theorem 5.2 is  $\mathcal{D}$  (source:  $D$ ). A  $D$  with a subscript is always a diagonal automorphism  $D_\lambda$  or  $D_\chi$ ; the unsubscripted  $D(x) = \sigma(x) - x$  in Lemma 11.1 is the displacement, as in the rigidity report.
- The real Puiseux field of Section 7 is  $\mathcal{P}$  (source:  $P$ ), leaving  $P_\theta$  for the phase twists.
- The automorphism in the non-density argument at the end of Section 7 is  $\psi$  (source:  $\tau$ ), since  $\tau$  always denotes an ordered automorphism of the value group.
- The sides of a cut are  $(\mathcal{L}, \mathcal{R})$  (source:  $(L, R)$ ), since  $L$  is the logarithmic modulus.

$T_s$  denotes the flows of Theorem 5.2; the maps  $T_r$  and  $T_{s_0, c}$  of the rigidity report are value-group maps, and for rational  $q > 0$  the dilation  $S_q$  of (4.4), restricted to  $\mathbf{No}$ , is that report's canonical field lift  $\widehat{T}_q$  (prop:hahn-lift). The coefficient  $\ell(g) = [g]_0$  of (4.5) is that report's layer coefficient  $[\gamma]_{s_0}$  at  $s_0 = 0$ . The letter  $H$  is local to three places: the kernel subgroup in the proof of Theorem 11.2, the real closure in Theorem 10.2, and the half-plane  $H_F$  of Section 2.

## 1.5 Relation to other reports of the collection

The source compared itself with the collection at its pin; that comparison is kept in Section 13. The relations below were checked against the collection as it stands after this report was placed.

**Exponential automorphism rigidity.** surreal/exponential-automorphism-rigidity studies exponential automorphisms of an ordered field with a total exponential and, in its Section 9,  $L$ -automorphisms of  $F(i)$  for real closed  $F$ ; this report studies plain field automorphisms of  $\mathbb{K}$ . The two meet in Section 11, whose results are that report's and are printed here once, with its proofs.

- Lemma 11.1 is its lem:amplify (Lemma 3.1), with the same probe.
- Theorem 11.2 is the faithfulness of its thm:cofinal and cor:faithful (Theorem 4.1, Corollary 4.2), by the route of lem:bridge and lem:bounded; the identification of the kernel with the finite surreals after it is prop:finite-log. That report proves more: cofinal and coinital displacement, coarsenings, a version needing one visible exponential scale, and an order-free detector.
- The display (11.2) and the sentence after it are its eq:main-no, thm:no and cor:question54 (Theorem 5.1, Corollary 5.2): every exponential 1-automorphism of  $\mathbf{No}$  is the identity, the negative answer to Question 5.4 of [5], arXiv version 3. The answer is that report's, not this one's, and the reading of the question rests on its check of the pinned PDF.
- Proposition 11.3 is its thm:dilation (Theorem 8.4), by the second, multiplication-based of its two proofs, with an arbitrary positive infinite  $X$  in place of  $\omega$ .

- Theorem 11.4 is its `thm:L-classification` with `lem:range` and the kernel clause of `thm:complex-kernel` and `cor:surcomplex` (Lemma 9.1, Theorems 9.2 and 9.4, Corollary 9.5), specialized to  $F = \mathbf{No}$ . That report proves it for every real closed  $F$  with an ordered exponential, and on every invariant nontrivial convex value quotient.

What this report adds on the surcomplex side is the plain-field picture around that classification. Theorem 2.2 is the field-level counterpart:  $G_{\mathbf{No}} \simeq \text{Aut}(\mathbf{No}) \times C_2$ , with  $\text{Aut}(\mathbf{No})$  in place of  $\text{Aut}_{\text{exp}}(\mathbf{No})$ . That report remarks that a shorter route to its classification, assuming commutation with conjugation and preservation of the exponential on the real axis from the start, proves strictly less; Theorem 2.2 is the exponential-free first step of that shorter route. By (1.2),  $\text{Aut}_L(\mathbb{K})$  is a proper subgroup of  $G_{\mathbf{No}}$ , which in turn is a proper subgroup of  $G_v$  and of  $\text{Aut}(\mathbb{K})$  (9.2). By the kernel clause of Theorem 11.4, an  $L$ -automorphism that fixes every value is `id` or `c`. The explicit maps  $D_\lambda$ ,  $P_\theta$ ,  $\Phi_d$  and  $T_s$  are value-fixing and fix  $i$ , so a nonidentity one is never an  $L$ -automorphism; nor is  $S_q$  for rational  $q \neq 1$ , by (1.2). For other  $a$  this report does not decide whether  $S_a|_{\mathbf{No}}$  is exponential. The open question on the image of exponential automorphisms in the value group (Section 14) is that report's `q:image` (Question 13.1).

**Foundations and analysis.** The pair construction of  $\mathbb{K}$  and its algebraic closedness are `found:prop:complex of foundations-and-computation/foundations`. Proposition 7.2 is `found:thm:discrete` there, items (1)–(2), with the absence of isolated points; `FORMALIZATION.md` records Lean proofs of those clauses for the actual surcomplex carrier, with lower-universe smallness. The fine derivative of Section 8 is `found:eq:derivative`. The collection's earlier zero-derivative witnesses are locally constant: the indicator of the finite ring in `found:rem:fourtypes` and in `b:L of surcomplex/analysis`, with a Lean proof recorded for the former. The dilation  $S_2$  of Theorem 8.2 is a different kind of witness: it is injective, so it is constant on no open set with more than one point, and the full field has no isolated points.

**Diagonal character twists.** The automorphisms  $D_\chi = M_{\text{id}, \text{id}, \chi}$  are the character twists that the collection already uses on set-sized Hahn fields as tools: `spec:lem:finiteindex of surcomplex/spectral-theory`, whose Galois group  $\text{Hom}(\Lambda/\Gamma, \mu_q)$  acts by them; `hnd:lem:supportdegree of surcomplex/hidden-negative-hermitian-directions`; and the twists built from `bst:lem:characters in surreal/transcendence-over-bounded-support`. There the characters take roots of unity or other values on a set-sized group; here  $\chi$  is an arbitrary homomorphism  $\Gamma = \mathbf{No} \rightarrow \mathbb{C}^\times$ , and the twists are placed in the decomposition of Theorem 6.1.

**Unit circle and rotations.** The Cayley map in the proof of Theorem 2.4,  $u = (x - i)/(x + i)$ , is a reparametrization of `trigonometry:thm:cayley of surcomplex/trigonometry` ( $u = C(-1/x)$  for  $x \neq 0$ , while  $x = 0$  gives  $u = -1$ ); that report identifies the unit circle with  $\text{SO}(2, \mathbf{No})$  in Section `trigonometry:sec:circle`. Rotations there, and in `surreal/euclidean-three-space`, placed in the same batch, are  $\mathbf{No}$ -linear isometries, not field automorphisms. The contrast is exact: multiplication by a unit-circle element preserves every modulus, whereas by Theorem 2.3 the only field automorphisms preserving every modulus are `id` and `c`.

**Lean interfaces.** Besides the two modules the source inspected (Section 13), the collection has `Surreal/Surcomplex/Basic.lean`, whose `conj` is conjugation as a ring automorphism of the concrete surcomplex field, and `Surreal/HahnSeries/ComplexNumbers.lean`, whose `coefficientEquiv` lifts a ring equivalence of coefficients to Hahn series over an ordered exponent monoid, keeping supports: the case  $M_{\rho, \text{id}, 1}$  of (4.1) in a workspace. `Surreal/Algebra/SigmaDerivation.lean` proves the valued-field forms of the rigidity report's displacement results. The later `Surreal/Algebra/ExponentialProfile.lean` proves

its ordered displacement lemma (`displacement_cofinal`) and the value-fixing exponential rigidity implication (`eq_id_of_commute`). The modules `Surreal/Algebra/LogModulusClassification.lean` and `Surreal/Algebra/ComplexValuationKernel.lean` now also prove the generic logarithmic-modulus classification and valuation-kernel theorem. They work over an ordered field with nonnegative square roots and an injective `OrderedExp`; injectivity is an explicit hypothesis, not a field of that structure. The kernel theorem additionally uses a nontrivial convex valuation. The classification includes the direct-product isomorphism (`autLEquiv`); the kernel module uses separate definitions of `log`, `L` and the `L`-automorphism predicate, without a formal bridge to the classification module. Its coarsened action is expressed relationally, not through a constructed ordered quotient. These generic results cover the corresponding arguments here; their mappings use the rigidity report's labels. No dedicated `saut`: implementation mapping is added. Instantiation at the actual surreal exponential, rational non-lifting and the classification for the actual surcomplex logarithmic modulus remain pending; no class automorphism-group object follows merely from a theorem about individual type-based maps.

**Finite symmetry descent (Sections 16–30).** [merge] A second source, added in batch 32, classifies the involutions in  $G_v$  that preserve  $\mathbf{Oz}[i]$ : they are conjugate, by an explicit strong norm section, to coefficientwise lifts of involutions of  $\mathbb{C}$ , and their conjugacy types correspond to the real closed fields of transcendence degree  $c$ . This answers the valued case of the real-form classification of Section 14. Its relations to `surreal/omnific-preserving-automorphisms` (whose automatic strongness makes the source's strongness hypothesis redundant), `surreal/omnific-diophantine-geometry` and `foundations-and-computation/surreal-fields-across-universes` are in Section 28.4.

**Gaussian-omnific real forms (Sections 31–40).** [merge] A third source, added in batch 34, re-derives the classification of the second independently and adds an explicit strong inverse of the norm conjugator by formal inversion [6], a descent theorem for strong finite actions in every characteristic, and the uniform finite arithmetic of the fixed rings. Its relations to the omnific-preserving, omnific Diophantine, set-sized-quotient, definable-surreal and first- $\kappa$  reports are in Section 40.2.

## 2 Conjugation and the real-axis stabilizer

The results of this section apply to any real closed ordered field  $F$ , with  $K = F(i)$ . No infinite summation or specifically surreal theorem is needed.

**Lemma 2.1.** *Every field automorphism of a real closed field is order preserving. It fixes its real algebraic numbers pointwise.*

*Proof.* In a real closed field,  $x > 0$  is equivalent to  $x = y^2$  for some  $y \neq 0$ . A field automorphism preserves this condition and hence preserves order. It fixes  $\mathbb{Q}$ . Each real algebraic number is isolated from the other real roots of its minimal polynomial by a rational interval, so it is fixed too.  $\square$

**Theorem 2.2** (Centralizer and stabilizer). *For  $K = F(i)$  with  $F$  real closed,*

$$\mathrm{Aut}(K/F) = \{\mathrm{id}, c\}, \quad (2.1)$$

$$\begin{aligned} \{\alpha \in \mathrm{Aut}(K) : \alpha(F) = F\} &= C_{\mathrm{Aut}(K)}(c) = N_{\mathrm{Aut}(K)}(\langle c \rangle) \\ &\simeq \mathrm{Aut}(F) \times C_2. \end{aligned} \quad (2.2)$$

*The pair  $(\sigma, e)$ , with  $e \in \{0, 1\}$ , acts by*

$$(x + iy) \mapsto \sigma(x) + (-1)^e i \sigma(y). \quad (2.3)$$

*Proof.* The roots of  $X^2 + 1$  in  $K$  are exactly  $i, -i$ . Thus an automorphism fixing  $F$  has precisely the two possibilities in (2.1). If  $\alpha$  preserves  $F$ , its restriction is a field automorphism  $\sigma$  and its value on  $i$  determines (2.3). Conversely, this formula respects addition and the multiplication rule

$$(x + iy)(u + iv) = (xu - yv) + i(xv + yu),$$

and has an inverse of the same form. The lift  $\widehat{\sigma}$  fixing  $i$  commutes with  $c$ , so the product is direct.

If  $\alpha$  commutes with  $c$ , it carries the fixed field  $\text{Fix}(c) = F$  onto itself. The converse follows from (2.3). Finally, normalizing the order-two subgroup  $\{\text{id}, c\}$  means fixing its unique nonidentity member under conjugation, which is the centralizer condition.  $\square$

For surcomplex numbers,  $\text{Aut}(\mathbf{No})$  is very far from trivial. In Section 4, exponent dilation will give a real-axis-preserving automorphism sending  $\omega$  to  $\omega^2$ . Thus even retaining conjugation leaves much more than the two classical-looking maps.

## 2.1 Equivariance is not exact preservation

For  $\alpha = \widehat{\sigma}c^e$  one has

$$N(\alpha z) = \sigma(Nz), \quad |\alpha z| = \sigma(|z|). \quad (2.4)$$

The second identity follows because  $\sigma$  preserves nonnegativity and square roots. In particular, balls are taken to balls, but their radii may change. Calling every such map an isometry would be misleading: it is *semilinearly* compatible with the surreal-valued modulus.

**Theorem 2.3** (Exact norm rigidity). *A field automorphism  $\alpha$  of  $F(i)$  satisfying  $N(\alpha z) = N(z)$  for every  $z$  is either  $\text{id}$  or  $c$ . The same conclusion holds under exact modulus preservation. No real-axis-preservation hypothesis is needed.*

*Proof.* The identity

$$N(z + 1) - N(z) - 1 = 2 \text{Re } z$$

and  $\alpha(z + 1) = \alpha z + 1$  imply  $\text{Re}(\alpha z) = \text{Re } z$ . For  $x \in F$ , write  $\alpha x = x + iy$ . Exact norm preservation gives  $x^2 + y^2 = x^2$ , hence  $y = 0$ . Thus  $\alpha$  fixes  $F$  pointwise and theorem 2.2 applies. Exact modulus preservation implies exact norm preservation by squaring.  $\square$

## 2.2 The unit circle remembers the real axis

Let  $U_F = \{z \in F(i) : N(z) = 1\}$ . Merely preserving this set, without preserving individual norms, already recovers the real-axis stabilizer.

**Theorem 2.4.** *For a field automorphism  $\alpha$  of  $F(i)$ ,*

$$\alpha(U_F) = U_F \iff \alpha(F) = F.$$

*Proof.* The reverse implication follows from (2.4). For the forward implication use the Cayley parametrization

$$u = \frac{x - i}{x + i}, \quad x = i \frac{1 + u}{1 - u}.$$

For every  $x \in F$ , the first expression lies in  $U_F \setminus \{1\}$ . For every  $u \in U_F \setminus \{1\}$ , the second expression is real: conjugating it and using  $\bar{u} = u^{-1}$  leaves it unchanged. Applying  $\alpha$  to the inverse expression changes  $i$  at most to  $-i$ , so  $\alpha x \in F$ . Apply the same reasoning to  $\alpha^{-1}$  to obtain equality of the axes.  $\square$

Let  $H_F = \{z : \operatorname{Im} z > 0\}$  be the chosen upper half-plane. If a field automorphism preserves  $H_F$  setwise, it also preserves  $-H_F$ , so it preserves their complement  $F$ . Since  $i \in H_F$ , it must fix  $i$ . Conversely, a lift  $\widehat{\sigma}$  preserves  $H_F$  by order preservation. Thus its stabilizer consists exactly of the lifts  $\widehat{\sigma}$ . There is no compatible ordered-field structure on  $F(i)$  itself, since  $i^2 = -1$ .

### 3 Normal forms, valuation, and summability

Conway normal form identifies the ordered field  $\mathbf{No}$  with real Hahn series having surreal exponents and set-sized well-ordered support. Taking real and imaginary parts coefficientwise gives

$$\mathbf{No} = \mathbb{R}((t^\Gamma)), \quad \mathbb{K} = \mathbb{C}((t^\Gamma)), \quad \Gamma = (\mathbf{No}, +, <). \quad (3.1)$$

The displayed equalities mean canonical identifications. A surcomplex normal form is

$$z = \sum_{g \in S} z_g t^g, \quad z_g \in \mathbb{C}, \quad S \subseteq \Gamma \text{ a well-ordered set.}$$

For  $z \neq 0$ , define

$$v(z) = \min \operatorname{supp}(z), \quad \operatorname{lc}(z) = z_{v(z)}, \quad \operatorname{lt}(z) = \operatorname{lc}(z) t^{v(z)},$$

and set  $v(0) = \infty$ . Multiplication is convolution, with only finitely many contributions to any fixed exponent. Its basic laws are

$$v(zw) = v(z) + v(w), \quad v(z + w) \geq \min\{v(z), v(w)\}.$$

Conjugation acts coefficientwise. Since  $\operatorname{lc}(z\bar{z}) = |\operatorname{lc}(z)|_{\mathbb{C}}^2 > 0$ ,

$$v(\bar{z}) = v(z), \quad v(Nz) = 2v(z), \quad v(|z|) = v(z). \quad (3.2)$$

The valuation ring and its maximal ideal are

$$\mathcal{O} = \{z : v(z) \geq 0\}, \quad \mathfrak{m} = \{z : v(z) > 0\}.$$

They are exactly the finite surcomplex numbers and the infinitesimals, respectively. Every  $z \in \mathcal{O}$  has a unique decomposition  $z = a + \varepsilon$  with  $a \in \mathbb{C}$ ,  $\varepsilon \in \mathfrak{m}$ , and  $\operatorname{st}(z) = a = z_0$ . Thus  $\mathcal{O}/\mathfrak{m} \simeq \mathbb{C}$ . The analogous real residue field is  $\mathbb{R}$ .

**Definition 3.1** (Strong summation). A set-indexed family  $(z_j)_{j \in J}$  is Hahn-summable if the union of its supports is well-ordered and every exponent occurs in only finitely many of the supports. Its sum is then defined coefficientwise by finite sums. An additive map is *strongly additive* when it carries summable families to summable families and commutes with their sums. An automorphism is *strong* here when it and its inverse are strongly additive. A strongly  $\mathbb{C}$ -linear map is a strongly additive  $\mathbb{C}$ -linear map; for an automorphism we also require its inverse to be strongly additive. For field automorphisms,  $\mathbb{C}$ -linearity is equivalent to fixing  $\mathbb{C}$  pointwise.

This definition is a support condition, not a limit of finite partial sums. For example,  $\sum_{n \geq 0} t^n$  is a legitimate Hahn sum and equals  $(1 - t)^{-1}$  even in a setting where its partial sums do not converge in the full fine topology. Section 7 explains that distinction.

**Lemma 3.2** (Support control used below). *If  $S$  is a well-ordered subset of an ordered abelian group and  $\delta > 0$ , then  $S + \mathbb{N}\delta$  is well-ordered, and each exponent has only finitely many representations  $s + n\delta$  with  $s \in S$  and  $n \in \mathbb{N}$ . More generally, the sum of two well-ordered subsets is well-ordered and has finitely many decompositions of any fixed element.*

*Proof.* Every infinite sequence in a well-order has an infinite nondecreasing subsequence. A strictly decreasing sequence of sums, by successively taking such subsequences for its two coordinates, would have a nondecreasing subsequence, a contradiction. Thus the sum is well-ordered. Infinitely many distinct representations of one value would give infinitely many distinct first coordinates. Choosing a strictly increasing subsequence of these forces a strictly decreasing sequence of second coordinates, again impossible. Apply this to  $S$  and  $\mathbb{N}\delta$ .  $\square$

These finite-decomposition arguments justify the rearrangements in the constructions below. More general support-controlled formal exponentials and derivations are developed in [6]; we will need only explicit fixed-shift cases, which can be checked directly.

## 4 Explicit monomial automorphisms and phase twists

Let  $\rho \in \text{Aut}(\mathbb{C})$ , let  $\tau \in \text{Aut}_{\text{ord}}(\Gamma)$ , and let  $\chi : (\Gamma, +) \rightarrow (\mathbb{C}^\times, \cdot)$  be a group homomorphism. Define

$$M_{\rho, \tau, \chi} \left( \sum_g z_g t^g \right) = \sum_g \rho(z_g) \chi(g) t^{\tau(g)}. \quad (4.1)$$

Here  $\chi$  need not arise from an exponential of an additive complex-valued map. Arbitrary multiplicative characters are allowed.

**Theorem 4.1** (Monomial construction). *Formula (4.1) defines a strong field automorphism of  $\mathbb{K}$ , with*

$$v(Mz) = \tau(vz), \quad \text{lc}(Mz) = \rho(\text{lc } z) \chi(vz).$$

*For composition in the order  $M_1 M_2$ , the parameters are*

$$\rho = \rho_1 \rho_2, \quad \tau = \tau_1 \tau_2, \quad \chi(g) = \rho_1(\chi_2(g)) \chi_1(\tau_2(g)). \quad (4.2)$$

*The map preserves the real axis exactly when*

$$\rho \in \{\text{id}_{\mathbb{C}}, c|_{\mathbb{C}}\}, \quad \chi(g) \in \mathbb{R}_{>0} \text{ for every } g. \quad (4.3)$$

*Proof.* The order bijection  $\tau$  takes a well-ordered support to a well-ordered support. All transformed coefficients on the support are nonzero. Additivity is coefficientwise. For multiplication, each coefficient is a finite sum,  $\tau(g+h) = \tau(g) + \tau(h)$ , and  $\chi(g+h) = \chi(g)\chi(h)$ , so convolution is preserved. The same formula with inverse parameters gives a two-sided inverse; explicitly one can use  $\rho^{-1}, \tau^{-1}$  and  $g \mapsto \rho^{-1}(\chi(\tau^{-1}g)^{-1})$ . Support unions and their finite incidence multiplicities are preserved, proving strongness in both directions. Substitution proves (4.2).

If the real axis is preserved, the image of each real constant is a real constant, so  $\rho(\mathbb{R}) = \mathbb{R}$ . A field automorphism of  $\mathbb{R}$  is the identity, because order and rational cuts determine each real number. Consequently  $\rho$  is identity or conjugation on  $\mathbb{C}$ . Also  $M(t^g)$  is real, so  $\chi(g) \in \mathbb{R}^\times$ . Divisibility of  $\Gamma$  gives  $\chi(g) = \chi(g/2)^2 > 0$ . Conversely, (4.3) makes the image of every real-coefficient series real, and the same holds for the inverse.  $\square$

### 4.1 Rescaling all valuation exponents

For every positive surreal  $a$  let

$$S_a \left( \sum_g z_g t^g \right) = \sum_g z_g t^{ag}. \quad (4.4)$$

Multiplication by  $a$  is an ordered additive automorphism of  $\Gamma$ , so theorem 4.1 applies. We obtain

$$S_a S_b = S_{ab}, \quad S_a^{-1} = S_{1/a}, \quad S_a(t) = t^a, \quad S_a(\omega) = \omega^a.$$

Every  $S_a$  fixes ordinary  $\mathbb{C}$  pointwise, commutes with  $c$ , and preserves order on  $\mathbf{No}$ . It fixes individual valuation values only for  $a = 1$ . For instance,

$$S_2 \left( 3 + 2t^{-1} + (1+i)t^{1/2} \right) = 3 + 2t^{-2} + (1+i)t.$$

This proves the first strict inclusion in (1.1).

Notice that  $S_a$  generally does *not* commute with the omega-map. Indeed,  $S_a(1) = 1$  whereas  $S_a(\omega^1) = \omega^a \neq \omega^{S_a(1)}$  when  $a \neq 1$ . The omega-map must not be mistaken for an exponent reindexing convention.

## 4.2 Positive real characters

For an additive map  $\lambda : \Gamma \rightarrow \mathbb{R}$ , define

$$D_\lambda \left( \sum_g z_g t^g \right) = \sum_g z_g e^{\lambda(g)} t^g.$$

These are strongly  $\mathbb{C}$ -linear real-axis-preserving automorphisms fixing all valuation values. Their group law is  $D_\lambda D_\mu = D_{\lambda+\mu}$ . They generally change leading coefficients, so they are not leading-term-fixing automorphisms.

There are explicit nonzero  $\lambda$ . Regard an exponent  $g$  itself as a surreal normal form and set

$$\ell(g) = [g]_0, \tag{4.5}$$

its real coefficient at exponent zero. This is an additive map defined on all of  $\mathbf{No}$ , and  $\ell(r) = r$  for every real  $r$ . It is not a standard-part map on all surreals: for an infinite  $g$ , coefficient extraction still makes sense even though standard part does not.

## 4.3 Complex phases move the real axis

For  $\theta \in \mathbb{R}$  use the same  $\ell$  to define

$$P_\theta \left( \sum_g z_g t^g \right) = \sum_g z_g \exp(i\theta\ell(g)) t^g. \tag{4.6}$$

Only the ordinary complex exponential is used in the coefficient.

**Theorem 4.2** (A faithful phase action). *The maps  $P_\theta$  form a faithful action of the additive group  $\mathbb{R}$  by strongly  $\mathbb{C}$ -linear, exactly value-preserving automorphisms of  $\mathbb{K}$ . For  $\theta \neq 0$ ,  $P_\theta$  does not preserve  $\mathbf{No}$ . Moreover,*

$$P_\theta P_\phi = P_{\theta+\phi}, \quad c P_\theta c = P_{-\theta}. \tag{4.7}$$

In particular,

$$P_{\pi/2}(t) = it.$$

*Proof.* The character identity follows from additivity of  $\ell$ , so the monomial theorem gives all field and summability assertions. If  $P_\theta = \text{id}$ , then  $e^{i\theta r} = 1$  for every real  $r$ , because  $\ell(r) = r$ . For  $\theta \neq 0$ , taking  $r = \pi/\theta$  contradicts this. Thus the action is faithful. Its parameter is not an angle modulo  $2\pi$ : all real exponents occur, not just integer exponents.

If  $\theta \neq 0$ , the choice  $r = \pi/(2\theta)$  yields  $P_\theta(t^r) = it^r$ , which is not real. Finally, conjugation reverses every coefficient phase, giving (4.7).  $\square$

This provides the second strict inclusion in (1.1) without a choice of transcendence basis or a nonconstructive extension. It also proves that none of the following conditions by itself recovers the real axis: fixing ordinary  $\mathbb{C}$ , strong  $\mathbb{C}$ -linearity, exact valuation preservation, or even all three together.

*Related (batch 34).* Adding preservation of the Gaussian omnific integers does not recover it either. The report `surreal/omnific-preserving-automorphisms` prints the phase twist  $P_{-2\pi}$ , with  $\ell$  as in (4.5), as a strongly additive automorphism of  $\mathbb{K}$  that fixes  $\mathbb{C}$ , preserves  $\mathbf{Oz}[i]$  and does not commute with  $c$  (`opa:es:prop:character`). For  $n \neq 0$  the fixed field of its  $n$ th power consists of the series supported on the exponents  $g$  with  $n\ell(g) \in \mathbb{Z}$ , and it is not algebraically closed.

Valuation preservation is much weaker than exact modulus preservation. For example,

$$|1+t| = 1+t, \quad |P_{\pi/2}(1+t)| = |1+it| = \sqrt{1+t^2} \neq 1+t.$$

Likewise,  $P_{\pi/2}$  does not preserve the distinguished unit circle, by theorem 2.4. Algebraic multiplication survives while the chosen Euclidean geometry changes.

#### 4.4 A continuum of explicit conjugations

Set

$$j_\theta = P_\theta c P_\theta^{-1} = P_{2\theta} c, \quad F_\theta = P_\theta(\mathbf{No}). \quad (4.8)$$

Then  $j_\theta^2 = \text{id}$ ,  $\text{Fix}(j_\theta) = F_\theta$ , and  $\mathbb{K} = F_\theta(i)$ . Each  $F_\theta$  is a real closed subfield isomorphic to  $\mathbf{No}$ , but the actual embedded real axes are pairwise distinct. Indeed, equality  $F_\theta = F_\phi$  would make  $P_{\theta-\phi}$  preserve  $\mathbf{No}$ , forcing  $\theta = \phi$ . The fixed-field condition is especially concrete:

$$z \in F_\theta \iff z_g e^{-i\theta\ell(g)} \in \mathbb{R} \quad \text{for all } g. \quad (4.9)$$

These are different *placements* of an isomorphic real form. We will also construct a real form not isomorphic to  $\mathbf{No}$  in Section 10.

### 5 Leading-term kernels and infinitesimal motion

Define

$$\mathcal{U} = \{\alpha \in \text{Aut}(\mathbb{K}) : v(\alpha z - z) > v(z) \text{ for all } z \neq 0\}. \quad (5.1)$$

Here  $v(0) = \infty$ , so fixed inputs satisfy the inequality. Equivalently,  $\alpha$  fixes the complete leading term of every nonzero series. Such maps are often called 1-automorphisms. They can still change very substantial lower-order data. Let  $\mathcal{U}_{\mathbb{R}}$  denote the corresponding leading-term-fixing subgroup of  $\text{Aut}(\mathbf{No})$ .

#### 5.1 A real automorphism need not fix ordinary reals

An order-preserving automorphism  $\sigma$  of  $\mathbf{No}$  fixes every rational. For  $r \in \mathbb{R}$ , rational comparisons imply only

$$\sigma(r) - r \in \mathfrak{m} \cap \mathbf{No},$$

not, for a general  $r$ ,  $\sigma(r) = r$ . For an *irrational* real  $r$ , every infinitesimal perturbation has the same strict rational comparisons: each nonzero ordinary distance  $|r - q|$ ,  $q \in \mathbb{Q}$ , dominates every infinitesimal. Rational  $r$  are fixed already, and real algebraic  $r$  are fixed by their polynomial and order data (theorem 2.1). The following construction makes the distinction explicit.

**Theorem 5.1** (Taylor motion of coefficient constants). *Let  $d : \mathbb{C} \rightarrow \mathbb{C}$  be a field derivation and let  $\delta \in \Gamma$  be positive. Then*

$$\Phi_d \left( \sum_g z_g t^g \right) = \sum_{g, n \geq 0} \frac{d^n(z_g)}{n!} t^{g+n\delta} \quad (5.2)$$

*defines a strong automorphism belonging to  $\mathcal{U}$ , with inverse  $\Phi_{-d}$ . It fixes all monomials  $t^g$ . If  $d(\mathbb{R}) \subseteq \mathbb{R}$  and  $d(i) = 0$ , it preserves **No** and commutes with conjugation. There exist choices for which a real transcendental  $b$  satisfies*

$$\Phi_d(b) = b + t^\delta. \quad (5.3)$$

*Proof.* For an input with support  $S$ , all output exponents lie in  $S + \mathbb{N}\delta$ . Lemma 3.2 proves well-ordering and finite incidence at every exponent. The same argument works for a summable family of inputs and for all double sums needed in a composition. The iterated Leibniz rule

$$d^n(ab) = \sum_{k=0}^n \binom{n}{k} d^k(a) d^{n-k}(b)$$

proves multiplicativity after division by  $n!$ . Additivity is immediate. In  $\Phi_d \Phi_{-d}$ , the coefficient multiplying the  $k$ th derivative at total shift  $k\delta$  is

$$\sum_{n=0}^k \frac{(-1)^n}{n!(k-n)!},$$

which is 1 for  $k = 0$  and 0 otherwise. This proves the inverse formula. Every term with  $n > 0$  has exponent strictly larger than the original minimum; the  $n = 0$  term retains its original leading coefficient. Hence  $\Phi_d \in \mathcal{U}$ . Since  $d(1) = 0$ , each monomial is fixed. For any  $z \neq 0$ , the correction in fact satisfies  $v(\Phi_d(z) - z) \geq v(z) + \delta$ , with  $v(0) = \infty$ . Also  $d(i) = 0$  is automatic from  $d(i^2 + 1) = 2i d(i) = 0$ ; preservation of the real coefficients is the extra condition in the real-axis assertion.

To obtain (5.3), choose a transcendence basis  $B$  of  $\mathbb{R}/\mathbb{Q}$  and an element  $b \in B$ . Define a derivation on  $\mathbb{Q}(B)$  by  $d(b) = 1$  and  $d(b') = 0$  for  $b' \neq b$ . In characteristic zero it extends uniquely through an algebraic extension: for a minimal polynomial  $p(X)$  and a root  $a$ ,

$$d(a) = -\frac{p^d(a)}{p'(a)},$$

where  $p^d$  means differentiate the coefficients. Extend first to  $\mathbb{R}$ , then to  $\mathbb{C}$  by  $d(i) = 0$ . Now  $d^2(b) = 0$ , giving (5.3). All real coefficients stay real, and the construction commutes with  $c$ .  $\square$

Thus even a strongly additive, leading-term-fixing surreal automorphism can move an ordinary real constant. It is not strongly  $\mathbb{R}$ -linear. Confusing strong additivity with coefficient linearity would erase this entire type of automorphism.

The construction uses choice only for a derivation on the set  $\mathbb{R}$ ; no class-sized basis is selected. Its action on a given normal form is then an explicit support-controlled formula. It is generally noncomputable from ordinary numerical names of the real coefficients, as should be expected from its use of a transcendence basis.

## 5.2 Strongly complex-linear flows

There are also abundant nontrivial members of  $\mathcal{U}$  fixing all constants. Let  $\lambda : \Gamma \rightarrow \mathbb{C}$  be additive, and fix  $\delta > 0$ . Define

$$\mathcal{D} \left( \sum_g z_g t^g \right) = \sum_g z_g \lambda(g) t^{g+\delta}. \quad (5.4)$$

The positive shift is important:  $g - \delta$  would generally lower valuation, not raise it.

**Theorem 5.2** (A fixed-shift flow). *The map  $\mathcal{D}$  is a strongly  $\mathbb{C}$ -linear derivation, and for every  $s \in \mathbb{C}$ ,*

$$T_s = \sum_{n \geq 0} \frac{s^n \mathcal{D}^n}{n!} \quad (5.5)$$

*is a strongly  $\mathbb{C}$ -linear automorphism in  $\mathcal{U}$ . The group law is  $T_s T_r = T_{s+r}$ . If  $\lambda$  is real-valued and  $s \in \mathbb{R}$ , then  $T_s$  commutes with  $c$  and preserves **No**.*

*Proof.* Additivity of  $\lambda$  proves the Leibniz rule on monomials, and finite convolution proves it on all series. Iterating gives

$$\mathcal{D}^n(t^g) = \left( \prod_{k=0}^{n-1} \lambda(g + k\delta) \right) t^{g+n\delta}.$$

All supports in (5.5) lie in  $S + \mathbb{N}\delta$ . The support lemma justifies summation and the finite regroupings at each coefficient. The iterated Leibniz identity proves multiplicativity, and the binomial formula gives  $T_s T_r = T_{s+r}$ . Thus  $T_{-s}$  is the inverse. Nonconstant terms of the operator exponential strictly increase every input's valuation, so the leading term is unchanged. Constants are killed by  $\mathcal{D}$  because  $\lambda(0) = 0$ . The real-coefficient assertion is immediate.  $\square$

For a particularly transparent example choose  $\delta = 1$  and  $\lambda = \ell$  from (4.5). Then

$$\mathcal{D}(t) = t^2, \quad \mathcal{D}^n(t) = n! t^{n+1}, \quad T_s(t) = \frac{t}{1 - st}.$$

Consequently  $T_1$  is a nonidentity strongly  $\mathbb{C}$ -linear, real-axis-preserving 1-automorphism. These explicit flows illustrate the automorphism–derivation connection without requiring a blanket assertion that every formally written operator exponential is summable.

### 5.3 What monomial prescriptions do not automatically give

For a strongly  $\mathbb{C}$ -linear element  $\alpha \in \mathcal{U}$ , the assignments

$$u_g = t^{-g} \alpha(t^g) \in 1 + \mathfrak{m}$$

form a group homomorphism  $\Gamma \rightarrow 1 + \mathfrak{m}$ . Formally one would like to write

$$\alpha \left( \sum_g z_g t^g \right) = \sum_g z_g t^g u_g.$$

This is valid for an existing strong automorphism. Conversely, an arbitrary homomorphism  $g \mapsto u_g$  does not by itself certify that this formula is summable for *every* well-ordered support or that it is onto. One must check the transformed support family and construct a compatible inverse. The fixed-shift proofs above supply exactly those missing checks. This is a genuine obstacle in turning a group decomposition into a complete explicit parametrization of its largest kernel. [merge] The inverse half is supplied in general by formal inversion once the forward map is strongly  $\mathbb{C}$ -linear (Remark 32.6); forward summability remains to be checked.

## 6 Decomposing the valued automorphism group

Define

$$G_v = \{\alpha \in \text{Aut}(\mathbb{K}) : \alpha(\mathcal{O}) = \mathcal{O}\}.$$

Preserving the valuation here means preserving its comparison structure, not necessarily each value. Such an  $\alpha$  induces

$$\rho_\alpha \in \text{Aut}(\mathbb{C}), \quad \tau_\alpha \in \text{Aut}_{\text{ord}}(\Gamma), \quad v(\alpha z) = \tau_\alpha(vz).$$

The residue action is  $\rho_\alpha(\text{st } z) = \text{st}(\alpha z)$  for finite  $z$ . This does not assert  $\alpha(\mathbb{C}) = \mathbb{C}$ : the coefficient-motion examples have identity residue action while sending constants out of the constant field.

**Theorem 6.1** (Four layers for surcomplex valued automorphisms). *With  $\mathcal{U}$  as in (5.1), every  $\alpha \in G_v$  factors uniquely as*

$$\alpha = u D_\chi M_{\rho, \tau}, \tag{6.1}$$

where  $u \in \mathcal{U}$ ,  $\chi \in \text{Hom}(\Gamma, \mathbb{C}^\times)$ , and

$$D_\chi \left( \sum z_g t^g \right) = \sum z_g \chi(g) t^g, \quad M_{\rho, \tau} \left( \sum z_g t^g \right) = \sum \rho(z_g) t^{\tau(g)}.$$

(In the notation of (4.1),  $D_\chi = M_{\text{id}, \text{id}, \chi}$  and  $M_{\rho, \tau} = M_{\rho, \tau, 1}$ ; the maps  $D_\lambda$  of Section 4 are the  $D_\chi$  with  $\chi = \exp \circ \lambda$ .) In group notation,

$$G_v \simeq (\mathcal{U} \rtimes \text{Hom}(\Gamma, \mathbb{C}^\times)) \rtimes (\text{Aut}(\mathbb{C}) \times \text{Aut}_{\text{ord}}(\Gamma)). \tag{6.2}$$

The outer action on the character factor is

$$((\rho, \tau) \cdot \chi)(g) = \rho(\chi(\tau^{-1}g)). \tag{6.3}$$

The other actions are conjugation by the displayed lifts.

*Proof.* Stabilizing  $\mathcal{O}$  also stabilizes its units and its maximal ideal, so it induces the residue automorphism  $\rho_\alpha$ . Define directly  $\tau_\alpha(g) = v(\alpha(t^g))$ . If  $v(z) = g$ , then  $z/t^g$  is a unit, so  $v(\alpha z) = \tau_\alpha(g)$ . This proves independence from the chosen representative. Multiplication proves additivity; membership of quotients in  $\mathcal{O}$  proves order preservation; the same construction for  $\alpha^{-1}$  supplies the inverse. Thus no quotient whose elements would be proper classes needs to be formed. The maps  $M_{\rho, \tau}$  give a section of the induced-action homomorphism, by the monomial construction. Let  $I$  be its kernel and put  $\beta = \alpha M_{\rho, \tau}^{-1} \in I$ .

For  $\beta \in I$ , define

$$\chi_\beta(g) = \text{lc}(\beta(t^g)).$$

Since  $\beta$  fixes values, multiplicativity of leading coefficients gives  $\chi_\beta(g+h) = \chi_\beta(g)\chi_\beta(h)$ . For an arbitrary nonzero  $z$ , write  $z = at^g(1+\varepsilon)$  with  $a = \text{lc } z$  and  $v\varepsilon > 0$ . Identity residue action gives  $\text{lc}(\beta a) = a$ , and  $\beta(1+\varepsilon)$  still has leading coefficient 1. Hence

$$\text{lc}(\beta z) = \text{lc}(z)\chi_\beta(vz). \tag{6.4}$$

It follows that  $\beta \mapsto \chi_\beta$  is a group homomorphism on  $I$ , with kernel exactly  $\mathcal{U}$ . The maps  $D_\chi$  split it. Thus  $u = \beta D_{\chi_\beta}^{-1}$  gives (6.1), and all parameters are forced by the preceding operations. More explicitly, the character in that factorization is  $\chi(g) = \text{lc}(\alpha(t^{\tau^{-1}(g)}))$ ; it is indexed after undoing the value action. The subgroup  $\mathcal{U}$  is normal in  $G_v$ , since conjugation by a valued automorphism transports its strict valuation inequality through the ordered map  $\tau$ . Evaluating  $M_{\rho, \tau} D_\chi M_{\rho, \tau}^{-1}$  on a monomial gives (6.3).  $\square$

This direct argument is the full-Hahn instance of the valued-field structure studied in [3]. It also clarifies its limits: (6.2) is not a decomposition of *all* surcomplex field automorphisms. We will exhibit automorphisms outside  $G_v$  in Section 9.

The same proof works with the strong subgroup, replacing  $\mathcal{U}$  by its strong part. All explicit sections are strong, so strength of an automorphism is concentrated in the residual kernel factor. The factor  $\mathcal{U}$  itself is not replaced by a conveniently finite list of coefficients. A strong field map is determined by its action on *both* ordinary coefficients and monomials, using  $\alpha(\sum z_g t^g) = \sum \alpha(z_g) \alpha(t^g)$ . Monomial images alone suffice when it also fixes  $\mathbb{C}$  pointwise. Strongness alone does not remove the coefficient data: the nonidentity Taylor motions theorem 5.1 fix every monomial. Without strongness, the displayed reconstruction is not licensed: Section 7 gives an automorphism that fixes every summand of a Hahn family but moves its sum.

## 6.1 The real reduction

Every field automorphism of  $\mathbf{No}$  preserves order, hence preserves the natural valuation ring. Its residue action on  $\mathbb{R}$  is an ordered-field automorphism and therefore identity. Applying the same proof over  $\mathbb{R}$  gives

$$\mathrm{Aut}(\mathbf{No}) \simeq (\mathcal{U}_{\mathbb{R}} \rtimes \mathrm{Hom}(\Gamma, \mathbb{R}_{>0})) \rtimes \mathrm{Aut}_{\mathrm{ord}}(\Gamma). \quad (6.5)$$

Every character into  $\mathbb{R}^\times$  is positive because  $\Gamma$  is divisible. Applying the ordinary logarithm pointwise identifies its group with  $\mathrm{Hom}(\Gamma, \mathbb{R})$  under addition.

Combining (6.5) with the real-axis theorem gives a structural description of  $G_{\mathbf{No}}$ :

$$G_{\mathbf{No}} \simeq [(\mathcal{U}_{\mathbb{R}} \rtimes \mathrm{Hom}(\Gamma, \mathbb{R}_{>0})) \rtimes \mathrm{Aut}_{\mathrm{ord}}(\Gamma)] \times C_2. \quad (6.6)$$

In particular, “residue action is identity” must not be shortened to “all embedded real constants are fixed.” Theorem 5.1 is an explicit counterexample to that shortcut.

## 6.2 One level deeper: the ordered additive value group

Because  $\Gamma = (\mathbf{No}, +, <)$  is itself the ordered additive group of real Hahn series, one can further analyze  $\mathrm{Aut}_{\mathrm{ord}}(\Gamma)$ . Let  $\mathcal{U}_{\mathrm{add}}$  consist of its additive automorphisms fixing each leading term. For an order bijection  $\zeta : \Gamma \rightarrow \Gamma$  and positive real scalars  $(a_g)_{g \in \Gamma}$ , the formula

$$A_{\zeta, a} \left( \sum_g r_g t^g \right) = \sum_g a_g r_g t^{\zeta(g)} \quad (6.7)$$

defines an ordered additive automorphism. Here  $\zeta$  need not be additive: this is an additive-group construction, not a field construction.

An ordered additive automorphism  $\tau$  permutes the Archimedean classes in order. Its parameters can be read directly as  $\zeta(g) = v(\tau(t^g))$  and  $a_g = \mathrm{lc}(\tau(t^g)) > 0$ . For fixed  $g$ , comparison of  $rt^g$  with rational multiples of  $t^g$  shows that the leading coefficient of  $\tau(rt^g)$  is  $a_g r$  for every  $r \in \mathbb{R}^\times$ , at value  $\zeta(g)$ . A tail of value greater than  $g$  is infinitesimal relative to  $t^g$  in the ordered additive group, and its image has value greater than  $\zeta(g)$ . Thus the leading term of the image of any input with leading term  $rt^g$  is  $a_g r t^{\zeta(g)}$ . This uses only order and additivity, not strong summation or a quotient whose elements would be proper classes. Removing (6.7) leaves exactly  $\mathcal{U}_{\mathrm{add}}$ , and the parameters are unique. This is the additional kernel–skeleton factorization, parallel to [4].

It is important to retain the reindexing action in its group law. Direct calculation gives

$$A_{\zeta, a} A_{\eta, b} = A_{\zeta \eta, d}, \quad d_g = a_{\eta(g)} b_g. \quad (6.8)$$

Thus these parameters have a semidirect-product law, not componentwise multiplication of an unrelated permutation and weight family. The ordinary logarithm turns positive weights into additive real weights but does not remove their reindexing. Together with (6.5), this identifies five structural layers on the real side: field leading-term motion, characters, additive leading-term motion, exponent-order permutations, and Archimedean coefficient weights.

## 7 Fine topology does not mean strong summation

The fine topology on  $\mathbb{K}$  is generated by the balls

$$B(a, r) = \{z : |z - a| < r\}, \quad r \in \mathbf{No}_{>0}.$$

By (3.2), it is the same topology as the valuation topology with neighborhoods

$$V_\gamma(a) = \{z : v(z - a) > \gamma\}, \quad \gamma \in \Gamma.$$

For example,  $V_{v(r)}(a) \subseteq B(a, r)$ , while a ball of radius  $t^{\gamma+1}$  is contained in  $V_\gamma(a)$ . These inclusions suffice to identify the neighborhood systems.

**Proposition 7.1.** *Every element of  $G_v$  is a fine homeomorphism and a uniform equivalence. In particular, every real-axis-preserving surcomplex field automorphism is fine-continuous.*

*Proof.* If  $\alpha$  induces  $\tau$  on  $\Gamma$ , then

$$v(\alpha z - \alpha a) = \tau(v(z - a)).$$

The order bijection  $\tau$  takes the neighborhood basis to itself: for a target threshold  $\eta$ , use the input threshold  $\tau^{-1}(\eta)$ . This threshold is independent of the base point. Applying the same equation to pairs of points proves uniform continuity of both  $\alpha$  and its inverse. The real-axis subgroup lies in  $G_v$  by theorem 2.1 and (2.4).  $\square$

This is an inclusion in the continuous automorphism group, not a classification of that whole group. In particular, we do not infer a converse merely from continuity. The chosen valuation ring contains additional information beyond the neighborhood calculation used here.

The phase twists are therefore continuous even though they do not preserve the real axis. Lifting any ordinary field automorphism  $\rho$  of  $\mathbb{C}$  coefficientwise gives further examples in  $G_v$ . There is no contradiction with the familiar classification of continuous automorphisms of ordinary  $\mathbb{C}$ : the topology induced on  $\mathbb{C} \subset \mathbb{K}$  is *discrete*, not its usual Archimedean topology. A ball of infinitesimal radius around a constant contains no other ordinary complex constant.

### 7.1 Why small nets cannot test the full topology

**Proposition 7.2.** *Every set-sized subset of the full surcomplex field is closed and discrete. Every convergent set-indexed net is eventually equal to its limit. Nevertheless, the full field has no isolated points.*

*Proof.* For a set  $S \subseteq \mathbb{K}$  and a point  $a$ , the values  $\{v(s - a) : s \in S, s \neq a\}$  form a set of surreal numbers. Every set of surreals has an upper bound, so choose  $\gamma$  strictly above all of them. Then  $V_\gamma(a)$  misses  $S \setminus \{a\}$ . This proves both closedness and discreteness. Apply the same argument to the range of a set-indexed net together with its limit to get eventual equality. Finally,  $a + t^{\gamma+1} \neq a$  belongs to  $V_\gamma(a)$ , so no point of the full field is isolated.  $\square$

For example,  $t^n$  does not tend to zero in the full surreal fine topology: none of these terms lies in  $V_\omega(0)$ . In the fixed field  $\mathbb{C}((t^\mathbb{Q}))$ , by contrast, the integers are cofinal in its value group and  $t^n \rightarrow 0$  in its intrinsic valuation topology. A set-sized Hahn workspace inside  $\mathbb{K}$  acquires a discrete subspace topology from the full field, even when its intrinsic valuation topology is nondiscrete. The choice of topology is part of the theorem, not merely a presentational convention.

## 7.2 A continuous automorphism that fails to preserve a Hahn sum

We give an explicit support-correct witness. We will use the standard Newton–Puiseux fact that

$$\mathcal{P} = \bigcup_{m \geq 1} \mathbb{R}((t^{1/m}))$$

is real closed. Indeed,  $\mathcal{P}(i) = \bigcup_{m \geq 1} \mathbb{C}((t^{1/m}))$  is algebraically closed by Newton–Puiseux [19]; an ordered field with algebraically closed quadratic extension by  $i$  is real closed. This set-sized Puiseux field is also discussed among the special cases in [3, Section 5.5]. Define

$$a = \sum_{n \geq 2} t^{1-1/n}, \quad \varepsilon = t^\omega. \quad (7.1)$$

The support  $\{1/2, 2/3, 3/4, \dots\}$  is a well-ordered increasing sequence. Its denominators are unbounded, so  $a \notin \mathcal{P}$ . Since  $\mathcal{P}$  is real closed,  $a$  is transcendental over  $\mathcal{P}$ .

For each  $p \in \mathcal{P}$ , the nonzero difference  $a - p$  has rational support and hence rational valuation. Thus  $\varepsilon$  is smaller than  $|a - p|$ . It follows that  $a$  and  $a + \varepsilon$  have the same cut over  $\mathcal{P}$ . An ordered isomorphism  $\mathcal{P}(a) \rightarrow \mathcal{P}(a + \varepsilon)$  fixing  $\mathcal{P}$  and moving  $a$  therefore exists; real closedness of  $\mathcal{P}$  ensures that signs of rational functions are determined by this cut. Extend it through real closures and then to an automorphism  $\sigma$  of  $\mathbf{No}$  by the ordered extension theorem proved in Section 9.

Every summand in (7.1) belongs to  $\mathcal{P}$  and is fixed. But

$$\sigma(a) = a + \varepsilon \neq a = \sum_{n \geq 2} \sigma(t^{1-1/n}).$$

The lift of  $\sigma$  to  $\mathbb{K}$  fixes ordinary  $\mathbb{C}$ , commutes with conjugation, and is fine-continuous, but is not strong. We have not claimed that this particular extension fixes every value or every leading term; those stronger assertions are unnecessary here.

There is also a finite-observation obstruction to approximation by strong maps. Any real surreal automorphism fixing  $t$  fixes all its positive rational powers, by uniqueness of positive roots. If it is strong, it fixes  $a$ . Hence the neighborhood specified by

$$\psi(t) = t, \quad \psi(a) = a + \varepsilon$$

in the pointwise-equality topology on  $\text{Aut}(\mathbf{No})$  contains  $\sigma$  and contains no strong automorphism. This proves non-density in that particular automorphism-group topology, not in every possible topology on the group.

The same finite observations exclude strong maps in the full  $\text{Aut}(\mathbb{K})$ . If  $\psi(t) = t$  and  $q_n = 1 - 1/n$ , then  $\psi(t^{q_n})^n = t^{n-1}$ , so  $\psi(t^{q_n}) = \zeta_n t^{q_n}$  with  $\zeta_n$  an ordinary complex  $n$ th root of unity. Strongness would force  $\psi(a) = \sum_{n \geq 2} \zeta_n t^{q_n}$ , whose support is contained in  $\mathbb{Q}$  and cannot contain the exponent  $\omega$  of  $a + \varepsilon$ . The lifted witness satisfies both observations, so the strong subgroup is not dense there either. This argument requires neither preservation of the real axis nor fixation of all ordinary complex constants by  $\psi$ .

## 8 Differentiability: a surprising zero-derivative phenomenon

For a function  $f : \mathbb{K} \rightarrow \mathbb{K}$ , its fine derivative at  $z$  is the punctured-neighborhood limit

$$f'(z) = \lim_{h \rightarrow 0, h \neq 0} \frac{f(z+h) - f(z)}{h},$$

when this limit exists in the full fine topology. The limit quantifies over all small surcomplex increments; it is not tested only on an ordinary sequence. This is the same difference-quotient convention encoded in the repository's generic `FineHasDerivAt` interface [13].

**Theorem 8.1** (Derivative dichotomy for field automorphisms). *If a field automorphism  $\alpha$  of  $\mathbb{K}$  has a fine derivative at one point, it has the same derivative at every point. If that derivative is nonzero, then  $\alpha = \text{id}$  and the derivative is 1. Nonidentity differentiable automorphisms are nevertheless possible: they must have derivative 0 everywhere.*

*Proof.* Additivity makes every difference quotient equal to  $q(h) = \alpha(h)/h$ , independently of the base point. For a fixed  $b \neq 0$ ,

$$q(bh) = \frac{\alpha(b)}{b} q(h).$$

If  $q(h) \rightarrow L$ , multiplication by  $b$  is a homeomorphism of the punctured neighborhood of zero, so the left-hand side also tends to  $L$ . Hausdorff uniqueness gives  $L = \alpha(b)L/b$ . When  $L \neq 0$ , cancellation forces  $\alpha(b) = b$  for every nonzero  $b$ . Thus  $\alpha = \text{id}$ , whose quotient is constantly 1.  $\square$

**Theorem 8.2** (Exponent dilations have derivative zero). *For the maps  $S_a$  of (4.4),*

$$S'_a \equiv 0 \quad (a > 1), \quad S'_1 \equiv 1.$$

*For  $0 < a < 1$ ,  $S_a$  has no  $\mathbb{K}$ -valued fine derivative at any point. In particular,  $S_2$  is a nonidentity field automorphism and a fine homeomorphism with derivative zero everywhere.*

*Proof.* For every nonzero  $h$ ,

$$v\left(\frac{S_a(h)}{h}\right) = (a-1)v(h). \tag{8.1}$$

If  $a > 1$ , given any  $\eta \in \Gamma$ , taking  $v(h) > \eta/(a-1)$  forces the quotient into  $V_\eta(0)$ . Thus the quotient tends to zero. This argument works for positive infinitesimal  $a-1$  as well as for ordinary positive  $a-1$ . If  $0 < a < 1$ , theorem 8.1 would force any derivative to be zero, since  $S_a(t) = t^a \neq t$ . But for every proposed input threshold  $\gamma_0$ , choose  $\gamma > \max\{\gamma_0, 0\}$  and put  $h = t^\gamma$ . Then  $h \in V_{\gamma_0}(0)$ , while the quotient has value  $(a-1)\gamma < 0$  and lies outside  $V_0(0)$ . This rules out a zero limit using the full neighbourhood criterion, without a set-indexed cofinal net.  $\square$

Conjugating the input does not change its valuation. Thus  $S_a c$  also has zero derivative for  $a > 1$ , even though  $c$  by itself is not surcomplex-differentiable. For  $c$ , real and imaginary increments give quotients 1 and  $-1$  at arbitrarily small scales.

**Corollary 8.3.** *If a field automorphism and its inverse are both fine-differentiable, the automorphism is identity.*

*Proof.* Differentiability implies continuity by writing an increment of the function as its difference quotient times the input increment. If both maps were nonidentity and differentiable, both derivatives would be zero. Put  $k = \alpha(h)$ ; continuity implies  $k \rightarrow 0$  as  $h \rightarrow 0$ . The product

$$\frac{\alpha(h)}{h} \frac{\alpha^{-1}(k)}{k} = 1$$

would then tend to zero, a contradiction. Injectivity ensures  $k \neq 0$  for  $h \neq 0$ , so the inverse's punctured limit applies. The identity satisfies both conditions.  $\square$

These results do not assert that  $S_2$  is representable by a locally convergent power series with fixed coefficients. They concern the fine difference-quotient definition. In this setting, “everywhere differentiable” cannot be used as a synonym for ordinary complex analytic rigidity, and a zero derivative does not imply local constancy:  $S_2$  is injective and the full field has no isolated points. The Berarducci–Mantova scalar derivation discussed later is a different operation again.

## 9 The unrestricted field: homogeneity and fixed fields

The preceding explicit formulas already disprove real-axis rigidity. The full pure-field picture is even less rigid. To discuss it without misusing a set-sized extension theorem on a proper class, we give the extension argument directly.

**Theorem 9.1** (Class back-and-forth). *Assume global choice. Let  $K_1, K_2$  be proper-class algebraically closed fields of the same characteristic. Every isomorphism between set-sized subfields of  $K_1$  and  $K_2$  extends to an isomorphism  $K_1 \rightarrow K_2$ . In particular, every isomorphism between set-sized subfields of  $\mathbb{K}$  extends to a field automorphism of  $\mathbb{K}$ .*

*Proof.* Fix a set-like global well-order and the resulting class enumerations of  $K_1$  and  $K_2$ , indexed by **On**. Recursively maintain an isomorphism  $f_\alpha : A_\alpha \rightarrow B_\alpha$  between *set-sized* subfields. At a forth step, take the next element  $x$  of  $K_1$ . If  $x$  is algebraic over  $A_\alpha$ , send it to a root in  $K_2$  of its transported irreducible polynomial, extending the field isomorphism. If  $x$  is transcendental, choose a transcendental element over  $B_\alpha$ . This is possible because the elements of  $K_2$  algebraic over a set-sized subfield form a set: there are set many polynomials and each has finitely many roots. A proper-class field cannot be exhausted by it.

At each successor stage perform both a forth step and a back step, the latter for the next element of  $K_2$  using the inverse partial map; skip an element already in the relevant partial field. Always select the least admissible witness in the fixed global well-order. At a limit ordinal take unions. There are only set many earlier stages at any given ordinal, so the partial domains and ranges remain sets. The class union of all partial maps is a field isomorphism; forth steps exhaust  $K_1$  and back steps exhaust  $K_2$ . Each partial map is a set. The deterministic rule has a unique recursion of every set ordinal length, with the ambient fields and well-order as class parameters; these compatible recursions define the class union. The candidate witnesses can form a proper class, but its least member is selected by the global well-order. No recursion whose stage value is itself a proper class, or class-sized transcendence basis, is needed.  $\square$

*Related (batch 34).* The report `surreal/set-sized-quotients-of-omnific-integers` prints the forth half of this argument with target  $\mathbb{K}$  as `osq:cr:thm:universalfield`: every set or class field of characteristic zero embeds in  $\mathbb{K}$ , and every embedding of a set-sized subfield extends. It uses this to map the class residue fields of omnific rings into  $\mathbb{K}$ . Its corollary `osq:cr:cor:acfiso`, that every proper-class algebraically closed field of characteristic zero is isomorphic to  $\mathbb{K}$ , is the case of prime subfields of [theorem 9.1](#), as that report says.

**Theorem 9.2** (Ordered surreal extension). *Under the same foundational convention, every ordered-field isomorphism between set-sized subfields of **No** extends to an automorphism of **No**.*

*Proof.* First extend the map to real closures inside **No**; uniqueness of ordered real closures supplies the extension. Run a back-and-forth as above, now keeping the partial fields real closed. An element  $x$  outside a

partial real closed field  $A$  is transcendental over it. Its comparisons with  $A$  form a set-sized cut  $(\mathcal{L}, \mathcal{R})$  with  $\mathcal{L} \cup \mathcal{R} = A$ . Transport this cut to the current range  $B$ . The surreal cut property supplies an element  $y$  strictly between every element of its left and right sides. It cannot belong to  $B$ , since every element of  $B$  occurs on one of those sides. Thus  $y$  is transcendental over  $B$ .

The substitution  $x \mapsto y$  gives an ordered isomorphism of rational function fields. To see preservation of signs, factor polynomials over the real closed coefficient field into linear factors and quadratic factors without real roots; their signs are determined by the cut and leading coefficient. Extend again to real closures. At limits, unions of increasing real closed fields are real closed, since each polynomial has its finitely many coefficients in one stage. Forth and back exhaust **No**.  $\square$

These proofs explain why field homogeneity is so strong and why it should not be confused with simplicity preservation. The selected separator  $y$  need not have the same birthday or sign-sequence complexity as  $x$ .

## 9.1 Orbits over ordinary complex constants

Every element of  $\mathbb{K} \setminus \mathbb{C}$  is transcendental over  $\mathbb{C}$ , since  $\mathbb{C}$  is algebraically closed. Therefore, for  $x, y \notin \mathbb{C}$ , the substitution  $x \mapsto y$  is an isomorphism  $\mathbb{C}(x) \rightarrow \mathbb{C}(y)$  and extends by [theorem 9.1](#). We obtain

$$\text{Aut}(\mathbb{K}/\mathbb{C}) \text{ is transitive on } \mathbb{K} \setminus \mathbb{C}. \quad (9.1)$$

The analogous statement for finite algebraically independent tuples follows by the same rational-function argument. Thus a pure-field automorphism fixing every ordinary complex number can send an infinite surreal to an infinitesimal, or to an element off the real axis.

**Proposition 9.3.** *For every set  $A \subseteq \mathbb{K}$ , there are automorphisms fixing  $A \cup \mathbb{C}$  pointwise which do not preserve **No**. There are also such automorphisms which do not preserve  $\mathcal{O}$ . Consequently, neither the distinguished real axis nor the natural valuation ring is invariant under all pure-field automorphisms over any prescribed set of parameters.*

*Proof.* Let  $E$  be the algebraic closure inside  $\mathbb{K}$  of the field generated by  $A \cup \mathbb{C}$ . It is a set. Choose a positive infinite  $x \in \mathbf{No} \setminus E$ ; for instance choose  $x$  larger than the moduli of all elements of  $E$ . Then  $x$  is transcendental over  $E$ . Both maps

$$E(x) \longrightarrow E(ix), \quad x \mapsto ix, \quad E(x) \longrightarrow E(x^{-1}), \quad x \mapsto x^{-1}$$

are field isomorphisms fixing  $E$ . Extend each to  $\mathbb{K}$ . The first moves a real element off the real axis. The second sends an element outside  $\mathcal{O}$  into its maximal ideal, so cannot stabilize  $\mathcal{O}$ .  $\square$

In particular, these structures are not first-order definable in the pure field with parameters: definable sets are invariant under automorphisms fixing their parameters. This conclusion is stronger than merely saying that no preferred coordinate system has yet been chosen. Combining the proposition with the phase example gives the proper chain

$$G_{\mathbf{No}} \subsetneq G_v \subsetneq \text{Aut}(\mathbb{K}). \quad (9.2)$$

The first strictness has an explicit strong,  $\mathbb{C}$ -linear, value-fixing witness; the second has a class back-and-forth witness.

## 9.2 Fixed fields of the main collections

Let  $\mathbb{Q}^{\text{rc}} \subset \mathbb{R}$  be the field of real algebraic numbers and let  $\overline{\mathbb{Q}} = \mathbb{Q}^{\text{rc}}(i) \subset \mathbb{C}$ .

**Proposition 9.4.** *In the class-map interpretation,*

$$\begin{aligned} \text{Fix}(\text{Aut}(\mathbb{K})) &= \mathbb{Q}, & \text{Fix}(\text{Aut}(\mathbb{K}/\mathbb{C})) &= \mathbb{C}, \\ \text{Fix}(\text{Aut}(\mathbf{No})) &= \mathbb{Q}^{\text{rc}}, & \text{Fix}(G_{\mathbf{No}}) &= \mathbb{Q}^{\text{rc}}. \end{aligned}$$

*The fixed field of the  $i$ -fixing lifts of all surreal automorphisms is  $\overline{\mathbb{Q}}$ .*

*Proof.* A nonrational algebraic element has a distinct conjugate in characteristic zero, and its conjugacy isomorphism extends by [theorem 9.1](#). A transcendental element can similarly be moved by a rational-function isomorphism. This gives the first assertion. Transitivity (9.1) gives the second.

Every surreal automorphism fixes  $\mathbb{Q}^{\text{rc}}$  by order. For  $x \notin \mathbb{Q}^{\text{rc}}$ ,  $x$  is transcendental over this real closed field. Choose a positive  $\varepsilon$  smaller than all nonzero distances  $|x - a|$  for  $a \in \mathbb{Q}^{\text{rc}}$ . This is possible for a set of distances by the surreal cut property. Then  $x$  and  $x + \varepsilon$  have the same cut, so the ordered extension theorem moves  $x$ . This proves the third assertion. For the last two assertions apply the coordinate formula (2.3); allowing  $c$  kills the imaginary fixed coordinate, whereas forbidding  $c$  leaves both coordinates in  $\mathbb{Q}^{\text{rc}}$ .  $\square$

## 9.3 Why a transcendence-basis slogan is not a classification

For a set-sized algebraically closed field, specifying a transcendence basis is useful for producing automorphisms: first choose compatible images of transcendental generators and then extend through algebraic elements. But an automorphism need not permute one fixed basis, and its action on algebraic elements is additional data. Even there, “the automorphism group is the symmetric group on a transcendence basis” is false.

For the full surcomplex field, the corresponding basis would be a proper class and requires additional foundational care. The set-stage back-and-forth proof is sufficient for all existential statements here. It gives neither a computational procedure on arbitrary surcomplex inputs nor a canonical parametrization of all class automorphisms. The much more explicit decomposition in Section 6 succeeds precisely because it retains a valuation and a normal-form presentation.

## 10 Involutions and genuinely different real forms

An *involution* in this section means an automorphism of order exactly two. A *real form* of  $\mathbb{K}$  means a real closed subfield  $F \subset \mathbb{K}$  such that  $\mathbb{K} = F(i)$ . Its conjugation is the involution fixing  $F$  and sending  $i$  to  $-i$ . The fields  $F_\theta$  from (4.8) already give many examples, all isomorphic to the distinguished  $\mathbf{No}$ .

We use the classical Artin–Schreier theorem: if an algebraically closed field is a proper finite extension of a subfield, that subfield is real closed, the characteristic is zero, and the extension has degree two. A proof and the finite-automorphism application are given in [8]. We apply these ordinary set-field theorems inside invariant set-sized algebraically closed subfields, as follows.

**Theorem 10.1** (Finite subgroups and real forms). *Every nontrivial finite subgroup of  $\text{Aut}(\mathbb{K})$  has order two. Every nonidentity finite-order automorphism is an involution  $j$ , its fixed field  $F_j$  is real closed, and  $\mathbb{K} = F_j(i)$ . In particular,  $j(i) = -i$ . Consequently  $\text{Aut}(\mathbb{K}/\mathbb{C})$  and the leading-term kernel  $\mathcal{U}$  are torsion-free. Within  $G_{\mathbf{No}}$ , the only nonidentity torsion element is  $c$ .*

*Proof.* For a nontrivial finite indexed group  $H$ , choose for each nonidentity element a moved point. Adjoin  $i$  and close this finite set under  $H$ , then take the relative algebraic closure in  $\mathbb{K}$  of its generated field. The result is a set-sized,  $H$ -stable algebraically closed field  $E$ , and the restricted action is faithful. Artin’s fixed-field theorem and Artin–Schreier give  $[E : E^H] = |H| = 2$ . Applying this to a finite cyclic action shows that every nonidentity finite-order automorphism is an involution.

For such an involution  $j$ , the same construction can include any prescribed set of elements of  $\text{Fix}(j)$ . Each  $E^j$  is real closed, and these fields form a directed family with union  $F_j = \text{Fix}(j)$ ; hence  $F_j$  is real closed. Because  $i \in E$  and  $E^j$  is real closed,  $j(i) = -i$ . For every  $z \in \mathbb{K}$ ,

$$z = \frac{z + jz}{2} + i \frac{z - jz}{2i},$$

and both displayed coordinates are fixed by  $j$ . Thus  $\mathbb{K} = F_j(i)$ . An automorphism fixing  $\mathbb{C}$  cannot be such an involution. A member of  $\mathcal{U}$  must fix  $i$ , since changing it to  $-i$  changes its leading coefficient, and is therefore also torsion-free. Finally, an order-preserving bijection of a linear order has no nontrivial finite order: a moved point has a strictly monotone finite orbit segment. Use  $G_{\mathbf{No}} \simeq \text{Aut}(\mathbf{No}) \times C_2$ .  $\square$

Two involutions are conjugate in  $\text{Aut}(\mathbb{K})$  exactly when their fixed real fields are isomorphic. One direction follows by restricting a conjugating automorphism. Conversely an isomorphism  $F_j \rightarrow F_k$  extends by  $x + iy \mapsto f(x) + if(y)$ , and this extension conjugates  $j$  to  $k$ . Thus classifying involutions is a real-form classification problem, not merely the observation that  $i$  has two possible images.

*Related (batch 34).* The report `surreal/omnific-preserving-automorphisms` classifies the set groups that act faithfully on  $\mathbb{K}$  by automorphisms commuting with  $c$ : they are the left-orderable groups and the products  $G_0 \times C_2$  with  $G_0$  left-orderable, and the same holds for ring actions on  $\mathbf{Oz}[i]$  commuting with  $c$  (`opa:es:thm:complexmain`). The admissible groups for  $\mathbf{No}$  and for  $\mathbf{Oz}$  are exactly the left-orderable ones (`opa:es:cor:classification`). A second source rederives the classification without parameters in `opa:us:sec:complex`. Its consequence that such finite groups have order at most two is a special case of [theorem 10.1](#), which does not assume commutation with  $c$ . These results concern actions that commute with the named conjugation; they classify neither real forms nor the automorphisms of the pure field.

## 10.1 A real form not isomorphic to the surreal field

The following construction shows that not all involutions of  $\mathbb{K}$  are conjugate to its distinguished conjugation.

**Theorem 10.2.** *Assuming the class-theoretic framework above,  $\mathbb{K}$  has a real form with a countable cofinal subset. It is not isomorphic as a field to  $\mathbf{No}$ . The associated involution is not conjugate to  $c$ .*

*Proof.* Form the rational function field  $\mathbf{No}(T)$  with  $T$  transcendental, ordered so that  $T$  is larger than every surreal coefficient. The sign of a nonzero rational function is determined by the ratio of its leading coefficients. Let

$$H = \mathbf{No}(T)^{\text{rc}}$$

be an ordered real closure. To construct this in class foundations, choose a set-like enumeration of  $\mathbf{No}(T)$  and increasing set-sized subfields  $A_\alpha$  exhausting it. For each  $\alpha$  choose a set-coded ordered real closure  $R_\alpha$  of  $A_\alpha$ , using the fixed global well-order. For  $\alpha \leq \beta$ , there is a unique ordered embedding  $R_\alpha \rightarrow R_\beta$  extending  $A_\alpha \subseteq A_\beta$ . Uniqueness follows by preserving the order of the finitely many real roots of each polynomial, and makes these embeddings compatible under composition. Take the direct limit: represent an element by the least equivalent stage–element pair in the set-like global well-order. Thus the elements are sets, not proper-class equivalence classes. Every finite tuple occurs in one stage, so the resulting class field is real closed; every

element is algebraic over the embedded  $\mathbf{No}(T)$ . Identifying that canonical image with  $\mathbf{No}(T)$  gives the required extension. This is a coherent set-stage construction, not an application of a class Zorn lemma.

The powers  $T, T^2, T^3, \dots$  are cofinal in  $\mathbf{No}(T)$ . Indeed, each rational function has finite degree and each nonzero coefficient and its reciprocal is bounded in absolute value by  $T$ . Increasing an integer power of  $T$  therefore dominates the function. The same powers remain cofinal in  $H$ : if  $h$  is algebraic, a monic polynomial for  $h$  has finitely many coefficients, and the usual algebraic root bound

$$|h| \leq 1 + \max_{k < n} |a_k| \quad \text{when } h^n + a_{n-1}h^{n-1} + \dots + a_0 = 0$$

is valid in any ordered field. Its right-hand side is bounded by a power of  $T$ .

The field  $H$  is proper-class-sized because it contains  $\mathbf{No}$ . Its complexification  $H(i)$  is algebraically closed of characteristic zero. By [theorem 9.1](#), it is isomorphic as a pure field to  $\mathbb{K}$ . Transport  $H$  and its conjugation across this isomorphism. The transported real form has a countable cofinal subset. By contrast, every set-sized subset of  $\mathbf{No}$  is bounded above, so  $\mathbf{No}$  has no countable cofinal subset. A field isomorphism between real closed fields preserves their unique orders; hence these real forms cannot be isomorphic. The conjugacy criterion for involutions proves the final assertion.  $\square$

This separates two phenomena. The phase family gives distinct embedded copies of the *same* real form. [Theorem 10.2](#) gives a different isomorphism type of real form. Neither result supplies a classification of all real closed class fields whose complexification is  $\mathbb{K}$ . [\[merge\] Section 23](#) classifies the real forms whose involution preserves the valuation ring and  $\mathbf{Oz}[i]$  ([Theorem 16.2](#)); the real form of [Theorem 10.2](#) is not of that kind, even up to conjugacy.

## 11 Simplicity, the omega-map, and exponential structure

The large automorphism groups above forget much of what makes a surreal number canonical. Adding back suitable structure can impose rigidity, but different added structures impose different conditions.

### 11.1 Simplicity restores the canonical surreal numbers

Let  $<_s$  denote proper-prefix simplicity on ordinal-length surreal sign sequences. A bijection preserving both numerical order and simplicity is identity. For completeness, fix the empty sign sequence. At a successor stage, the two immediate simplicity extensions of a fixed sequence lie on opposite numerical sides, so both are fixed. At a limit stage, the unique sign sequence whose proper initial segments are the already fixed chain is fixed. Transfinite induction proves the assertion.

Thus a member of  $G_{\mathbf{No}}$  whose real restriction preserves simplicity is either identity or conjugation; marking  $i$  eliminates conjugation. This is rigidity of an enriched structure, not rigidity of the real closed field reduct. A transformation such as  $S_2$  is allowed to change birthdays and simplicity and therefore does not contradict it.

### 11.2 The omega-map detects exact value preservation

Suppose  $\sigma \in \text{Aut}(\mathbf{No})$  commutes with  $x \mapsto \omega^x$  and fixes valuation values pointwise. Since  $v(\omega^x) = -x$ ,

$$-\sigma(x) = v(\omega^{\sigma(x)}) = v(\sigma(\omega^x)) = v(\omega^x) = -x.$$

Therefore  $\sigma = \text{id}$ . The qualifier “pointwise” is indispensable: all surreal field automorphisms preserve the natural valuation *up to* an ordered group automorphism, and that weaker fact does not give the last equality. Without pointwise fixation, the displayed computation instead says that the induced value-group action equals  $\sigma$  itself on the underlying surreal elements. That is a compatibility condition, not a proof of triviality.

### 11.3 Exponential faithfulness: the existing repository result

The following finite argument is taken from the repository's *Growth-Scale Rigidity* manuscript [10]. We give the proof because it is short, powerful, and avoids exchanging an automorphism with an infinite sum.

**Lemma 11.1** (Displacement cannot be globally bounded). *Let  $F$  be an ordered field and  $\sigma : F \rightarrow F$  a nonidentity unital field endomorphism. Then  $\{\sigma(x) - x : x \in F\}$  is cofinal and coinitial in  $F$ .*

*Proof.* Write  $D(x) = \sigma(x) - x$ . The correct multiplication identity is

$$D(ab) = \sigma(a)D(b) + bD(a). \quad (11.1)$$

Choose  $a$  with  $d = D(a) \neq 0$ . Given  $B > 0$ , put

$$b = \frac{2B(1 + |\sigma(a)|)}{|d|}.$$

If both  $|D(b)|$  and  $|D(ab)|$  were at most  $B$ , then

$$2B(1 + |\sigma(a)|) = |bd| \leq |D(ab)| + |\sigma(a)||D(b)| \leq B(1 + |\sigma(a)|),$$

a contradiction. Thus absolute displacements have no upper bound. Since  $D(-x) = -D(x)$ , both signs occur with arbitrarily large magnitude.  $\square$

**Theorem 11.2** (Faithful value-group action). *Let  $F$  be an ordered field with an increasing group isomorphism  $E : (F, +) \rightarrow (F_{>0}, \cdot)$ , and let  $w : F^\times \rightarrow \Lambda$  be a nontrivial convex valuation. If an automorphism  $\sigma$  commutes with  $E$  and fixes all  $w$ -values, then  $\sigma = \text{id}$ . Hence the exponential automorphisms stabilizing the valuation ring act faithfully on its value group.*

*Proof.* The homomorphism

$$\nu : F \rightarrow \Lambda, \quad \nu(x) = w(E(x))$$

is additive, with image exactly  $w(F^\times)$ : for  $a \neq 0$  use  $x = E^{-1}(|a|)$  and  $w(-1) = 0$  to get  $\nu(x) = w(a)$ . Thus it is onto the actual value group, even if  $\Lambda$  was presented as a larger ambient group. Nontriviality of  $w$  makes its kernel  $H$  a proper convex additive subgroup:  $E$  is increasing and a convex valuation reverses the order on positive elements. A proper convex additive subgroup is bounded above in  $F$ ; if  $b > 0$  lies outside it, then every  $h \in H$  satisfies  $|h| < b$ .

Commutation and pointwise value fixation give, for every  $x$ ,

$$\nu(\sigma x - x) = w(E(\sigma x)) - w(E(x)) = w(\sigma(E(x))) - w(E(x)) = 0.$$

Thus all displacements lie in the bounded subgroup  $H$ . Lemma 11.1 forces  $\sigma = \text{id}$ . Taking the kernel of the induced-action homomorphism proves faithfulness.  $\square$

For the usual surreal exponential, the kernel  $H$  is exactly the finite real-surreal elements. Indeed, bounded  $x$  gives bounded  $\exp x$  and bounded reciprocal by comparison with  $e^n$ ; conversely, if  $\exp x$  and its reciprocal are bounded, comparison with integers and application of log bound  $x$ . Consequently,

$$\ker(\text{Aut}_{\exp}(\mathbf{No}) \rightarrow \text{Aut}_{\text{ord}}(\Gamma)) = \{\text{id}\}. \quad (11.2)$$

In particular, a nonidentity surreal 1-automorphism cannot preserve the total real exponential. This settles the precise mathematical alternative posed in Question 5.4 of the inspected arXiv version of [5], by the repository proof above; it is not a new result of this article. Faithfulness does not say that the exponential automorphism group is trivial. It says that no nonidentity member can hide all its motion below the valuation scale.

### 11.4 A simple obstruction to exponential lifting

An ordered automorphism of the value group always has the Hahn lift  $M_{\text{id}, \tau}$ . It need not have *any* lift preserving the total surreal exponential. One elementary instance, also covered in [10], is worth spelling out.

**Proposition 11.3.** *For  $q \in \mathbb{Q}_{>0}$  with  $q \neq 1$ , the value-group dilation  $g \mapsto qg$  is not induced by any exponential automorphism of  $\mathbf{No}$ .*

*Proof.* If  $\sigma$  were such an automorphism, the homomorphism  $v(x) = v(\exp x)$  would satisfy  $v(\sigma x) = qv(x) = v(qx)$ , using the  $\mathbb{Q}$ -linearity of the additive map  $v$ . Thus  $\sigma(x) - qx$  would be finite for every  $x$ . Choose a positive infinite  $X$ , and write

$$\sigma(X) = qX + e, \quad \sigma(X^2) = qX^2 + f$$

with  $e, f$  finite. Multiplicativity yields

$$q^2 - q + \frac{2qe}{X} + \frac{e^2 - f}{X^2} = 0.$$

The last two terms are infinitesimal. Taking standard part gives  $q^2 - q = 0$ , contradicting  $q > 0, q \neq 1$ .  $\square$

Thus even the very simple field automorphism  $S_2$  cannot be repaired by composing with a value-fixing automorphism to obtain an exponential lift of its value action.

### 11.5 A global surcomplex operation that remembers real exponential

There is no need to postulate a total complex logarithm. The operation

$$L : \mathbb{K}^\times \rightarrow \mathbf{No} \subset \mathbb{K}, \quad L(z) = \log |z| \tag{11.3}$$

is already globally defined, using only the usual real surreal logarithm. Say that  $\alpha$  preserves  $L$  when  $\alpha(L(z)) = L(\alpha(z))$  for every nonzero  $z$ .

**Theorem 11.4** (Logarithmic-modulus automorphisms). *For  $\mathbb{K} = \mathbf{No}(i)$ ,*

$$\text{Aut}_L(\mathbb{K}) \simeq \text{Aut}_{\text{exp}}(\mathbf{No}) \times C_2. \tag{11.4}$$

*Its action on the natural value group has kernel exactly  $\{\text{id}, c\}$ . Marking  $i$  makes that action faithful.*

*Proof.* The range of  $L$  is all of  $\mathbf{No}$ , because  $L(\exp x) = x$ . A commuting automorphism and its inverse therefore preserve  $\mathbf{No}$ . Write it as  $\widehat{\sigma}c^e$ . For  $r > 0$  real-surreal,

$$\sigma(\log r) = \alpha(L(r)) = L(\sigma r) = \log(\sigma r).$$

Thus  $\sigma$  commutes with logarithm and hence with exponential. Conversely, if  $\sigma$  is exponential, the norm equivariance (2.4) proves that both lifts commute with  $L$ . The kernel statement follows from theorem 11.2 and the fact that conjugation fixes every valuation value.  $\square$

Theorem 11.4 is another result already present in [10]. Notice that  $L$  is required to transform *equivariantly*:  $L(\alpha z) = \alpha(Lz)$ . Demanding instead  $L(\alpha z) = Lz$  pointwise would impose a stronger condition.

## 11.6 Local exponentials and derivations are different structures

For  $\varepsilon \in \mathfrak{m}$ , the Hahn series

$$\text{Exp}(\varepsilon) = \sum_{n \geq 0} \frac{\varepsilon^n}{n!}, \quad \text{Log}(1 + \varepsilon) = \sum_{n \geq 1} \frac{(-1)^{n+1} \varepsilon^n}{n}$$

are well-defined by positive-support summability. Any strongly  $\mathbb{C}$ -linear element of  $G_v$  commutes with these operations, simply by applying strong summation and the field laws. This includes the phase twists, the exponent dilations, and the constant-fixing flows above. For finite  $z = a + \varepsilon$ , such an automorphism also respects the finite exponential  $e^a \text{Exp}(\varepsilon)$ . None of this implies compatibility with the total real surreal exponential on infinite arguments, as  $S_2$  and [theorem 11.3](#) demonstrate.

The Berarducci–Mantova construction equips **No** with a nonzero derivation compatible with substantial exponential and transseries structure [7]. Fix one such derivation  $\partial$  and extend it to  $\mathbb{K}$  by

$$\partial(x + iy) = \partial x + i\partial y.$$

For real-axis-preserving automorphisms, a direct coordinate calculation gives

$$\{\alpha \in G_{\mathbf{No}} : \alpha\partial = \partial\alpha\} \simeq \text{Aut}(\mathbf{No}, \partial) \times C_2.$$

This is a reduction, not a classification of the differential group or of all differential automorphisms without a marked real axis. In particular, the scalar derivation  $\partial$ , formal differentiation of a power series, and the fine derivative of a function are three distinct notions. The zero-derivative automorphism in [theorem 8.2](#) concerns the third.

## 12 Consequences for algebra, geometry, and symbolic computation

Automorphisms transport identities, but a transported identity must include all its coefficients and all its chosen structures. This elementary rule is especially useful when applying surcomplex theory.

### 12.1 Polynomial and finite-dimensional invariants

If  $p(X) = \sum_{k=0}^n a_k X^k$  and  $\alpha$  is a field automorphism, put  $p^\alpha(X) = \sum_{k=0}^n \alpha(a_k) X^k$ . Then

$$\alpha(p(z)) = p^\alpha(\alpha z).$$

Roots, root multiplicities, common-root relations, resultants, discriminants, and finite matrix determinants therefore transform algebraically. When all coefficients are fixed, the root multiset is permuted. Without coefficient fixation, it is the roots of the *transformed* polynomial that are obtained.

For  $\alpha \in G_v$ , root separations satisfy

$$v(\alpha r_j - \alpha r_k) = \tau_\alpha(v(r_j - r_k)).$$

Thus finite valuation-cluster trees retain their combinatorial incidence, although the labels can be reindexed. For  $\alpha \in \mathcal{U}$ , every nonzero separation even retains its leading term. These statements follow from additivity and the valuation laws; they require no compactness or ordinary analytic convergence.

For  $\alpha \in G_{\mathbf{No}}$ , Hermitian algebra is also transported, because  $\alpha$  commutes with conjugation. The identity  $A^* = \overline{A}^T$  transforms to  $(\alpha A)^* = \alpha(A^*)$ , and positive real scalars remain positive. Under a phase twist not preserving the axis, the same conclusion holds only after conjugation is also transported to  $j_\theta$ . Keeping the original conjugation while changing the scalars is not an invariant operation.

## 12.2 Automorphism tests as checks on a proposed canonical definition

Suppose a construction is claimed to depend only on the pure field  $(\mathbb{K}, +, \cdot)$  and on a specified set of scalar parameters. It must commute with every automorphism fixing those parameters. Proposition 9.3 then immediately rules out pure-field definitions of the distinguished real axis and finite ring. A purported canonical modulus on the bare field would similarly determine extra real-form information and cannot be the chosen modulus without additional structure.

This does not make those constructions invalid. It identifies which data must be declared: conjugation or a real form for Hermitian geometry, a valuation for scales, monomials and a summation structure for normal-form algorithms, and an exponential for logarithmic growth. An automorphism counterexample is often the quickest way to detect an omitted hypothesis.

## 12.3 What a symbolic implementation can realistically support

A symbolic system can implement (4.1) exactly on a declared Hahn workspace once it can evaluate the coefficient automorphism, exponent map, and character, and certify that the exponent map is an ordered additive bijection. For finite supports, the arithmetic checks reduce to finite convolution. For symbolic infinite families, well-order and finite-incidence certificates remain necessary.

The maps  $S_a$ ,  $D_\lambda$ , and  $P_\theta$  are particularly transparent on normal-form expressions with accessible exponent coefficients. The fixed-shift flow can be represented by its operator formula or by closed expressions such as  $t \mapsto t/(1 - st)$ . The coefficient-motion example requires an abstract derivation on real constants and should not be advertised as an algorithm on arbitrary numerical real inputs. Likewise, the back-and-forth existence theorems are not executable algorithms on unrestricted surcomplex names.

## 13 Relation to the Surreal repository and a formalization plan

This section is the source’s comparison, as of its pin. It was checked for this report and is accurate at the pin; Section 1.5 adds the relations it did not draw, and Appendix C.2 records what has changed since.

The inspection in this article uses the repository revision

dcf86662b574988e75d08d515d3a2a5ba7bbc0d7.

The root documentation, the report catalogue, the exponential-rigidity manuscript and its audit, the normal-form bridge note, and the two specific Lean modules below were read through the GitHub connector. This is a selected-source inspection, not a claim to have searched every file or to have rebuilt the project.

The inspected documentation distinguishes research arguments from checked Lean coverage. In particular, the current bridge note reports an ordered field equivalence for small normal forms, transfer of real closedness, actual surcomplex algebraic closedness, and valuation/standard-part preservation [11]. These are *repository-reported* implementation facts. This article supplies no independent build result for them.

### 13.1 Existing interfaces actually inspected

Surreal/Algebra/Complexify.lean defines the quadratic algebra  $F[i]$ , its imaginary unit, coordinate multiplication, conjugation, normSq, and the identity mul\_conj. Its field construction uses an ordered coefficient field and explicitly does not, by itself, assert algebraic closedness [12]. These interfaces are sufficient foundations for a generic development of theorems 2.2 to 2.4.

`Surreal/Algebra/FineDerivative.lean` defines `FineHasDerivAt` by punctured-neighborhood difference quotients and contains uniqueness, continuity, algebraic differentiation rules, and the polynomial derivative theorem [13]. It is an appropriate interface for theorem 8.1; the exponent-dilation example additionally requires a proved normal-form automorphism and its exact valuation formula.

The documentation’s statements about smallness must retain their universe indices. A theorem that every lower-universe subset is closed and discrete does not say that every subset available in a larger universe is discrete. The class theory in this article and a universe-relative Lean realization are compatible approaches, but they are not textually interchangeable.

### 13.2 Suggested implementation order

A useful order is to separate finite algebra from normal-form and global existence arguments.

1. **Generic real-form algebra.** Construct the lift of a base-field ring equivalence to  $F[i]$ . Prove uniqueness from its restriction and its value on  $i$ , and identify the conjugation centralizer. Add exact norm rigidity and the Cayley-circle criterion. These proofs need no Hahn theory.
2. **Explicit Hahn automorphisms.** Package coefficient maps, ordered additive exponent equivalences, and characters in one construction. Prove the composition law (4.2) and valuation formula before introducing the phase action. Keep the nested coefficient extraction  $\ell$  on surreal *exponents* distinct from coefficient extraction on a surcomplex input.
3. **Support-certified flows.** Reuse a summable-family interface and prove the fixed-shift family is jointly summable. Establish inverse formulas for theorems 5.1 and 5.2 through coefficientwise finite identities. Merely entering a formal expression  $\exp(D)$  is not a proof of an automorphism.
4. **Topology and derivatives.** Prove the induced homeomorphism from a valuation reindexing, the nonzero-derivative rigidity theorem for a generic topological field, and finally  $S'_a \equiv 0$  for  $a > 1$  using (8.1). No mean-value theorem is needed or available for that last conclusion.
5. **Global foundations separately.** Treat class back-and-forth, real-form existence, and collection-level group factorizations in an explicit foundational layer or in bounded universes with the appropriate hypotheses. Do not silently replace a set-stage extension theorem by an unrestricted type-level statement about every model.

The names in this implementation order describe proposed work; they are not assertions that corresponding new modules already exist. In particular, no new Lean code or Lean-checking claim is included in the present package.

## 14 What remains to be classified

The central stabilizer question has a clean answer, and the valued subgroup has an exact kernel–quotient decomposition. Neither result gives a fully explicit description of the unrestricted automorphism collection. The following are useful research targets, with their status kept distinct.

**The image of exponential automorphisms on the value group.** Faithfulness (11.2) determines the kernel, not the image. An intrinsic necessary-and-sufficient criterion on an ordered additive  $\tau : \mathbf{No} \rightarrow \mathbf{No}$  for it to come from an exponential automorphism would be much stronger. Positive rational dilations already show that arbitrary Hahn lifting does not solve the problem. Any such criterion must use the interaction between the value group and real exponential growth, not just ordered-group structure. (This is Question 13.1, `q: image`, of the collection’s rigidity report, where it is also open.)

**Omega-map compatibility.** The nontrivial automorphisms preserving the omega-map, and extensions of order automorphisms of its generalized epsilon fixed points, are posed in Questions 5.6 and 5.7 of the inspected version of [5]. They are not resolved here. Exact value rigidity only describes one kernel and must not be mistaken for a solution of those broader extension problems.

**Effective descriptions inside the leading-term kernel.** The abstract support-and-invertibility conditions for strong Hahn substitutions are meaningful, but finite, compositional certificates for larger symbolic families would be useful for computation and formalization. The fixed-shift constructions are a tractable subclass, not an exhaustive normal form for  $\mathcal{U}$  or its strong part. This is a proposed computational research direction, not a claim that the general Hahn literature lacks abstract descriptions.

**Real-form and continuity classifications.** The phase family and the cofinality example show that real forms have both embedding freedom and isomorphism-type freedom. A detailed classification with additional valuation, summation, or simplicity conditions is a separate problem. Similarly, this article proves that  $G_v$  acts continuously but does not identify every fine-continuous pure-field automorphism. These are boundaries of the present account; they are not asserted to be newly formulated or historically unresolved in every related setting. *Status (batch 32; [merge]).* The real-form classification is *partly answered*. For involutions that preserve the valuation ring and  $\mathbf{Oz}[i]$  (summation is then automatic), Theorems 16.2 and 16.4 give the complete classification: conjugacy types correspond to real closed fields of transcendence degree  $c$ , exactly  $2^c$  of them. Real forms whose involution is not both valued and  $\mathbf{Oz}[i]$ -preserving (for instance one that is only valued, or only strong), the unrestricted real forms, and simplicity conditions remain unclassified, as does the continuity question.

*Remark 14.1* (Invariants that do not classify real forms; batch 37). Added in batch 37 (28 September 2026). Part VIII of `foundations-and-computation/surreal-fields-across-universes` (its Section 30, labels `univ:gs:`) limits what a classification of the unrestricted real forms can use. Under Global Choice, in suitable set-forcing extensions,  $\mathbb{K}$  has families of pairwise nonconjugate involutions, each with a fixed field isomorphic to the surreal field  $\mathbf{No}^M$  of a proper inner model  $M$ . In one kind all restrict to  $c$  on  $\mathbb{C}$ , the structures  $(\mathbb{K}, j)$  are elementarily equivalent, and the fixed fields have the same saturation spectrum (its Theorems 30.7 and 30.12, `univ:gs:thm:samesatinvolutions`, `univ:gs:thm:classinvolutions`). In another the fixed fields have one and the same full gap spectrum and are separated by their real traces (Theorems 30.3, 30.8 and 27.3, `univ:gs:thm:ampinvolutions`, `univ:gs:thm:boolinvolutions`, `univ:gs:thm:oldtrace`). So it is consistent that neither the restriction to  $\mathbb{C}$ , nor the theory of  $(\mathbb{K}, j)$ , nor the saturation or full gap spectrum of the fixed field is a complete conjugacy invariant. The involutions of the first kind agree with  $c$  on  $\mathbb{C}$ , but their fixed fields omit set cuts (as every such  $\mathbf{No}^M$  does) and  $\mathbf{No}$  does not, so none is conjugate to  $c$ ; by Proposition 23.1 none is both valued and  $\mathbf{Oz}[i]$ -preserving, as for the families listed after that proposition. Whether the real trace and the full gap spectrum together classify a broad family is that report's open Question 31.20 (`univ:gs:q:involutions`). The classification of this section stays open.

## 15 Source audit, proof status, and conclusions

### 15.1 Version-sensitive points

The version-specific literature anchor is Kaplan–Krapp–Serra, arXiv:2509.22374v3, submitted 23 April 2026, whose PDF is dated 27 April 2026. Its displayed formulas were checked against the PDF, not only the HTML rendering. Two local cautions affect example construction. The shift  $g - 1$  printed in Example 4.3 is not contracting for the convention  $v = \min \text{supp}$ ; our construction uses a positive shift. The series  $\sum_n \omega^{-1/n}$  printed

in Example 4.4 does not have decreasing Conway support; (7.1) uses a valid support instead. Our nonstrong example does not claim the stronger global leading-term property asserted for that source’s construction.

The semidirect actions in this article are specified by actual composition formulas, especially (4.2) and (6.8). A display listing factors should not be read as a componentwise direct product when exponent reindexing acts on coefficient weights.

The repository’s exponential-rigidity report is explicitly prior material at the pinned revision. Its appended source audit records an earlier catalogue/tree mismatch, illustrating why absence from a catalogue or failure of code search is not evidence of absence from the repository. The present article neither relies on that earlier absence claim nor makes an exhaustive nonduplication claim.

## 15.2 Evidence categories

| Category                | Treatment in this article                                                                                                                                                 |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Classical prerequisites | Real closedness and normal forms of <b>No</b> ; Newton–Puiseux; finite-group fixed-field theory and Artin–Schreier are imported with references.                          |
| Direct proofs           | Real-axis, circle, and norm stabilizers; explicit Hahn maps and flows; valued factorization; topology and derivative results; class extension and real-form consequences. |
| Repository prior work   | Exponential displacement faithfulness, rational non-lifting, and logarithmic-modulus reduction are attributed, with proofs included for their role here.                  |
| Implementation evidence | Selected source code and documentation were read; no independent repository build and no new Lean certification.                                                          |
| Historical novelty      | Not certified. Derived statements and useful examples are not automatically claimed to be first discoveries.                                                              |

## 15.3 Non-claims

The source states its limitations where they arise, and each is kept in place. The list collects them, with the place where each stands. Items 1–28 are the source’s; items 29–33 were added when the report joined the collection. [merge] The non-claims of the second source are listed separately in Section 30.4, and those of the third in Section 40.5.

1. No complete classification of the automorphisms of  $\mathbb{K}$ , and in particular none of the unrestricted ones, is claimed (title page, abstract, Section 14; the delivered README).
2. Historical priority is not claimed or certified; derived statements and useful examples are not automatically claimed to be first discoveries, and no exhaustive nonduplication claim is made (title page, Section 15; the delivered README).
3. No new Lean verification, no Lean code, and no independent build of the repository are claimed; the implementation order describes proposed work, not existing modules (title page, Section 13, Section 15).
4. The repository comparison is a selected-source inspection, not a search of every file; the bridge-note facts are repository-reported, not independently rebuilt (Section 13).
5. Exponential faithfulness, the rational non-lifting proposition and the logarithmic-modulus reduction are prior work of the repository, not results of this report; the settlement of Question 5.4 of [5] is by the repository’s proof and “is not a new result of this article” (abstract, Sections 1 and 11, Section 15).
6. Faithfulness does not say that the exponential automorphism group is trivial (Section 11).
7. The article does not rely on the earlier absence claim in the rigidity report’s audit (Section 15).

8.  $\text{Aut}(\mathbb{K})$  and the other groups are definable collections of class maps, not class objects of NBG; a group isomorphism is a unique factorization statement about individual automorphisms (Section 1).
9. Global homogeneity and the topology of the full field are not to be transferred to an arbitrary fixed set-sized workspace (Section 1).
10. The four-layer decomposition (6.2) is not a decomposition of all field automorphisms of  $\mathbb{K}$ , and its factor  $\mathcal{U}$  is not reduced to a finite list of coefficients (Section 6).
11. An arbitrary homomorphism  $g \mapsto u_g$  into  $1 + \mathfrak{m}$  does not by itself give a summable or surjective map; the fixed-shift flows illustrate the automorphism–derivation connection without asserting that every formal operator exponential is summable (Section 5).
12. The Taylor motions  $\Phi_d$  are generally noncomputable from numerical names of real coefficients and are not to be advertised as algorithms; the back-and-forth existence theorems are not executable algorithms and give no canonical parametrization of the class automorphisms (Sections 5, 9 and 12).
13. The maps of  $G_{\mathbf{N}_0}$  are semilinearly compatible with the modulus; calling them isometries would be misleading (Section 2).
14. Proposition 7.1 is an inclusion in the group of fine-continuous automorphisms, not a classification of it, and no converse is inferred from continuity; not every fine-continuous pure-field automorphism is identified (Sections 7 and 14).
15. The non-strong witness of Section 7 is not claimed to fix every value or every leading term, nor to have the global leading-term property asserted for the construction in [5]; the non-density conclusion concerns the pointwise-equality topology on automorphisms only (Sections 7 and 15).
16. The choice of topology is part of each topological theorem; the discreteness of set-sized subsets in the full fine topology says nothing about the intrinsic valuation topology of a fixed Hahn workspace (Section 7).
17.  $S_2$  is not asserted to be representable by a locally convergent power series; “everywhere differentiable” is not a synonym for analytic rigidity, and a zero derivative does not imply local constancy (Section 8).
18. The scalar Berarducci–Mantova derivation, formal differentiation of a power series and the fine derivative are three distinct notions; the differential reduction is not a classification of the differential group or of differential automorphisms without a marked real axis (Section 11).
19. The transcendence-basis description is not a classification, even for a set-sized algebraically closed field (Section 9).
20. Neither the phase family nor Theorem 10.2 classifies the real closed class fields whose complexification is  $\mathbb{K}$ ; the conjugacy criterion of Appendix B is structural, not an effective decision procedure (Section 10, Appendix B). [merge] Still true; the valued,  $\mathbf{Oz}[i]$ -preserving real forms are now classified (Section 23).
21. Simplicity rigidity is rigidity of an enriched structure, not of the real closed field reduct (Section 11).
22. Exact value rigidity for the omega-map needs pointwise value fixation; it describes one kernel and is not a solution of the omega-map extension problems, Questions 5.6 and 5.7 of [5], which are not resolved (Sections 11 and 14).
23. Compatibility with the local exponential and logarithm does not imply compatibility with the total real exponential on infinite arguments (Section 11).
24. The image of exponential automorphisms in the value group is not determined (Section 14).
25. Automorphism tests do not make the constructions they constrain invalid; they identify data that must be declared (Section 12).

26. Smallness statements keep their universe indices; the class theory here and a universe-relative Lean realization are compatible but not textually interchangeable (Section 13).
27. The fixed-shift constructions are a tractable subclass, not an exhaustive normal form for  $\mathcal{U}$ ; the research targets of Section 14 are not asserted to be newly formulated or historically unresolved in every setting, nor is it claimed that the Hahn literature lacks abstract descriptions (Section 14).
28. The delivered README reports symbolic finite checks of the norm product and polarization identities, the Cayley inverse, and the rational-flow composition and inverse identities, and says that such checks are not proofs of infinite summability or of the full surreal theory. No code or record of them was delivered, and nothing in this report rests on them. The delivery redistributes no third-party papers or font files.
29. (*Added.*) The answer to Question 5.4 of [5] rests on the rigidity report's check of the pinned version of that paper. The source's readings of its Examples 4.3 and 4.4 and of its Questions 5.6 and 5.7 are the source's and were not rechecked for this report.
30. (*Added.*) Whether  $\text{Aut}_{\text{exp}}(\mathbf{No})$  is nontrivial, equivalently whether  $\{\text{id}, c\} \subsetneq \text{Aut}_L(\mathbb{K})$ , is not decided; nor is whether  $S_a|_{\mathbf{No}}$  is exponential when  $a$  is not rational (Section 1.4).
31. (*Added; status updated.*) No dedicated `saut` mappings are supplied. Existing generic Lean proofs cover ordered displacement, exponential rigidity, logarithmic-modulus classification and the valuation kernel, as recorded in Section 1.5; their actual surreal exponential instantiation remains pending. The discreteness proof concerns lower-universe-small subsets of the Lean carrier, not an unqualified proper-class statement.
32. (*Added.*) The relations of Section 1.5 identify results by their content; no theorem is transferred between the full fine topology and a workspace topology, or between the fine derivative and another derivative notion.
33. (*Added.*) Observation (1.2) is an elementary consequence of results printed here; it is not claimed as a result of the source or of the rigidity report.

## 15.4 The main picture

The correct starting point is the strict separation

$$\{\text{id}, c\} = \text{Aut}(\mathbb{K}/\mathbf{No}) \subsetneq G_{\mathbf{No}} \subsetneq G_v \subsetneq \text{Aut}(\mathbb{K}).$$

Preserving the real axis gives  $\text{Aut}(\mathbf{No}) \times C_2$ ; preserving the unit circle gives the same group. Exact preservation of every modulus reduces it to identity and conjugation. Pure-field automorphisms can move the real axis, the finite ring, and even the isomorphism type of a chosen real form under a change of conjugation.

The valuation organizes a large explicit subgroup but does not enforce classical analytic behavior. Phase twists can move real monomials while fixing constants and values; leading-term automorphisms can move ordinary real constants by infinitesimals; fine-continuous maps need not preserve Hahn sums; and exponent dilation can be a differentiable field homeomorphism with derivative zero everywhere. Exponential structure imposes a different and genuinely strong constraint: its value-group action is faithful, even though the underlying valued-field kernel is large.

An automorphism theorem for surcomplex numbers is therefore meaningful only after its preserved structure, support convention, topology, and size framework have been made explicit. With those choices in place, the theory has both exact structural reductions and a rich supply of concrete examples.

## ADDITION FROM A SECOND SOURCE

**Finite symmetry descent and valued real forms**

Sections 16–30: manuscript 02 of batch 32, pinned to repository commit 343dc2c (Appendix C.3)

**16 Finite symmetry descent: statements and conventions**

Sections 16–30 print the second source of this report, *Finite Symmetry Descent and Real Forms of Gaussian Omnific Integers* (Appendix C.3). Its text is kept with the renamings of Section 16.1; sentences added when it was merged are marked [merge].

The field  $\mathbb{K} = \mathbf{No}(i)$  is algebraically closed, whereas  $\mathbf{No}$  is its standard real closed subfield. Standard conjugation  $c$  fixes  $\mathbf{No}$  and restricts on  $\mathbf{Oz}[i]$  to an involution with fixed ring  $\mathbf{Oz}$ . This suggests a natural uniqueness problem: how much additional structure must an involution preserve before its fixed pair is necessarily a copy of  $(\mathbf{No}, \mathbf{Oz})$ ?

There are several different questions here. An automorphism may preserve only the field operations, or additionally the natural Hahn valuation, the coefficient field, the integer-support ring, or all infinite normal-form sums. These conditions must not be conflated (Section 1.4). The literature develops detailed valuation and summation decompositions of Hahn automorphism groups [3, 5]. The collection already contains strong monomial and phase symmetries and general surcomplex real forms (Sections 4 and 10 and Appendix B of this report), and omnific-preserving stabilizers (surreal/omnific-preserving-automorphisms). In particular, the existence of a nonstandard real form, without the additional conditions imposed here, is not new (Theorem 10.2).

The contribution of the source is a simultaneous finite-symmetry normalization and the resulting precise classification in an explicitly restricted category. It also gives a strong negative answer to the uniqueness problem just posed: even a genuine integer-part fixed ring does not force the standard real form.

**16.1 Conventions for the added sections**

The conventions of Section 1.4 stay in force:  $v = \min \text{supp}$ ,  $t^\gamma = \omega^{-\gamma}$ , “strong” means that a map and its inverse preserve set-indexed Hahn sums, a *valued* automorphism maps the valuation ring onto itself and may move values, and a *value-fixing* one fixes every value. Several letters of the source already have a meaning in this report, and some of the source’s letters carry two meanings. The source’s symbols were therefore renamed as follows; nothing else changed, and no normalization changed.

| Source                                     | Here                                             | Reason                                                                                                                                                                            |
|--------------------------------------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\tau, \tau_g = g _k$                      | $\rho, \rho_g$                                   | $\tau$ is always an ordered automorphism of a value group (Section 1.4); $\rho$ is this report’s coefficient automorphism, and $g _k$ is the residue action $\rho_g$ of Section 6 |
| $\widehat{\tau}$ (coefficientwise lift)    | $M_\rho$                                         | the case $M_{\rho, \text{id}, 1}$ of (4.1); $\widehat{\sigma}$ of Theorem 2.2 is a different lift, from $F$ to $F(i)$                                                             |
| $u$ (induced map on $\Gamma$ )             | $\tau$                                           | it is an ordered automorphism of the value group                                                                                                                                  |
| $\sigma$ (involution of $\mathbf{No}[i]$ ) | $j$                                              | $\sigma$ is an automorphism of $\mathbf{No}$ in Sections 2 and 11; $j$ is the involution of Section 10                                                                            |
| $\mathcal{H}, H, H_\sigma, H_a$            | $\mathcal{H}, \mathcal{H}_j, \mathcal{H}_\alpha$ | one symbol for the norm conjugator; the italic $H$ stays local as in Section 1.4                                                                                                  |
| $\mathcal{G}$                              | $\mathcal{G}$                                    | now <i>valued</i> automorphisms preserving $\mathbf{Oz}[i]$ ; the source added “strong”, which is automatic (Proposition 18.5)                                                    |

| Source                                                | Here                                             | Reason                                                                                                       |
|-------------------------------------------------------|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| $A = \mathbf{Oz}[i]$                                  | $\mathbf{Oz}[i]$                                 | $A$ is a parameter set in Proposition 9.3                                                                    |
| $B = k + \mathcal{I}$                                 | $A_k$                                            | $B(a, r)$ is a fine ball in Section 7; $A_k$ is the source's own $A_D$ at $D = k$                            |
| $D$ (subring of $k$ ), $A_D$                          | $\mathfrak{o}, A_{\mathfrak{o}}$                 | unsubscripted $D$ is the displacement of Lemma 11.1                                                          |
| $B(a, r)$ (closed valuation ball)                     | $\mathbb{B}(a, r)$                               | $B(a, r)$ is the open fine ball of Section 7                                                                 |
| $L_F$                                                 | $\mathbf{No}_F$                                  | $L$ is the logarithmic modulus (11.3)                                                                        |
| cut sets $L, U$                                       | $\mathcal{L}, \mathcal{R}$                       | as in Section 9                                                                                              |
| $E$ (a second real closed field)                      | $F'$                                             | $E$ is reserved for an exponential                                                                           |
| $E/F$ (an immediate extension), $E_N$ (Baire sets)    | $K_1/K_0, Z_N$                                   | as for $E$ ; $F$ is the coefficient field; $K_1, K_2$ are generic fields as in Theorem 9.1                   |
| $T, T_0, T_1$ (transcendence basis)                   | $B, B_0, B_1$                                    | $T$ is the transcendental of Theorem 10.2 and $T_s$ the flows; $B$ is the transcendence basis of Theorem 5.1 |
| $U \subseteq T_1, F_U, \psi_U, c_U, \tau_U, \sigma_U$ | $Y \subseteq B_1, F_Y, \psi_Y, c_Y, \rho_Y, j_Y$ | $\mathcal{U}$ is the leading-term kernel                                                                     |
| $U$ (a conjugator)                                    | written out                                      | as above                                                                                                     |
| $P$ (finite group), $a, b$ (its actions)              | $\Theta, \alpha, \alpha'$                        | $P_\theta$ and $\mathcal{P}$ are the phase twists and the Puiseux field                                      |
| $u_p$ (cocycle)                                       | $\xi_p$                                          | $u$ is an element of $\mathcal{U}$ in (6.1)                                                                  |
| $\rho$ (conjugator in $\text{Aut}(\mathbb{C})$ )      | $\kappa$                                         | $\rho$ is now the coefficient involution                                                                     |
| $\ell_j$ (linear forms)                               | $f_j$                                            | $\ell$ is the exponent coefficient (4.5)                                                                     |
| $P$ (positive elements above $\mathbb{N}$ )           | $F_\infty$                                       | as for $P$ above                                                                                             |
| $a_\gamma$ (a character)                              | $\chi(\gamma)$                                   | characters are $\chi$ in (4.1)                                                                               |
| $P$ (automorphism of an example), $K_2$               | $\Psi, K_{\text{lex}}$                           | $K_2$ is a class field in Theorem 9.1                                                                        |

The symbols kept from the source are local to Sections 16–30:  $k$  is a set-sized coefficient field of characteristic zero,  $\Gamma$  is a set-sized divisible ordered abelian group or  $\mathbf{No}$  (the set-based reading of Section 1),  $K = k((t^\Gamma))$  is a full Hahn field (the blackboard  $\mathbb{K}$  remains  $\mathbf{No}(i)$ ),  $G$  is a finite group of automorphisms with elements  $g, h$  and  $n = |G|$  (exponents are  $\gamma, \delta$ , never  $g$ ),  $m_\gamma$  is the norm section,  $\mathcal{I}$  is the strictly negative-support part,  $O = k + \mathfrak{m}$  is the valuation ring of  $K$ ,  $\text{ct}$  is the constant coefficient,  $\mathcal{N}$  is a kernel group,  $F$  is a set-sized real closed *coefficient* field,  $\mathfrak{c} = 2^{\mathbb{N}_0}$  (not the conjugation  $c$ ), and  $\psi$  is a local isomorphism of fields. In Section 26 the letters  $z, u$  are the two monomials of  $K_{\text{lex}}$ , not a generic element.

The following readings are tempting and false.

- $\mathbf{No}_F = F((t^{\mathbf{No}}))$  is a field whose *coefficient* field is  $F$ ; the subscript is not an inequality as in  $\mathbf{No}_{<0}$ . It is isomorphic to  $\mathbf{No}$  only when  $F \cong \mathbb{R}$  (Theorem 24.4).
- $R_F = \mathbb{Z} + F((t^{\mathbf{No}_{<0}}))$  is discretely ordered with least positive element 1 for every  $F$ , but it is an integer part of  $\mathbf{No}_F$  only when  $F$  is Archimedean (Theorem 24.3). It is not “the omnific integers of  $\mathbf{No}_F$ ” in general.
- $F((t^{\mathbf{No}_{<0}}))$  is a support-restricted additive part, not a Hahn field:  $\mathbf{No}_{<0}$  is not a group.
- The coefficient field  $F = \mathbb{C}^p$  of an involution is a real closed field of transcendence degree  $\mathfrak{c}$  in its intrinsic order; it need not be  $\mathbb{R}$  and need not lie on the usual real axis, although the ambient coefficient field is named  $\mathbb{C}$  (Section 27).

- $F_j = \text{Fix}(j)$  of Section 10 is a class real form; for  $j \in \mathcal{G}$  it is isomorphic to  $\mathbf{No}_F$  with  $F = \mathbb{C}^{j|\mathbb{C}}$ , a set-sized field. The two letters  $F$  are never interchanged.
- $\text{ct}$  extends the residue map  $\text{st}$  of Section 3 to all of  $K$  but is not a ring homomorphism on  $K$ .
- A value-fixing automorphism need not fix leading terms: the conjugator  $\mathcal{H}$  is value-fixing, and generally not a 1-automorphism.

## 16.2 An explicit finite-symmetry normal form

Small monomials are used throughout:  $t^\gamma = \omega^{-\gamma}$  when  $\Gamma = \mathbf{No}$ . A Hahn support is therefore well ordered, the valuation is its least exponent, and purely infinite terms have *negative* exponents. Let

$$K = k((t^\Gamma)), \quad \mathcal{I} = \{x \in K : \text{supp}(x) \subseteq \Gamma_{<0}\}.$$

The phrase *strong automorphism* means preservation of set-indexed Hahn-summable families and their sums in both directions.

**Theorem 16.1** (Finite symmetry descent; main statement). *Let  $k$  be a set-sized field of characteristic zero. Let  $\Gamma$  be a set-sized divisible ordered abelian group, or let  $\Gamma = \mathbf{No}$  with set-supported series. Suppose a finite group  $G$  acts on  $K = k((t^\Gamma))$  by valued automorphisms and stabilizes  $k$  and  $\mathcal{I}$ . Write  $n = |G|$  and  $\rho_g = g|_k$ . There is a strong  $k$ -linear, value-fixing field automorphism  $\mathcal{H}$  such that*

$$\mathcal{H}(\mathcal{I}) = \mathcal{I}, \quad \text{ct} \circ \mathcal{H} = \text{ct}, \quad \mathcal{H}^{-1}g\mathcal{H} = M_{\rho_g} \quad (g \in G),$$

where  $M_{\rho_g}$  acts coefficientwise. One such automorphism is

$$\mathcal{H}\left(\sum_{\gamma} a_{\gamma} t^{\gamma}\right) = \sum_{\gamma} a_{\gamma} \prod_{g \in G} g(t^{\gamma/n}). \quad (16.1)$$

In particular, restriction to  $k$  is faithful on  $G$ .

[merge] The source assumes in addition that  $G$  acts by *strong* valued automorphisms. That hypothesis follows from the others (Proposition 18.5, which is `opa:as:thm:ctiso` of `surreal/omnific-preserving-automorphisms`, placed after the source's pin), so the theorem is printed without it; the proof in Section 20 uses strongness of the  $g$  only through that proposition. Source 03 (batch 34) proves the same conjugacy for strong actions that need not stabilize  $\mathcal{I}$ , over a field of any characteristic, with an explicit inverse (Theorem 33.1, Remark 33.2).

The formula is a multiplicative norm, but not a norm of an arbitrary element followed by an arbitrary root extraction. Dividing the exponent by  $n$  *before* taking the product supplies a coherent multiplicative section. This is what makes the construction a field homomorphism rather than merely a collection of invariant elements.

## 16.3 The Gaussian omnific classification

Put  $\mathbb{K} = \mathbf{No}[i] = \mathbb{C}((t^{\mathbf{No}}))$  and let

$$\mathcal{G} = \{\alpha \in G_v : \alpha(\mathbf{Oz}[i]) = \mathbf{Oz}[i]\}$$

be the valued automorphisms of  $\mathbb{K}$  that preserve  $\mathbf{Oz}[i]$  setwise. A valued automorphism is allowed to induce a nontrivial ordered automorphism of the exponent group; it need not initially fix every value. [merge] The source defines  $\mathcal{G}$  as the *strong* valued automorphisms preserving  $\mathbf{Oz}[i]$ . By `opa:as:thm:gaussian` of `surreal/omnific-preserving-automorphisms` (Proposition 18.5 and Remark 22.3 here), every valued automorphism preserving  $\mathbf{Oz}[i]$  is strong, so the two collections coincide.

**Theorem 16.2** (Classification of admissible involutions). *For every nonidentity involution  $j \in \mathcal{G}$ , its restriction  $\rho = j|_{\mathbb{C}}$  is a nonidentity involution of  $\mathbb{C}$ . There is a  $\mathbb{C}$ -linear, strong, value-fixing automorphism  $\mathcal{H}_j$  preserving  $\mathbf{Oz}[i]$  with*

$$j = \mathcal{H}_j M_\rho \mathcal{H}_j^{-1}, \quad \mathcal{H}_j(t^\gamma) = t^{\gamma/2} j(t^{\gamma/2}).$$

*Two such involutions are conjugate in  $\mathcal{G}$  if and only if their coefficient involutions are conjugate in  $\text{Aut}(\mathbb{C})$ . Equivalently, their coefficient fixed fields are isomorphic.*

*If  $F = \mathbb{C}^p$ , the fixed pair  $(\text{Fix}(j), \text{Fix}(j) \cap \mathbf{Oz}[i])$  is isomorphic to*

$$(\mathbf{No}_F, R_F) := \left( F((t^{\mathbf{No}})), \mathbb{Z} + F((t^{\mathbf{No}_{<0}})) \right). \quad (16.2)$$

*The possible coefficient fields are exactly the real closed fields of transcendence degree  $\mathfrak{c}$  over  $\mathbb{Q}$ , up to isomorphism.*

The notation  $F((t^{\mathbf{No}_{<0}}))$  denotes a support-restricted additive part, not a Hahn field:  $\mathbf{No}_{<0}$  is not a group. This convention will always be explicit when it occurs.

**Theorem 16.3** (Integer parts and the standard real form). *For any set-sized real closed field  $F$ ,  $R_F$  is discretely ordered with least positive element 1. Moreover,*

$$R_F \text{ is an integer part of } \mathbf{No}_F \iff F \text{ is Archimedean,}$$

and

$$\mathbf{No}_F \cong \mathbf{No} \iff F \cong \mathbb{R}.$$

*The field isomorphism in the second equivalence may be a plain field isomorphism: real closed fields carry an intrinsically determined order.*

**Theorem 16.4** (Maximal set-sized multiplicity). *There are exactly  $2^{\mathfrak{c}}$  conjugacy types of nonidentity involutions in  $\mathcal{G}$ . There is a set-indexed family of  $2^{\mathfrak{c}}$  representatives whose fixed fields are pairwise nonisomorphic and whose fixed rings are genuine integer parts. The representatives can all be chosen coefficientwise, hence strong and value-fixing. Excluding the standard type still leaves  $2^{\mathfrak{c}}$  types. These representatives are pairwise nonconjugate even under arbitrary plain field automorphisms of  $\mathbf{No}[i]$ .*

## 16.4 What is, and is not, being resolved

Theorems 16.1–16.4 solve the finite-action normalization and involution-classification problems *under their stated support and valuation hypotheses*. They are not asserted to settle a named unrestricted question in the published literature. In particular, the source does not prove that an arbitrary automorphism of  $\mathbf{Oz}[i]$  preserves the valuation or all Hahn sums. It does not classify all real forms of the plain class field  $\mathbf{No}[i]$ , and it makes no claim about exponential compatibility. [merge] Status against the current collection: a *valued* automorphism preserving  $\mathbf{Oz}[i]$  preserves all Hahn sums (opa:as:thm:gaussian); whether every automorphism of  $\mathbf{Oz}[i]$  is valued is opa:as:q:gaussian, still open. For this report, Theorems 16.2 and 16.4 answer the valued case of the real-form classification left open in Section 14; the unrestricted case stays open.

The proof is organized so that no proposed theorem from an unreviewed draft is an indispensable premise. The needed reconstruction observations are reproved. Classical maximality and algebraic-closure results are identified explicitly. The proper-class argument is local and requires no large-cardinal assumption.

## 17 Hahn fields, support rings, and size conventions

### 17.1 Normal forms and strong summation

Let  $\Gamma$  be an ordered abelian group and  $k$  a field. A Hahn series is a formal expression  $x = \sum_{\gamma \in S} x_\gamma t^\gamma$ , where  $S \subseteq \Gamma$  is well ordered and  $x_\gamma \in k$ . Its support is the subset on which  $x_\gamma \neq 0$ . Addition is coefficientwise and multiplication is convolution. If  $S, S'$  are well ordered, then  $S + S'$  is well ordered and every fixed exponent has only finitely many presentations as  $s + s'$  with  $s \in S, s' \in S'$  (Lemma 3.2; [14, Lemma 1]).

A set-indexed family  $(x_j)_{j \in J}$  is *Hahn summable* when the union of its supports is well ordered and every exponent occurs in the support of only finitely many members. Its sum is obtained coefficientwise. The definition is stronger than a claim about convergence of a sequence in a particular topology. Uncountable index sets are allowed, but never a proper-class-indexed sum.

**Definition 17.1.** An additive map is strong if it takes every Hahn-summable family to a Hahn-summable family and commutes with its sum. A strong automorphism is a field automorphism for which both it and its inverse are strong.

([merge] This is the definition of Section 3, for an arbitrary coefficient field and exponent group.) The two-direction requirement is intentional. It agrees with the care taken in the definition of strongly additive valued automorphisms in [3, Definition 4.0.5]. Below, strongness of the inverse will be proved, not inferred merely from bijectivity.

For  $x \neq 0$ , put  $v(x) = \min \text{supp}(x)$  and  $v(0) = \infty$ . A valued automorphism preserves the valuation ring and hence induces an ordered automorphism of  $\Gamma$ . A value-fixing automorphism  $\alpha$  satisfies  $v(\alpha x) = v(x)$  for all  $x$ .

### 17.2 Three support regions

Write

$$\begin{aligned} \mathcal{I} &= \{x : \text{supp}(x) \subseteq \Gamma_{<0}\}, & \mathfrak{m} &= \{x : \text{supp}(x) \subseteq \Gamma_{>0}\}, \\ A_k &= k + \mathcal{I}, & \mathcal{O} &= k + \mathfrak{m}. \end{aligned}$$

Then  $A_k$  and  $\mathcal{O}$  are rings,  $\mathcal{I}$  is an ideal of  $A_k$ ,  $\mathfrak{m}$  is the maximal ideal of  $\mathcal{O}$ , and

$$K = \mathcal{I} \oplus k \oplus \mathfrak{m} \tag{17.1}$$

as additive groups and as  $k$ -vector spaces. The zero coefficient is denoted  $\text{ct}(x)$ . It is a ring homomorphism on each of  $A_k$  and  $\mathcal{O}$ , but *not* on all of  $K$ : for example,  $\text{ct}(t^\gamma) = \text{ct}(t^{-\gamma}) = 0$  while  $\text{ct}(t^\gamma t^{-\gamma}) = 1$  when  $\gamma \neq 0$ .

For a unital subring  $\mathfrak{o} \subseteq k$ , define

$$A_{\mathfrak{o}} = \mathfrak{o} + \mathcal{I}.$$

The distinction between  $A_k$  and  $\mathcal{O}$  is essential: one has only nonpositive support, and the other only nonnegative support.

### 17.3 The surreal identification

Conway normal forms give the identifications (3.1),  $\mathbf{No} = \mathbb{R}((t^{\mathbf{No}}))$  and  $\mathbf{No}[i] = \mathbb{C}((t^{\mathbf{No}}))$  with set-sized supports. In this identification,

$$\mathbf{Oz} = \mathbb{Z} + \mathcal{I}_{\mathbb{R}}, \quad \mathbf{Oz}[i] = \mathbb{Z}[i] + \mathcal{I}_{\mathbb{C}}. \tag{17.2}$$

Here an omnific integer has no negative powers of  $\omega$ , and its constant coefficient is an ordinary integer. The standard foundations are Conway's and Gonshor's accounts [1, 2]; the overview [16] supplied with the source is useful for orientation, not as a substitute for those foundations.

For a real closed  $F$ , the field  $F((t^\Gamma))$  with divisible  $\Gamma$  is real closed. One way to see the precise input is to use [14, Corollary 4]:  $F(i)((t^\Gamma))$  is algebraically closed, and it equals  $F((t^\Gamma))(i)$ . The order is determined by the sign of the leading coefficient. For  $\Gamma = \mathbf{No}$ , each polynomial involves a set of exponents and lies in a set-sized divisible-group Hahn subfield, so the same conclusion follows locally.

## 17.4 Class conventions

Set-sized statements are in ZFC. The proper-class statements may be read in GBC with global choice (the framework of Section 1), with all displayed automorphisms understood as class maps and all supports and families understood to be sets. Only replacement for images of sets, elementary class comprehension, and set-sized field arguments are used in the construction. There is no appeal to a class-sized transcendence basis or to Zorn's lemma on a proper class.

The notation  $\mathcal{G}$  for a collection of class automorphisms is convenient second-order shorthand, like  $\text{Aut}(\mathbb{K})$  in Section 1. The counting theorem means that there is a uniform family indexed by a set of size  $2^c$  meeting every admissible conjugacy type exactly once. It does not assert that class automorphisms are sets or elements of an ordinary set-theoretic quotient group.

Every set of surreal exponents has an upper and a lower bound in  $\mathbf{No}$ . This elementary set-cut property will be used for denominator clearing. It must not be confused with an analogous assertion for a fixed set-sized ordered exponent group, where it can fail.

## 18 A coefficient-pairing test for summability

A technical point in the main proof is that a constructed bijective strong map need not automatically have a strong inverse. The following elementary criterion supplies the missing implication. The source includes a complete proof as a supporting tool and attaches no priority claim to the criterion. [merge] Theorem 18.2 is `opa:as:thm:detect` and Corollary 18.3 is the case  $\alpha = \text{id}$  of `opa:as:thm:ctiso`, both in `surreal/omnific-preserving-automorphisms` and both placed after the source's pin. That report proves more: it suffices to test countably supported  $y$  with coefficients in  $\{0, 1\}$ , it gives three routes (a triangular detector, a binary finite-row lemma, and an induced matching), and it treats isomorphisms between different Hahn fields that are semilinear over a coefficient isomorphism. The Baire-category route below is a fourth route. It quantifies over all  $y$ , so it proves the weaker form of the criterion.

**Lemma 18.1** (Row-finite uniformity). *Let  $k$  be a field and let  $(f_j)_{j \in \mathbb{N}}$  be linear functionals on  $k^{\mathbb{N}}$ , each depending on only finitely many coordinates. If for every  $b \in k^{\mathbb{N}}$  only finitely many  $f_j(b)$  are nonzero, then only finitely many  $f_j$  are nonzero functionals.*

*Proof.* Give  $k$  the discrete topology and  $k^{\mathbb{N}}$  the complete product metric in which two different sequences have distance  $2^{-r}$  when  $r$  is their first differing coordinate. For each  $N$ , the set

$$Z_N = \bigcap_{j \geq N} \ker f_j$$

is closed. The hypothesis says  $\bigcup_N Z_N = k^{\mathbb{N}}$ . By the Baire category theorem, some  $Z_N$  contains a nonempty basic cylinder, which fixes only coordinates  $0, \dots, r-1$ .

For  $j \geq N$ , varying any coordinate beyond this prefix inside the cylinder shows that  $f_j$  has zero coefficient at that coordinate. Thus all those  $f_j$  depend only on the same finite prefix. Apply the hypothesis separately to each of its coordinate vectors  $e_0, \dots, e_{r-1}$ . Beyond a common finite index, every  $f_j(e_s)$  vanishes. Such an  $f_j$  is the zero functional.  $\square$

**Theorem 18.2** (Constant-term pairing criterion). *Let  $K = k((t^\Gamma))$  be a full Hahn field, with either a set-sized exponent group or set-supported class exponents. A set-indexed family  $(x_i)_{i \in J}$  is Hahn summable if and only if*

$$\text{for every } y \in K, \quad \{i \in J : \text{ct}(x_i y) \neq 0\} \text{ is finite.} \quad (18.1)$$

*Proof.* If  $(x_i)$  is summable, multiplying it by the single series  $y$  gives a summable family. At exponent zero only finitely many members contribute, which proves (18.1).

Conversely, if a fixed exponent  $\gamma$  occurs in infinitely many  $x_i$ , use  $y = t^{-\gamma}$  to contradict the test. We may therefore assume point-finiteness of the support family. If the union  $S$  of the supports were not well ordered, it would contain a strictly decreasing sequence  $\gamma_0 > \gamma_1 > \dots$ . Each individual support meets this sequence only finitely often, since it is well ordered. Passing to a subsequence, choose distinct indices  $i_j$  with  $(x_{i_j})_{\gamma_j} \neq 0$ .

For arbitrary  $b = (b_j) \in k^{\mathbb{N}}$ , the expression

$$y_b = \sum_{j \geq 0} b_j t^{-\gamma_j}$$

is a Hahn series, because  $-\gamma_j$  is strictly increasing. For every selected row, the expression

$$f_r(b) = \text{ct}(x_{i_r}, y_b) = \sum_{j \geq 0} (x_{i_r})_{\gamma_j} b_j$$

is a finite linear form. There are infinitely many nonzero forms, since  $f_r$  has a nonzero coefficient at  $b_r$ . The pairing test says that every  $b$  makes only finitely many of them nonzero, contradicting Lemma 18.1. Thus  $S$  is well ordered, and the family is summable.  $\square$

**Corollary 18.3** (Constant-term preserving isomorphisms are strong). *Suppose  $T : K \rightarrow K$  is a  $k$ -linear field automorphism and  $\text{ct}(Tx) = \text{ct}(x)$  for every  $x \in K$ . Then both  $T$  and  $T^{-1}$  are strong.*

*Proof.* For all  $x, y$ ,

$$\text{ct}((Tx)y) = \text{ct}(T(xT^{-1}y)) = \text{ct}(xT^{-1}y).$$

The pairing criterion and the surjectivity of  $T$  show that  $(x_i)$  is summable exactly when  $(Tx_i)$  is summable. Put  $x = \sum_i x_i$  and  $z = \sum_i Tx_i$ . For every  $y$ ,

$$\text{ct}((Tx)y) = \text{ct}(xT^{-1}y) = \sum_i \text{ct}(x_i T^{-1}y) = \sum_i \text{ct}((Tx_i)y) = \text{ct}(zy).$$

The sums of coefficients here are finite. The pairing is nondegenerate: if  $w \neq 0$ , choose any nonzero coefficient  $w_\gamma$  and use  $y = t^{-\gamma}$ . Hence  $Tx = z$ . The same argument applies to  $T^{-1}$ .  $\square$

**Remark 18.4.** The test quantifies over *all* series  $y$ , not only monomials. Monomial tests detect point-finiteness but do not detect a descending sequence in the union of supports. That second obstruction is exactly what the row-finite argument handles. ([merge] `opa:as:ex:monomial tests` is an explicit instance.)

The corollary is stated for  $k$ -linear maps. [merge] Composed with a coefficientwise lift it gives a semilinear form, and with it the strongness hypothesis of Theorem 16.1 becomes redundant.

**Proposition 18.5** (Automatic strongness; [merge]). *Let  $K = k((t^\Gamma))$  be a full Hahn field,  $\Gamma$  a set or  $\mathbf{No}$  with set supports, and let  $g$  be a valued field automorphism of  $K$  with  $g(k) = k$  and  $g(\mathcal{I}) = \mathcal{I}$ . Put  $\rho = g|_k$ . Then  $\text{ct}(gx) = \rho(\text{ct}x)$  for every  $x$ ,  $g$  is  $\rho$ -semilinear, and  $g$  and  $g^{-1}$  are strong. No divisibility of  $\Gamma$  and no hypothesis on the characteristic is needed.*

*Proof.* Since  $g(\mathcal{O}) = \mathcal{O}$ , also  $g(\mathfrak{m}) = \mathfrak{m}$ . Writing  $x = p + a + \varepsilon$  according to (17.1),  $gx = gp + ga + g\varepsilon$  with  $gp \in \mathcal{I}$ ,  $ga = \rho(a) \in k$  and  $g\varepsilon \in \mathfrak{m}$ , so  $\text{ct}(gx) = \rho(\text{ct}x)$ ; semilinearity is  $g(ax) = \rho(a)g(x)$ . The coefficientwise lift  $M_\rho$  is a strong automorphism with  $\text{ct} \circ M_\rho^{-1} = \rho^{-1} \circ \text{ct}$ . Hence  $T = M_\rho^{-1}g$  is a  $k$ -linear field automorphism with  $\text{ct} \circ T = \text{ct}$ , and Corollary 18.3 makes  $T$  and  $T^{-1}$  strong. So are  $g = M_\rho T$  and  $g^{-1} = T^{-1}M_\rho^{-1}$ .  $\square$

This is `opa:as:thm:ctiso` with  $F = g$  and  $\alpha = \rho$ , together with the three-region argument of the proof of `opa:as:thm:gaussian`; it is printed here because the source's hypotheses and Question 29.1 depend on it. It is an observation of the merge, not a result of the source.

## 19 Maximality and the local treatment of class exponents

### 19.1 The set-sized maximality input

A valued-field extension is immediate if it has the same value group and residue field. A maximally complete valued field has no proper immediate extension. Full Hahn fields are maximally complete [14, Theorem 1]. For clarity, the source gives the standard support proof in the form needed here.

**Lemma 19.1** (Spherical completeness of full Hahn fields). *For set-sized  $k$  and  $\Gamma$ , every nest of nonempty closed valuation balls in  $k((t^\Gamma))$  has nonempty intersection.*

*Proof.* A ball  $\mathbb{B}(a, r) = \{x : v(x - a) \geq r\}$  prescribes the coefficients at all exponents  $\gamma < r$  to equal those of  $a$ . In a nest, these prescriptions are compatible. Define  $x_\gamma$  by a prescription whenever some ball prescribes that exponent, and set it to zero otherwise.

Its nonzero support  $S$  is well ordered. Indeed, given a nonempty subset  $S' \subseteq S$ , choose  $\gamma_0 \in S'$ . Some ball prescribes the coefficient at  $\gamma_0$ , so its radius is greater than  $\gamma_0$ . Every nonzero prescribed coefficient at an exponent at most  $\gamma_0$  then agrees with the center of that ball. The nonempty set  $S' \cap (-\infty, \gamma_0]$  is a subset of its well-ordered support, and its least element is also the least element of  $S'$ . Thus  $x$  is a Hahn series. By construction it belongs to every ball in the nest.  $\square$

**Lemma 19.2.** *A spherically complete valued field has no proper immediate extension. Consequently, a value-fixing self-embedding of a full set-sized Hahn field which induces the identity on its residue field is surjective.*

*Proof.* Suppose  $K_1/K_0$  were proper and immediate, and choose  $a \in K_1 \setminus K_0$ . For every  $w \in K_0$ , immediacy provides  $d \in K_0$  with  $v(a - d) > v(a - w)$ : first match the value of  $a - w$  by an element of  $K_0$ , then match its leading residue. Thus the set  $\{v(a - w) : w \in K_0\}$  has no maximum.

The balls  $\mathbb{B}(w, v(a - w))$  in  $K_0$  form a nest. If  $b$  lay in their intersection, the ultrametric inequality would give  $v(a - b) \geq v(a - w)$  for all  $w$ , contradicting the absence of a maximum. This proves maximality. For the final assertion, the image of the self-embedding is isometric to the full Hahn field and hence maximal; the ambient field is an immediate extension of that image and cannot be larger.  $\square$

## 19.2 Why a separate class argument is necessary

A proper-class Hahn field must not be declared maximal by repeating a set-sized proof with an unexamined class of balls. The following lemma avoids that issue entirely.

**Lemma 19.3** (Invariant set-sized exponent closure). *Let a finite group  $G$  act strongly and valuedly on  $k((t^{\mathbf{No}}))$ , stabilizing  $k$ . For any set  $S \subseteq \mathbf{No}$ , there is a set-sized divisible subgroup  $\Delta \subseteq \mathbf{No}$  containing  $S$  such that every  $g \in G$  restricts to an automorphism of  $k((t^\Delta))$ .*

*Proof.* The induced action of  $G$  on the ordered group  $\mathbf{No}$  is trivial: a finite-order increasing permutation of a linear order fixes every point. Indeed,  $\tau(\gamma) > \gamma$  or  $\tau(\gamma) < \gamma$  would yield a strict chain under iteration which could not return to  $\gamma$ .

Let  $\Delta_0$  be the divisible subgroup generated by  $S$ . Given the set-sized divisible subgroup  $\Delta_r$ , let  $\Delta_{r+1}$  be the divisible subgroup generated by

$$\Delta_r \cup \bigcup_{g \in G, \gamma \in \Delta_r} \text{supp}(g(t^\gamma)).$$

Replacement and union show that it is still a set. The union  $\Delta = \bigcup_{r < \omega} \Delta_r$  is set-sized and divisible. Every  $g(t^\gamma)$  with  $\gamma \in \Delta$  has support in  $\Delta$ . Strongness and  $k$ -stability therefore give  $g(k((t^\Delta))) \subseteq k((t^\Delta))$ . Since  $g^{-1}$  is also in  $G$ , equality holds.  $\square$

The closure is adapted to the specified finite action and to the element under consideration. It does not assert the existence of a single set-sized subfield containing the whole class field or all its automorphisms. [merge] When  $G$  also stabilizes  $\mathcal{I}$ , strongness is automatic (Proposition 18.5).

## 20 Proof of finite symmetry descent

We now prove Theorem 16.1. By Proposition 18.5 every  $g \in G$  is strong. Let  $G$  have order  $n$  and write

$$\rho_g = g|_k, \quad m_\gamma = \prod_{g \in G} g(t^{\gamma/n}).$$

The trivial-group case is included: then  $n = 1$  and  $m_\gamma = t^\gamma$ .

### 20.1 The invariant section

**Lemma 20.1.** *For all  $\gamma, \delta \in \Gamma$  and  $h \in G$ ,*

$$m_0 = 1, \quad m_{\gamma+\delta} = m_\gamma m_\delta, \quad v(m_\gamma) = \gamma, \quad h(m_\gamma) = m_\gamma.$$

*Proof.* Every element of the induced finite group of ordered automorphisms of  $\Gamma$  is the identity, by the argument in Lemma 19.3. Thus every  $g$  fixes values. Divisibility and torsion-freeness give the unique exponent  $\gamma/n$ . Multiplicativity follows by applying each  $g$  to  $t^{(\gamma+\delta)/n} = t^{\gamma/n} t^{\delta/n}$  and multiplying. The valuation of the product is the sum of its  $n$  values, hence  $\gamma$ . Finally, left multiplication by  $h$  permutes  $G$ , so it permutes the factors of  $m_\gamma$ .  $\square$

## 20.2 Diagonal summability

**Lemma 20.2.** *For every well-ordered set  $S \subseteq \Gamma$ , the family  $(m_\gamma)_{\gamma \in S}$  is Hahn summable.*

*Proof.* For a fixed  $g$ , the family  $(t^{\gamma/n})_{\gamma \in S}$  is summable, and strongness of  $g$  makes  $(g(t^{\gamma/n}))_{\gamma \in S}$  summable. Taking the product of these finitely many summable families gives a summable family indexed by  $S^G$ . This uses the ordinary finite-product Hahn support lemma.

The family  $(m_\gamma)_{\gamma \in S}$  is its diagonal subfamily, corresponding to tuples all of whose coordinates are  $\gamma$ . Every subfamily of a summable family is summable.  $\square$

Define

$$\mathcal{H}(x) = \mathcal{H}\left(\sum_{\gamma} x_{\gamma} t^{\gamma}\right) := \sum_{\gamma} x_{\gamma} m_{\gamma}. \quad (20.1)$$

The preceding lemma makes the definition meaningful for every  $x$ . It also proves forward strongness. More explicitly, for a summable family  $(x_j)$ , put  $S = \bigcup_j \text{supp}(x_j)$ . The family  $(m_\gamma)_{\gamma \in S}$  is summable, and for each  $\gamma$  only finitely many coefficients  $(x_j)_\gamma$  are nonzero. Consequently, the doubly indexed family  $((x_j)_\gamma m_\gamma)_{j,\gamma}$  is summable. Regrouping it proves both summability of  $(\mathcal{H}(x_j))$  and the equality  $\mathcal{H}(\sum_j x_j) = \sum_j \mathcal{H}(x_j)$ .

## 20.3 Field embedding and value preservation

**Lemma 20.3.** *The map  $\mathcal{H}$  is a strong  $k$ -linear field embedding, fixes  $k$  pointwise, and satisfies  $v(\mathcal{H}x) = v(x)$  for every  $x$ .*

*Proof.* Linearity and  $\mathcal{H}(1) = 1$  are immediate. For products, the ordinary Hahn convolution and Lemmas 20.1 and 20.2 justify

$$\mathcal{H}(x)\mathcal{H}(y) = \sum_{\gamma, \delta} x_{\gamma} y_{\delta} m_{\gamma} m_{\delta} = \sum_{\gamma, \delta} x_{\gamma} y_{\delta} m_{\gamma+\delta} = \mathcal{H}(xy).$$

If  $\gamma_0 = v(x)$ , the term  $x_{\gamma_0} m_{\gamma_0}$  has value  $\gamma_0$ , whereas every other term has strictly larger value. Its nonzero leading coefficient cannot cancel. Thus  $v(\mathcal{H}x) = \gamma_0$ , which also proves injectivity. Since  $m_0 = 1$ ,  $\mathcal{H}$  fixes  $k$ .  $\square$

The leading coefficient of  $m_\gamma$  need not be 1. Thus  $\mathcal{H}$  is value-fixing, but it need not be a leading-term-fixing automorphism. This minor distinction is necessary even for ordinary phase characters (Section 26).

## 20.4 Surjectivity: sets first, then classes

For a set-sized  $\Gamma$ , the image  $\mathcal{H}(K)$  is a maximally complete valued field by Lemma 19.2. It has value group  $\Gamma$  and contains  $k$ , with residue field  $k$ . The extension  $K/\mathcal{H}(K)$  is immediate; hence  $\mathcal{H}(K) = K$ .

For  $\Gamma = \mathbf{No}$ , fix a target  $y \in K$  and apply Lemma 19.3 to  $S = \text{supp}(y)$ . On  $k((t^\Delta))$ , all the automorphisms  $g$  restrict, and the very same formula (20.1) defines a value-fixing self-embedding of that full set-sized Hahn field. By the preceding paragraph it is surjective. In particular,  $y$  has a preimage in  $k((t^\Delta))$ . Since  $y$  was arbitrary, the class map  $\mathcal{H}$  is surjective.

This step is the reason for using a full Hahn field. The construction of a summable invariant section by itself proves an embedding, not yet an isomorphism on an arbitrary smaller series field.

## 20.5 The negative-support part and the inverse

If  $\gamma < 0$ , then  $t^{\gamma/n} \in \mathcal{I}$  and each  $g(t^{\gamma/n}) \in \mathcal{I}$ . The product  $m_\gamma$  is again in  $\mathcal{I}$ . By strong summation,

$$\mathcal{H}(\mathcal{I}) \subseteq \mathcal{I}. \quad (20.2)$$

Value preservation and surjectivity give  $\mathcal{H}(\mathfrak{m}) = \mathfrak{m}$ . To prove equality in (20.2), write the preimage of  $p \in \mathcal{I}$  as  $x = q + a + \varepsilon$  according to (17.1). Then

$$p = \mathcal{H}(q) + a + \mathcal{H}(\varepsilon), \quad \mathcal{H}(q) \in \mathcal{I}, \quad \mathcal{H}(\varepsilon) \in \mathfrak{m}.$$

The direct-sum decomposition forces  $a = 0$  and  $\mathcal{H}(\varepsilon) = 0$ . Thus  $x \in \mathcal{I}$ , proving  $\mathcal{H}(\mathcal{I}) = \mathcal{I}$ .

The same decomposition now shows  $\text{ct}(\mathcal{H}x) = \text{ct}(x)$  for every  $x$ . By Corollary 18.3, the inverse  $\mathcal{H}^{-1}$  is strong. This proves all the structural properties asserted for  $\mathcal{H}$ .

## 20.6 Intertwining and fixed points

For  $h \in G$ , strongness and invariance of  $m_\gamma$  give

$$h\mathcal{H}\left(\sum_{\gamma} a_{\gamma} t^{\gamma}\right) = \sum_{\gamma} \rho_h(a_{\gamma}) m_{\gamma} = \mathcal{H} M_{\rho_h} \left(\sum_{\gamma} a_{\gamma} t^{\gamma}\right).$$

Therefore  $\mathcal{H}^{-1} h \mathcal{H} = M_{\rho_h}$ . If  $\rho_h = \text{id}$ , this identity and surjectivity imply  $h = \text{id}$ . This completes the proof of Theorem 16.1.

**Corollary 20.4** (Fixed fields and support rings). *Under the hypotheses of Theorem 16.1, suppose also that  $G$  stabilizes a unital subring  $\mathfrak{o} \subseteq k$ . Then*

$$K^G = \mathcal{H}(k^G((t^{\Gamma}))), \quad A_{\mathfrak{o}}^G = \mathcal{H}(\mathfrak{o}^G + \mathcal{I}_{k^G}).$$

*The conjugating map fixes  $\mathfrak{o}$  pointwise and preserves  $A_{\mathfrak{o}}$  setwise.*

*Proof.* A series is fixed by every coefficientwise  $M_{\rho_g}$  exactly when each of its coefficients belongs to  $k^G$ . Intersecting with  $\mathfrak{o} + \mathcal{I}$  restricts the constant coefficient to  $\mathfrak{o}^G$ . The properties of  $\mathcal{H}$  proved above transport these identities to the original action.  $\square$

## 21 Finite lifts, conjugacy, and torsion

### 21.1 All finite lifts of a given coefficient action

Let  $\mathcal{N}$  be the strong  $k$ -linear, value-fixing automorphisms of  $K$  preserving  $\mathcal{I}$ . ([merge] By Corollary 18.3, “strong” is automatic here: such a map preserves  $\mathcal{I}$ ,  $k$  and  $\mathfrak{m}$ , hence  $\text{ct}$ .) A coefficient automorphism  $\rho \in \text{Aut}(k)$  has a canonical strong lift  $M_{\rho}$ , which normalizes  $\mathcal{N}$ .

**Corollary 21.1** (Conjugacy of finite lifts). *Let  $\Theta$  be a finite group and let  $\rho : \Theta \rightarrow \text{Aut}(k)$  be an action. Suppose  $\alpha, \alpha' : \Theta \rightarrow \text{Aut}(K)$  are valued actions stabilizing  $k$  and  $\mathcal{I}$ , each inducing  $\rho$  on  $k$ . Then  $\mathcal{H}_{\alpha'} \mathcal{H}_{\alpha}^{-1} \in \mathcal{N}$  and*

$$\alpha'_p = (\mathcal{H}_{\alpha'} \mathcal{H}_{\alpha}^{-1}) \alpha_p (\mathcal{H}_{\alpha'} \mathcal{H}_{\alpha}^{-1})^{-1} \quad (p \in \Theta).$$

*If  $\rho$  is faithful, these are embeddings of  $\Theta$  and all lifts of the specified coefficient action are conjugate by  $\mathcal{N}$ .*

*Proof.* The norm proof applies to a finite acting group even when the action is not faithful: the product is taken over its  $|\Theta|$  elements, with repetitions of the same automorphism allowed. It gives  $\mathcal{H}_\alpha, \mathcal{H}_{\alpha'} \in \mathcal{N}$  such that  $\alpha_p = \mathcal{H}_\alpha M_{\rho_p} \mathcal{H}_\alpha^{-1}$  and  $\alpha'_p = \mathcal{H}_{\alpha'} M_{\rho_p} \mathcal{H}_{\alpha'}^{-1}$ ; compose.  $\square$

**Corollary 21.2** (Torsion-free coefficient kernel). *Every finite-order valued automorphism of  $K$  stabilizing  $k$  and  $\mathcal{I}$  and fixing  $k$  pointwise is the identity. In particular,  $\mathcal{N}$  is torsion-free.*

(As in Theorem 16.1, the source’s word “strong” in the two corollaries is dropped by Proposition 18.5; [merge].) For  $K = \mathbb{K}$  and  $k = \mathbb{C}$ , Corollary 21.2 is contained in Theorem 10.1, which proves  $\text{Aut}(\mathbb{K}/\mathbb{C})$  torsion-free with no valuation hypothesis ([merge]); for other coefficient fields the statement is the source’s.

There is also a precise cohomological formulation, without introducing an extra existence assumption. Let  $\Theta$  act on  $\mathcal{N}$  by conjugation with  $M_{\rho_p}$ . A cocycle  $\xi_p \in \mathcal{N}$  satisfies

$$\xi_{pq} = \xi_p (M_{\rho_p} \xi_q M_{\rho_p}^{-1}).$$

Then  $\alpha_p = \xi_p M_{\rho_p}$  is a finite action of the required kind. The theorem gives  $\mathcal{H} \in \mathcal{N}$  with

$$\xi_p = \mathcal{H} M_{\rho_p} \mathcal{H}^{-1} M_{\rho_p}^{-1}.$$

Thus the associated nonabelian  $H^1(\Theta, \mathcal{N})$  is trivial. For class fields, this is a shorthand assertion about each set-indexed finite cocycle, not an unexamined construction of a set of all class cocycles.

## 21.2 What the theorem does not say about infinite groups

A finite orbit permits an invariant product with exactly  $n$  factors and a coherent exponent division by  $n$ . Neither operation has an automatic analogue for an infinite group. The theorem therefore gives no classification of infinite subgroup actions, no assertion that the entire automorphism group is conjugate to a coefficient group, and no vanishing theorem for cohomology of arbitrary infinite groups.

## 22 Recovering coefficients from the Gaussian omnific ring

To apply the descent theorem to  $\mathbf{Oz}[i]$ , coefficient stability and  $\mathcal{I}$ -stability should not be imposed silently. They follow from  $\mathbf{Oz}[i]$  itself. The reconstruction package is already present in the collection’s omnific reports:  $\mathcal{I} = \bigcap_n nA_0$  is `opa:lem:divisible` of `surreal/omnific-preserving-automorphisms`, the multiplier identity and its units are `odg:def:thm:multiplier`, and  $\text{Frac } \mathbf{Oz} = \mathbf{No}$  is `odg:thm:fractions`, both of `surreal/omnific-diophantine-geometry`. The short arguments below are included to make the application independent of their proof status; the source claims no novelty for them.

**Proposition 22.1** (Intrinsic support ideal and coefficient field). *Let  $k$  be a set-sized characteristic-zero field and let  $\mathfrak{o} \subseteq k$  be either  $\mathbb{Z}$ , or  $\mathbb{Z}[i]$  with  $k$  containing  $i$ . For  $K = k((t^{\mathbf{No}}))$ ,  $\mathcal{I} = \mathcal{I}_k$ , and  $A_0 = \mathfrak{o} + \mathcal{I}$ , one has*

$$\text{Frac}(A_0) = K, \tag{22.1}$$

$$\mathcal{I} = \bigcap_{n \geq 1} nA_0, \tag{22.2}$$

$$A_k = k + \mathcal{I} = (\mathcal{I} : \mathcal{I}) := \{x \in K : x\mathcal{I} \subseteq \mathcal{I}\}, \tag{22.3}$$

$$k = A_k^\times \cup \{0\}. \tag{22.4}$$

*Every unital automorphism of  $A_0$  extends uniquely to  $K$  and preserves  $\mathcal{I}$ ,  $A_k$ , and  $k$  setwise.*

*Proof.* Given  $x \in K$ , its support is a set of surreal exponents. Choose  $\delta > 0$  larger than every exponent in that support. Then  $t^{-\delta} \in \mathcal{I}$  and  $t^{-\delta}x \in \mathcal{I}$ , so  $x = (t^{-\delta}x)/t^{-\delta}$  lies in  $\text{Frac}(A_{\mathfrak{o}})$ .

The ideal  $\mathcal{I}$  is divisible by every nonzero ordinary integer. Conversely, if  $a + p \in \bigcap_{n \geq 1} nA_{\mathfrak{o}}$ , with  $a \in \mathfrak{o}$ , then  $a \in n\mathfrak{o}$  for every  $n$ . The only such integer or Gaussian integer is zero. This proves (22.2).

An element supported at nonpositive exponents multiplies strictly negative support into strictly negative support. If  $x$  has a nonzero coefficient at an exponent  $\epsilon > 0$ , choose  $0 < \delta < \epsilon$ . The element  $t^{-\delta} \in \mathcal{I}$  makes  $xt^{-\delta}$  have a nonzero coefficient at  $\epsilon - \delta > 0$ , so  $x \notin (\mathcal{I} : \mathcal{I})$ . This proves (22.3).

For nonzero  $x, y \in A_k$  with  $xy = 1$ , both  $v(x)$  and  $v(y)$  are nonpositive, and their sum is zero. Hence both valuations are zero. A series supported at nonpositive exponents with minimum exponent zero is a constant. Thus  $A_k^{\times} = k^{\times}$ , proving (22.4). All four constructions are preserved by a ring automorphism, after its unique extension to the fraction field.  $\square$

*Remark 22.2* (The valuation is not recovered by this proposition). The proposition recovers the coefficient field and the strictly negative-support ideal. It does *not* assert that every abstract ring automorphism preserves the Hahn valuation ring  $\mathcal{O}$ . The source adds: nor does it assert strongness, and both remain explicit hypotheses in the definition of  $\mathcal{G}$ . [merge] The second half is now stale: once the valuation ring is preserved, strongness follows (Remark 22.3). Preservation of  $\mathcal{O}$  remains a hypothesis (`opa:as:q:gaussian`).

*Remark 22.3* (Valued Gaussian automorphisms are strong; [merge]). Let  $\alpha$  be a valued automorphism of  $K = k((t^{\mathbb{N}_0}))$  preserving  $A_{\mathfrak{o}}$ , with  $\mathfrak{o}$  as in Proposition 22.1. By that proposition  $\alpha(k) = k$  and  $\alpha(\mathcal{I}) = \mathcal{I}$ , and Proposition 18.5 makes  $\alpha$  strong and  $\alpha|_k$ -semilinear. For  $k = \mathbb{C}$  and  $\mathfrak{o} = \mathbb{Z}[i]$  this is `opa:as:thm:gaussian`, whose proof is the same. Hence  $\mathcal{G}$  is the class the source calls  $\mathcal{G}$ . For arbitrary characteristic-zero  $k$  and  $\mathfrak{o} \in \{\mathbb{Z}, \mathbb{Z}[i]\}$  it is a small piece of information on `opa:as:q:otherrings`, which asks when preserving such a ring forces strongness; there the valuation is not assumed, and the question is not answered.

**Corollary 22.4.** *For set-sized real closed fields  $F, F'$ , any unital ring isomorphism  $R_F \cong R_{F'}$  induces an isomorphism  $F \cong F'$ . Conversely, an isomorphism  $F \cong F'$  extends coefficientwise to an isomorphism of the pairs  $(\mathbf{No}_F, R_F) \cong (\mathbf{No}_{F'}, R_{F'})$ . Thus*

$$R_F \cong R_{F'} \iff F \cong F'.$$

*Proof.* Apply Proposition 22.1 with  $\mathfrak{o} = \mathbb{Z}$  on each side. The converse uses the coefficientwise extension and the identity on exponents.  $\square$

This corollary is useful even when the coefficient fields are non-Archimedean: the abstract fixed ring remembers the whole coefficient field, not merely its natural residue field.

## 23 Classification of Gaussian-preserving involutions

Let  $\mathbb{K} = \mathbf{No}[i] = \mathbb{C}((t^{\mathbb{N}_0}))$  and let  $c$  be standard conjugation. Recall that  $\mathcal{G}$  consists of the valued automorphisms preserving  $\mathbf{Oz}[i]$ . It is closed under composition and inverse.

### 23.1 Reduction to coefficient conjugation

Let  $j \in \mathcal{G}$  have order two. By Proposition 22.1,  $j$  stabilizes  $\mathbb{C}$  and  $\mathcal{I}$ . The finite descent theorem applies to  $\{1, j\}$ . It yields

$$\mathcal{H}_j(t^{\gamma}) = t^{\gamma/2} j(t^{\gamma/2}), \quad j = \mathcal{H}_j M_{\rho} \mathcal{H}_j^{-1}, \quad \rho = j|_{\mathbb{C}}. \quad (23.1)$$

If  $\rho$  were the identity, so would be  $j$ . Thus  $\rho$  is a nontrivial involution of  $\mathbb{C}$ .

The classical Artin–Schreier theorem, applied in the ordinary set field  $\mathbb{C}$ , says that  $F = \mathbb{C}^\rho$  is real closed and  $\mathbb{C} = F(i)$ , with  $\rho(i) = -i$  [8]. Consequently  $\mathbb{Z}[i]^\rho = \mathbb{Z}$ , and Corollary 20.4 identifies the fixed pair with  $(\mathbf{No}_F, R_F)$  in (16.2).

No extension of the Artin–Schreier theorem to a proper-class Galois theory is needed in this step. Once the coefficient theorem is known, the class fixed field is obtained directly, coefficient by coefficient. In particular, every  $z \in \mathbb{K}$  can also be written uniquely as  $x + iy$  with  $x, y \in \text{Fix}(j)$  after transporting the coefficientwise decomposition through  $\mathcal{H}_j$ . ([merge] That  $\text{Fix}(j)$  is a real form also follows from Theorem 10.1, for every involution.)

## 23.2 Conjugacy is exactly coefficient conjugacy

**Proposition 23.1.** *For nonidentity involutions  $j_1, j_2 \in \mathcal{G}$ , the following are equivalent:*

- (i)  $j_1$  and  $j_2$  are conjugate in  $\mathcal{G}$ ;
- (ii)  $\rho_1 = j_1|_{\mathbb{C}}$  and  $\rho_2 = j_2|_{\mathbb{C}}$  are conjugate in  $\text{Aut}(\mathbb{C})$ ;
- (iii)  $\mathbb{C}^{\rho_1} \cong \mathbb{C}^{\rho_2}$ ;
- (iv)  $\mathbf{Oz}[i]^{j_1} \cong \mathbf{Oz}[i]^{j_2}$  as unital rings.

*Proof.* A conjugating element of  $\mathcal{G}$  preserves  $\mathbb{C}$  by reconstruction, so (i) implies (ii). If  $\rho_2 = \kappa \rho_1 \kappa^{-1}$ , then

$$\mathcal{H}_{j_2} M_\kappa \mathcal{H}_{j_1}^{-1} \in \mathcal{G}$$

conjugates  $j_1$  to  $j_2$ . Indeed, every automorphism of  $\mathbb{C}$  fixes  $\mathbb{Z}$  and sends  $i$  to  $i$  or  $-i$ , so its coefficientwise lift preserves  $\mathbb{Z}[i] + \mathcal{I}$ . This proves (ii) implies (i).

A coefficient conjugacy restricts to an isomorphism of fixed fields. Conversely, an isomorphism  $F_1 \rightarrow F_2$  of the fixed fields extends to  $F_1(i) \rightarrow F_2(i)$  by sending  $i$  to  $i$ ; these are both  $\mathbb{C}$ . That extension conjugates the two involutions. This proves (ii) iff (iii). Finally, the fixed-pair description and Corollary 22.4 prove (iii) iff (iv).  $\square$

In particular, involutions with the *same* coefficient restriction are conjugate by a  $\mathbb{C}$ -linear value-fixing automorphism preserving  $\mathbf{Oz}[i]$ . If the coefficient restriction is ordinary complex conjugation, this conjugates the involution to the standard one.

[merge] The criterion is the valued,  $\mathbf{Oz}[i]$ -preserving refinement of the plain criterion of Section 10, by which two involutions are conjugate in  $\text{Aut}(\mathbb{K})$  exactly when their fixed real fields are isomorphic. Three consequences for material already in the collection:

- The twisted conjugations  $j_\theta = P_\theta c P_\theta^{-1}$  of (4.8) lie in  $\mathcal{G}$  ( $P_\theta$  is strong, value-fixing, and fixes  $\mathbb{Z}[i]$  and every support) and restrict to  $c$  on  $\mathbb{C}$ , so the proposition predicts their conjugacy to  $c$  inside  $\mathcal{G}$ ; their norm conjugator is  $P_\theta$  itself (Section 26).
- The real form of Theorem 10.2 has a countable cofinal subset, whereas every set of elements of  $\mathbf{No}_F$  is bounded above by some  $t^{-\delta}$ . So its involution is not conjugate, even by a plain field automorphism, to any member of  $\mathcal{G}$ .
- An involution of  $\mathbb{K}$  that agrees with  $c$  on  $\mathbb{C}$  and lies in  $\mathcal{G}$  is conjugate to  $c$ . The forcing-sensitive involutions of foundations-and-computation/surreal-fields-across-universes that agree with  $c$  on  $\mathbb{C}$  (univ:cor:agreement) are not conjugate to  $c$ ; hence none of them is both valued and  $\mathbf{Oz}[i]$ -preserving. The same holds for the proper-class family of pure-field involutions that the report surcomplex/first-kappa-coefficients receives in batch 32 (fkc:sb:thm:involutions): they agree with  $c$  on  $\mathbb{C}$  and none is conjugate to  $c$ .

### 23.3 Which coefficient fields occur?

**Proposition 23.2.** *A set-sized field occurs, up to isomorphism, as  $\mathbb{C}^\rho$  for a nonidentity involution  $\rho$  of  $\mathbb{C}$  if and only if it is real closed and has transcendence degree  $\mathfrak{c}$  over  $\mathbb{Q}$ .*

*Proof.* Necessity of real closedness is Artin–Schreier. Since  $\mathbb{C}/F$  is algebraic,  $\text{trdeg}_{\mathbb{Q}} F = \text{trdeg}_{\mathbb{Q}} \mathbb{C} = \mathfrak{c}$ . The last equality follows from the standard classification of algebraically closed fields, or directly from cardinality: the algebraic closure of a field generated over the countable field  $\mathbb{Q}$  by an infinite set  $X$  has cardinal  $\max(\aleph_0, |X|)$ .

Conversely, a real closed  $F$  of transcendence degree  $\mathfrak{c}$  has algebraically closed  $F(i)$  of characteristic zero and transcendence degree  $\mathfrak{c}$ . Choosing a transcendence-basis bijection and extending across algebraic closures gives an isomorphism  $\psi : F(i) \rightarrow \mathbb{C}$ . Transport conjugation on  $F(i)$  through  $\psi$ . Its fixed field is  $\psi(F) \cong F$ .  $\square$

The standard algebraically closed field classification used here is purely set-sized; see, for example, [15]. The choices of coefficient isomorphisms do not enter the explicit norm construction once an action is given.

**Corollary 23.3.** *Every nontrivial finite subgroup of  $\mathcal{G}$  has order two. The kernel of restriction  $\mathcal{G} \rightarrow \text{Aut}(\mathbb{C})$  has no nontrivial torsion.*

*Proof.* [merge] Both assertions are contained in Theorem 10.1, which proves them for all of  $\text{Aut}(\mathbb{K})$  and for  $\text{Aut}(\mathbb{K}/\mathbb{C})$ . The source’s route: restriction is faithful on any finite subgroup by descent. A nontrivial finite group of automorphisms of the algebraically closed set field  $\mathbb{C}$  has order two by Artin–Schreier and Artin’s fixed-field theorem [8, Section 4]. The second assertion follows from Corollary 21.2.  $\square$

The order-two restriction is a classical algebraic phenomenon, not a new claim specific to surcomplex numbers. The new work is the simultaneous support-preserving conjugation and the resulting fixed-ring classification.

## 24 Integer parts, missing cuts, and the standard surreal field

Let  $F$  be a set-sized real closed field. Give  $\mathbf{No}_F = F((t^{\mathbf{No}}))$  its leading-coefficient order and put  $R_F = \mathbb{Z} + I_F$ .

### 24.1 Discreteness is weaker than the integer-part property

**Definition 24.1.** A subring  $R$  of an ordered field  $L$  is an integer part if it is discretely ordered with least positive element 1 and for every  $x \in L$  there is an  $r \in R$  satisfying  $r \leq x < r + 1$ .

**Lemma 24.2.** *The ring  $R_F$  is discretely ordered with least positive element 1. Every nonzero  $p \in I_F$  has absolute value greater than every positive element of  $F$ .*

*Proof.* The leading exponent of a nonzero  $p \in I_F$  is negative. Therefore its absolute value dominates every constant, independently of whether the coefficient field itself is Archimedean. A positive element  $n + p$  of  $R_F$  is consequently either an ordinary positive integer or is larger than all ordinary integers. No element lies strictly between 0 and 1.  $\square$

**Theorem 24.3** (Exact integer-part criterion). *The ring  $R_F$  is an integer part of  $\mathbf{No}_F$  if and only if  $F$  is Archimedean. In that case write  $x = p + a + \varepsilon$  according to (17.1), and let  $n \in \mathbb{Z}$  satisfy  $n \leq a < n + 1$  in  $F$ . Its unique  $R_F$ -floor is*

$$\lfloor x \rfloor_{R_F} = \begin{cases} p + n - 1, & a = n \text{ and } \varepsilon < 0, \\ p + n, & \text{otherwise.} \end{cases} \quad (24.1)$$

*Proof.* If  $F$  is Archimedean, every element of  $F$  lies between consecutive ordinary integers. A nonzero constant of  $F$  dominates every infinitesimal in  $\mathfrak{m}_F$ . Thus adding  $\varepsilon$  cannot cross an integer boundary unless  $a$  is already that integer, and in that case it crosses downward exactly when  $\varepsilon < 0$ . Formula (24.1) follows. Discreteness gives uniqueness.

If  $F$  is non-Archimedean, choose  $a \in F$  with  $a > n$  for every ordinary integer  $n$ . No ordinary integer is within one below  $a$ . An  $r = n + p$  with  $p \neq 0$  is either larger than every constant in  $F$ , or its successor is smaller than every constant in  $F$ , by Lemma 24.2. Such an  $r$  cannot satisfy  $r \leq a < r + 1$  either. Hence  $R_F$  is not an integer part.  $\square$

The negative-infinitesimal correction in (24.1) is not optional. In particular, the floor of  $-t^\delta$ , for  $\delta > 0$ , is  $-1$ . The formula specializes to the usual omnific floor when  $F = \mathbb{R}$ ; that specialization is already part of the collection's omnific foundations.

## 24.2 When does the fixed field itself equal a surreal field?

The order on a real closed field is definable from its field operations: a nonzero element is positive exactly when it is a square. Every field isomorphism between real closed fields therefore preserves order (Lemma 2.1).

Recall the set-cut interpolation property (Appendix B): for sets  $\mathcal{L}, \mathcal{R}$  in an ordered class field with every element of  $\mathcal{L}$  below every element of  $\mathcal{R}$ , there is an element strictly between all of  $\mathcal{L}$  and all of  $\mathcal{R}$ . The surreal field has this property by its cut construction [1]. Explicit omitted cuts are used below rather than a broad universality characterization.

**Theorem 24.4** (Standard-form criterion). *For set-sized real closed  $F$ ,*

$$F((t^{\mathbf{No}})) \cong \mathbf{No} \iff F \cong \mathbb{R}.$$

*Proof.* An isomorphism  $F \cong \mathbb{R}$  extends coefficientwise and proves the reverse implication. For the forward implication, first suppose that  $F$  is non-Archimedean. Let  $F_\infty \subseteq F$  be the set of positive elements larger than every ordinary integer. Consider the set cut

$$\mathbb{N} < F_\infty.$$

No  $x \in \mathbf{No}_F$  realizes it. A realizing  $x$  would be positive. If  $v(x) < 0$ , then  $x$  is larger than every positive coefficient constant, contrary to  $x < F_\infty$ . If  $v(x) > 0$ , then  $x < 1$ . If  $v(x) = 0$ , write  $x = a + \varepsilon$  with  $a \in F_{>0}$  and  $\varepsilon \in \mathfrak{m}_F$ . When  $a$  is bounded by an ordinary integer, so is  $x$ . Otherwise  $a \in F_\infty$ , and  $a/2 \in F_\infty$  while  $x > a/2$ , again a contradiction. Thus  $\mathbf{No}_F$  omits a set cut which  $\mathbf{No}$  realizes.

Now suppose  $F$  is Archimedean. Its rational-cut embedding identifies it with an ordered subfield of  $\mathbb{R}$ . If it is proper, choose  $r \in \mathbb{R} \setminus F$  and take the cut

$$\{q \in \mathbb{Q} : q < r\} < \{q \in \mathbb{Q} : q > r\}.$$

Any realizing series must be bounded by ordinary integers, hence has no negative support. Its constant coefficient  $a \in F$  has the same rational cut as  $r$ : any strict nonzero constant difference dominates its infinitesimal part. Thus  $a = r$ , impossible. This is again an omitted set cut. Therefore  $F$  cannot be a proper Archimedean coefficient subfield, and  $F \cong \mathbb{R}$  is necessary.  $\square$

**Corollary 24.5** (Standard conjugacy in the full plain field group). *An involution  $j \in \mathcal{G}$  is conjugate to standard conjugation under an arbitrary plain field automorphism of  $\mathbb{K}$  if and only if  $F = \mathbb{C}^{j|c} \cong \mathbb{R}$ . In that case it is already conjugate to standard conjugation within  $\mathcal{G}$ .*

*Proof.* A plain conjugacy induces an isomorphism  $\text{Fix}(j) \cong \mathbf{No}$ , which forces  $F \cong \mathbb{R}$  by Theorem 24.4. Conversely, an isomorphism  $F \cong \mathbb{R}$  extends to a coefficient conjugacy on  $\mathbb{C}$ ; Proposition 23.1 then gives a conjugacy in  $\mathcal{G}$ .  $\square$

[merge] Together with the conjugacy criterion Proposition B.1, the corollary says that for  $j \in \mathcal{G}$  the fixed field  $\text{Fix}(j) \cong \mathbf{No}_F$  has the set-cut interpolation property exactly when  $F \cong \mathbb{R}$ ; the proof of Theorem 24.4 exhibits the omitted set cut when  $F \not\cong \mathbb{R}$ . The two criteria are consistent.

### 24.3 The intrinsic residue field in the Archimedean case

**Proposition 24.6.** *If  $F$  is Archimedean, the natural valuation ring of  $\mathbf{No}_F$ , defined as the elements bounded in absolute value by an ordinary integer, is exactly  $\mathcal{O}_F = F + \mathfrak{m}_F$ . Its residue field is  $F$ . Consequently, if  $F, F'$  are Archimedean real closed fields and  $\mathbf{No}_F \cong \mathbf{No}_{F'}$  as plain fields, then  $F \cong F'$ .*

*Proof.* A series of negative valuation dominates all ordinary integers. A series of positive valuation is infinitesimal. A series of valuation zero has the form  $a + \varepsilon$  with  $a \in F$  bounded by an ordinary integer, and  $\varepsilon$  infinitesimal. This proves the ring assertion. Its infinitesimal ideal is  $\mathfrak{m}_F$ , and the constant-term residue is  $F$ . A field isomorphism preserves the intrinsic order and ordinary integers, therefore preserves the natural valuation ring and induces an isomorphism of residue fields.  $\square$

For non-Archimedean  $F$ , the outer Hahn valuation need not be the natural valuation of the ordered field. The proposition is deliberately not stated in that generality. The omitted-cut proof above handles that case without this identification.

## 25 Exactly $2^c$ types

### 25.1 A large family of Archimedean coefficient fields

**Lemma 25.1.** *There are  $2^c$  pairwise nonisomorphic Archimedean real closed fields of transcendence degree  $c$  over  $\mathbb{Q}$ . They may be chosen as subfields of  $\mathbb{R}$ .*

*Proof.* Choose a real transcendence basis  $B$  of  $\mathbb{R}/\mathbb{Q}$ . It has cardinal  $c$ , and it may be partitioned as  $B = B_0 \sqcup B_1$  with both pieces of cardinal  $c$ . For each  $Y \subseteq B_1$ , let

$$F_Y = \{x \in \mathbb{R} : x \text{ is algebraic over } \mathbb{Q}(B_0 \cup Y)\}. \quad (25.1)$$

This is the real closure inside  $\mathbb{R}$  of the indicated ordered field: its positive square roots and roots of odd-degree polynomials in  $\mathbb{R}$  remain algebraic over the base. Thus it is real closed and Archimedean, and it has transcendence degree  $c$  because it contains  $B_0$ .

Algebraic independence gives  $F_Y \cap B = B_0 \cup Y$ . Distinct  $Y$  therefore give distinct subfields. A field isomorphism between two real closed subfields of  $\mathbb{R}$  is order preserving and fixes  $\mathbb{Q}$ . Rational cuts uniquely determine real numbers, so such an isomorphism is pointwise the identity. Thus two different  $F_Y$  cannot be isomorphic.  $\square$

### 25.2 Transport to Gaussian-preserving symmetries

Choose an isomorphism  $\psi_Y : F_Y(i) \rightarrow \mathbb{C}$ . Let  $c_Y$  be conjugation on  $F_Y(i)$ , and define

$$\rho_Y = \psi_Y c_Y \psi_Y^{-1}, \quad j_Y = M_{\rho_Y} \quad \text{on } \mathbb{C}((t^{\mathbf{No}})).$$

Every  $j_Y$  is strong and value-fixing. Since every automorphism of  $\mathbb{C}$  sends  $i$  to  $\pm i$ , it preserves  $\mathbb{Z}[i] + \mathcal{I}$ . Hence  $j_Y \in \mathcal{G}$ . Its fixed pair is isomorphic to  $(\mathbf{No}_{F_Y}, R_{F_Y})$ .

**Theorem 25.2** (Large nonconjugate family). *The family just constructed consists of  $2^c$  involutions whose fixed fields are pairwise nonisomorphic, whose fixed rings are pairwise nonisomorphic integer parts, and which are pairwise nonconjugate even in the full plain field automorphism collection of  $\mathbf{No}[i]$ .*

*Proof.* By Proposition 24.6, an isomorphism between two fixed fields would induce  $F_Y \cong F_{Y'}$ , contrary to Lemma 25.1 when  $Y \neq Y'$ . The integer-part assertion is Theorem 24.3. The fixed-ring nonisomorphism also follows directly from Corollary 22.4. Finally, any conjugacy of involutions induces an isomorphism of their fixed fields, so no plain field conjugacy exists between different members.  $\square$

### 25.3 The upper bound and the standard exception

The set of all maps  $\mathbb{C} \rightarrow \mathbb{C}$  has cardinal

$$c^c = (2^{\aleph_0})^c = 2^c.$$

Thus  $\text{Aut}(\mathbb{C})$  has at most  $2^c$  elements and at most that many involution conjugacy classes. By Proposition 23.1, these classes classify all involutions in  $\mathcal{G}$ . Combined with Theorem 25.2, this proves the exact count in Theorem 16.4. ([merge] The coefficientwise lifts  $M_\rho$ , one for each conjugacy class of nonidentity involutions  $\rho$  of  $\mathbb{C}$ , form a family meeting every type exactly once.)

For  $Y = B_1$ , the field  $F_Y$  is  $\mathbb{R}$ , so this parameter gives the standard type. For every proper  $Y$ , at least one element of the transcendence basis is omitted, so  $F_Y \neq \mathbb{R}$ . Such a field is not even abstractly isomorphic to  $\mathbb{R}$ , by the rational-cut argument. Removing  $Y = B_1$  leaves  $2^c$  parameters and proves the asserted multiplicity of nonstandard types.

*Remark 25.3* (What is counted). The upper bound concerns the specified class  $\mathcal{G}$  of valued  $\mathbf{Oz}[i]$ -preserving automorphisms. It is not an upper bound for all involutions of the plain proper-class field. The lower-bound family remains nonconjugate in that larger collection, which is a stronger separation statement but a different assertion from an unrestricted upper bound.

## 26 Explicit examples and the content of the norm formula

### 26.1 A phase-character calculation

Suppose  $\rho$  is standard conjugation on  $\mathbb{C}$  and  $\chi : \Gamma \rightarrow \mathbb{C}^\times$  is a character with  $\chi(\gamma)\overline{\chi(\gamma)} = 1$ . Define

$$j \left( \sum_{\gamma} z_{\gamma} t^{\gamma} \right) = \sum_{\gamma} \overline{z_{\gamma}} \chi(\gamma) t^{\gamma},$$

that is ([merge]),  $j = D_{\chi}c$  in the notation of Theorem 6.1. This is a strong value-fixing involution preserving the set of negative-support series and  $\mathbb{Z}[i] + \mathcal{I}$ . The norm construction gives

$$\mathcal{H}_j(t^{\gamma}) = \chi(\gamma/2) t^{\gamma}.$$

Since  $\chi(\gamma/2)^2 = \chi(\gamma)$ , the section uses a coherent square root supplied by the divisible exponent group, without selecting unrelated square roots of individual complex numbers. The conjugacy identity follows from  $\chi(\gamma/2) = \chi(\gamma/2)^{-1}$ .

Phase twists themselves already occur in the collection: Theorem 4.2 here, and `odg:def:lem:twist` of `surreal/omnific-diophantine-geometry`. This calculation illustrates how the general theorem

normalizes them rather than claiming them as new examples. [merge] For  $\chi(\gamma) = e^{2i\theta\ell(\gamma)}$  the involution is the twisted conjugation  $j_\theta = P_{2\theta}c$  of (4.8), and  $\mathcal{H}_{j_\theta}(t^\gamma) = e^{i\theta\ell(\gamma)}t^\gamma$ : the norm conjugator is exactly  $P_\theta$ , and the descent identity  $j_\theta = \mathcal{H}_{j_\theta}c\mathcal{H}_{j_\theta}^{-1}$  is the definition of  $j_\theta$ .

## 26.2 A nonlinear two-scale involution

Take the set-sized exponent group  $\Gamma = \mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}$ , write  $z = t^{(1,0)}$  and  $u = t^{(0,1)}$ , and work in

$$K_{\text{lex}} = \mathbb{C}((u^{\mathbb{Q}}))((z^{\mathbb{Q}})) = \mathbb{C}((t^{\mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}})).$$

Here  $u$  is small, but every positive power of  $z$  is smaller than every power of  $u$ . For rational  $a$ , binomial powers of a principal unit are understood as their unique formal power series with constant term 1.

Define a  $\mathbb{C}$ -linear map  $\Psi$  on monomials by

$$\Psi(z^a u^b) = z^a u^b (1 + iu)^a. \quad (26.1)$$

It is a strong valued automorphism. One direct verification groups a series by its outer exponent  $a$ : it multiplies the  $a$ -coefficient by the invertible series  $(1 + iu)^a$ , while leaving the outer support unchanged. Within each fiber, multiplication preserves Hahn summability. The inverse uses  $(1 + iu)^{-a}$ , so it is strong as well.

The map preserves the negative-support part. At a negative outer exponent, all corrections still have negative lexicographic exponent. At outer exponent zero, the factor is 1, so the sign of the inner exponent is unchanged. Thus  $\Psi$  and  $\Psi^{-1}$  both preserve  $\mathcal{I}$ .

Let  $c$  conjugate coefficients and fix  $z, u$ . Then  $j = \Psi c \Psi^{-1}$  is an involution. Set

$$W(u) = \frac{1 + iu}{1 - iu}.$$

A calculation gives

$$j(z^a u^b) = z^a u^b W(u)^a, \quad j|_{\mathbb{C}} = c|_{\mathbb{C}}, \quad (26.2)$$

where  $\overline{W} = W^{-1}$ . In particular,

$$j(z) = z + 2izu - 2zu^2 - 2izu^3 + 2zu^4 + \cdots.$$

The norm conjugator is not simply  $\Psi$ :

$$\mathcal{H}_j(z^a u^b) = z^a u^b W(u)^{a/2}. \quad (26.3)$$

Nevertheless,

$$\mathcal{H}_j^{-1} j \mathcal{H}_j = c.$$

The difference between  $\Psi$  and  $\mathcal{H}_j$  is an automorphism commuting with  $c$ . A conjugating map need not be unique even when the norm formula selects one from the given monomial section.

## 26.3 What the finite verifier checks

The accompanying program (`code/02-finite-symmetry-descent-verify_finite.py`) works with exact rational complex coefficients in the finite quotient  $\mathbb{Q}(i)[u]/(u^N)$ . It checks binomial multiplicativity, inverses,  $\overline{W} = W^{-1}$ , the involution identity, invariance of the norm section, and the displayed conjugacy formulas for a specified finite range of rational exponents. Its checks are exact modulo  $u^N$ , not floating-point approximations.

It does not verify arbitrary Hahn summability, the maximality argument, proper-class replacement, the Baire lemma, or the cardinality theorem. Those are proof obligations met by the written arguments, not by finite testing. [merge] The recorded run at  $N = 12$  passes 3,751 assertions over the 23 exponents  $a/b$  with  $|a| \leq 4$ ,  $1 \leq b \leq 4$ . Of these, 3,047 are lexicographic comparisons of exponent pairs (the family `negative_support_finite_window`), which hold by the order itself; the 704 others are the identity checks listed above (529 of them multiplicativity of  $a \mapsto W^a$ ). Rerunning the program for this report reproduced the record (README).

## 27 Sharp hypotheses and invalid shortcuts

### 27.1 Without exponent divisibility, the coefficient kernel has torsion

**Example 27.1.** On  $k((t))$  in characteristic zero, the substitution  $t \mapsto -t$  fixes  $k$  pointwise, preserves the valuation and negative-support part, and is a strong automorphism of order two. It cannot be conjugate to a nontrivial coefficient action while fixing the coefficient restriction, because that restriction is the identity. Here  $\Gamma = \mathbb{Z}$  is not two-divisible, so  $t^{\gamma/2}$  is unavailable.

For an action of order  $n$ , the proof only requires division by  $n$  in the exponent group, together with the corresponding invariant subgroup closure in the class argument. Divisibility is the clean uniform hypothesis for all finite groups. No necessity theorem for every individual action is claimed.

### 27.2 Fullness is needed for the surjectivity step

On the proper subfield  $k(t) \subset k((t))$ , the embedding  $t \mapsto t + t^2$  fixes the coefficient field, the value group, and residue field, but is not onto:  $t$  satisfies  $X^2 + X - (t + t^2) = 0$  over the image and is not a rational function of  $t + t^2$ . For instance, the nonidentity involution  $t \mapsto -1 - t$  fixes  $t + t^2$ , so  $t$  cannot lie in that rational subfield.

This example is not asserted to arise from a finite norm action, and it need not preserve the negative-support ring. Its purpose is narrower: a value-fixing immediate self-embedding of a smaller field need not be surjective. The full Hahn maximality hypothesis in the proof cannot be replaced by that implication without additional work.

### 27.3 Discreteness alone does not provide floors

For non-Archimedean  $F$ , the ring  $R_F$  still has least positive element 1, but the coefficient  $a > \mathbb{N}$  in the proof of Theorem 24.3 has no  $R_F$ -floor. Calling every discretely ordered fixed ring an “integer part” would erase the central distinction of the source.

### 27.4 A bijective forward-strong map is not enough as a proof

Even when a particular definition in the literature packages strongness in terms of normal-form expansions, inverse summability must be checked under that definition. The proof obtains constant-term preservation only *after* establishing surjectivity and equality  $\mathcal{H}(\mathcal{I}) = \mathcal{I}$ . It then invokes the pairing criterion to prove inverse strongness. Reversing that logical order would be circular.

## 27.5 The coefficient action can be wild

The involutions  $\rho_Y$  constructed in Section 25 are not assumed continuous for the usual complex topology. Their fixed fields are ordinary Archimedean real closed fields only in their *intrinsic* order; their images inside  $\mathbb{C}$  need not lie on the usual real axis. It would be incorrect to replace  $F = \mathbb{C}^o$  by  $\mathbb{R}$  merely because the ambient coefficient field is named  $\mathbb{C}$ .

## 28 Position relative to existing work

### 28.1 Classical inputs

Conway normal forms and the identification of omnific integers supply the ambient objects [1, 2]. Hahn support multiplication, maximality, and algebraic closure are classical; the exact support and maximality inputs are available in [14]. The Artin–Schreier theorem accounts for real closed fixed coefficient fields and the order-two restriction [8]. The classification of algebraically closed set fields by characteristic and transcendence degree is standard [15].

Kuhlmann and Serra describe valued and strongly additive automorphisms of Hahn fields and their structural decompositions [3]. Kaplan, Krapp, and Serra treat the surreal automorphism setting [5]; the version checked for the source is arXiv v3, dated 23 April 2026. These provide context for the distinction between coefficient, value, leading-term, and summation conditions. The source does not attribute its finite-norm formula to those papers, nor infer its novelty merely because a search did not locate it.

### 28.2 The collection at the source’s pin

The source inspected the collection at 343dc2c: this report, the omnific-preserving and set-sized-quotient reports, the large-cardinal report, and, through those guides, the omnific Diophantine report. It records that the collection contains substantial prior work on surcomplex field automorphisms, real forms, Gaussian phase twists, omnific-preserving support stabilizers, and coefficient reconstruction. This report already states that some involutions are not conjugate to standard conjugation (Theorem 10.2) and relates standard conjugacy to the set-cut interpolation property (Proposition B.1). Those facts are not presented as new.

The refinement requires the involution to preserve the valuation, all Hahn sums in both directions, and  $\mathbf{Oz}[i]$ . Within this category the source supplies an explicit simultaneous conjugator, classifies all involutions by coefficient real closed fields, identifies their fixed rings, and exhibits  $2^c$  pairwise nonisomorphic fixed fields with genuine integer parts. The fixed-ring reconstruction in Section 22 is supporting material, not a separate novelty claim.

The collection also has work on embeddings across universes and critical-point defects, in the report foundations-and-computation/large-cardinal-embeddings-and-normal-forms, which the source cites under the title *Critical-Point Defects in Surreal Arithmetic*. The proper-class extension here uses only local closure under a finite family of class maps; it assumes no measurable cardinal, elementary embedding of the universe, or nontrivial critical point. These are different research directions.

### 28.3 Novelty and verification boundary

The proposed contribution is the *package*: the norm-section theorem with a complete support, surjectivity and inverse audit, its class-local version, and the resulting Gaussian fixed-pair classification and exact multiplicity. The argument uses familiar algebraic devices, including norms, finite-group conjugacy, and residue invariants.

It may have antecedents or formulations in a broader valuation-theoretic language not located by the focused search. Publication priority remains unverified.

The source comparison used selected repository guides and report descriptions rather than every proof in every report. Some material was described there as pending review or assembly. The source does not claim a complete nonduplication audit of that evolving collection.

No Lean code proving these theorems is delivered, and the repository was not built. The exact finite tests are regression checks for the worked example, not a machine proof of the main results. Independent scrutiny should focus first on diagonal summability, the use of set-sized maximality, the coefficient-pairing criterion, and the class-local closure.

## 28.4 Relations checked against the current collection

[merge] The source's pin 343dc2c predates the parts of surreal/omnific-preserving-automorphisms on automatic strongness and summation detection. Checked at the merge:

- *Printed once, credited.* Theorem 18.2 and Corollary 18.3 are `opa:as:thm:detect` and `opa:as:thm:ctiso` (the source's Baire route is a fourth route, for the weaker all- $\gamma$  form); Proposition 22.1 is `opa:lem:divisible`, `odg:def:thm:multiplicator` and `odg:thm:fractions`, as the source says; Corollary 23.3 is contained in Theorem 10.1; the phase twists of Section 26 are Theorem 4.2 and `odg:def:lem:twist`.
- *Stale in the source.* Its strongness hypotheses are redundant (Proposition 18.5, Remark 22.3, from `opa:as:thm:ctiso` and `opa:as:thm:gaussian`); its first question is answered (Question 29.1); its second question is `opa:as:q:gaussian` (Section 29).
- *New relative to the collection.* The norm-section conjugator, conjugacy within  $\mathcal{G}$  (Proposition 23.1), the integer-part criterion for  $R_F$  (Theorem 24.3), the criterion  $\mathbf{No}_F \cong \mathbf{No} \iff F \cong \mathbb{R}$  (Theorem 24.4) and the exact count  $2^c$  (Theorem 16.4). A search of the collection at the merge found no other report proving them; priority beyond the collection is not certified.
- *This report.* The plain conjugacy criterion of Section 10 and Proposition B.1 are refined, not replaced: within  $\mathcal{G}$ , conjugacy is coefficient conjugacy, and set-cut interpolation of  $\text{Fix}(j)$  holds exactly when  $F \cong \mathbb{R}$  (Corollary 24.5). The real-form paragraph of Section 14 is answered in the valued,  $\mathbf{Oz}[i]$ -preserving case only.
- *The omnific-preserving report.* Its statement that no nonconjugacy is claimed for automorphisms preserving  $\mathbf{Oz}[i]$  but not conjugation concerns its copies of  $\mathbf{No}$  (`opa:as:cor:complexcopies`). For involutions, Theorem 25.2 gives  $2^c$  members of  $\mathcal{G}$  that are pairwise nonconjugate under every field automorphism, in particular under every  $\mathbf{Oz}[i]$ -preserving one; its copies are not addressed.
- *Negative information, no answer.* `odg:def:q:realform` of the omnific Diophantine report asks which natural structure on  $\mathbf{Oz}[i]$  determines the real omnific subring without naming conjugation or  $\mathbf{Oz}$ . Requiring an involution to preserve the valuation, Hahn summation and  $\mathbf{Oz}[i]$  leaves  $2^c$  pairwise nonisomorphic fixed rings  $R_F$ ,  $2^c$  of them integer parts; so the real omnific subring is not recovered, even up to isomorphism, as the fixed ring of an involution singled out by those structures alone. `univ:q:compatible` of foundations-and-computation/surreal-fields-across-universes (which transported real forms can preserve a prescribed valuation structure) is related and not answered; the one consequence drawn is in Section 23.
- *Complementary.* The report `surcomplex/first-kappa-coefficients` receives, in the same batch, a proper-class family of pure-field involutions that agree with  $c$  on  $\mathbb{C}$  and are not conjugate to  $c$  (`fkf:sb:`

`thm:involutions`); by Proposition 23.1 none of them lies in  $\mathcal{G}$ . The two results are complementary: that family shows how far the pure field is from rigid, this one what the valuation and  $\mathbf{Oz}[i]$  force.

## 29 Further research questions of source 02

The following are proposed questions motivated by the source. Unless an external source is explicitly named, they are not represented as longstanding published open problems.

*Question 29.1* (Remove strongness for finite Gaussian symmetries). Must every finite-order valued automorphism of  $\mathbf{No}[i]$  preserving  $\mathbf{Oz}[i]$  be strong? More modestly, does the invariant product section remain summable under weaker hypotheses than strongness of the original action?

The norm formula still makes sense on individual monomials without strongness. The precise missing step is Lemma 20.2. A proof must control the union of supports, not just the valuations of the terms. Conversely, a counterexample must preserve the Gaussian ring; a plain non-strong field automorphism is not sufficient. *Status* (batch 32; [merge]). *Answered*, yes, for every valued automorphism preserving  $\mathbf{Oz}[i]$ , of finite order or not: `opa:as:thm:gaussian` (Remark 22.3). The second sentence is answered in the same sense for the general theorem: a valued action stabilizing  $k$  and  $\mathcal{I}$  is strong (Proposition 18.5), so the product section is summable under the other hypotheses alone. Actions that are not assumed valued are not addressed, and without valuedness  $v(m_\gamma) = \gamma$  is no longer available.

**Source 02's second question: recover the outer valuation from the Gaussian ring.** “Does an arbitrary unital automorphism of  $\mathbf{Oz}[i]$ , extended to its fraction field, preserve the standard Hahn valuation ring? Does the answer change for involutions or finite group actions?” This is `opa:as:q:gaussian` of `surreal/omnific-preserving-automorphisms` (asked there by three earlier sources), with the additional clause on involutions and finite groups; it is recorded here rather than as a new question. The source's comment: Proposition 22.1 recovers  $\mathbb{C}$  and  $\mathcal{I}$ , but not by itself  $\mathcal{O}$ . A useful first target is an intrinsic characterization of the positive-support part in terms of the recovered ring and coefficient field. Any proposed characterization should be tested against the coefficientwise involutions of Section 25. *Status*: open. By Remark 22.3, a positive answer would make every automorphism of  $\mathbf{Oz}[i]$  strong, and every involution of  $\mathbf{Oz}[i]$  subject to the classification of Section 23.

*Question 29.2* (Nonfull and bounded-support Hahn fields). For which strongly summation-closed subfields, Rayner fields, or cardinality-bounded support fields does the norm conjugator remain surjective and preserve the chosen field?

The finite product does not enlarge the number of original group factors, but the inverse need not inherit the same support bound from maximality. An exact inverse-closure criterion would extend the theorem beyond full Hahn fields and clarify its relation to the structural setting of [3]. *Related* ([merge]): `opa:as:prop:kappa` proves inverse strongness for constant-term-covariant isomorphisms of support-bounded fields with a regular uncountable bound; surjectivity of the norm conjugator there is not addressed.

*Question 29.3* (Partial divisibility and ramification). For an ordered group which is not  $n$ -divisible, classify the obstruction to a finite action of order  $n$  admitting an invariant multiplicative section with the original value group.

The map  $t \mapsto -t$  on  $k((t))$  shows that torsion characters can survive. A plausible program separates the obstruction on  $\Gamma/n\Gamma$  from the principal-unit part and determines when passing to the divisible hull removes it without destroying the support ring. [merge] Source 03 asks the same, with the same example (Section 39).

*Question 29.4* (Infinite and profinite descent). For an infinite group action, which compatible finite-level norm sections can be assembled into a single strong conjugating map?

This requires a genuine summability or inverse-limit theorem. Finite-level conjugacy alone does not provide compatible choices, and no formal infinite product over a group should be used without an explicit support condition.

*Question 29.5* (Additional surreal structure). Which involutions in  $\mathcal{G}$  preserve a named real exponential structure, a specified surreal derivation, a simplicity relation, or a distinguished monomial section? How far do these requirements reduce the  $2^c$  coefficient types?

The present counterexamples exploit wild coefficient real forms. Naming the standard real coefficient field pointwise, or imposing the usual coefficient conjugation, already reduces all admissible involutions to the standard conjugacy type. Exponential and simplicity conditions are stronger and must be investigated in their own languages, not silently added to this field language. *Related:* the simplicity rigidity of Section 11 and `univ:q:compatible`; neither answers the question. [merge] Source 03 asks the same for derivations, exponentials and Hahn-analytic functions (Section 39).

*Question 29.6* (Classify plain fixed fields for non-Archimedean coefficients). For non-Archimedean real closed  $F, F'$  of transcendence degree  $c$ , when are  $F((t^{\mathbb{N}^0}))$  and  $F'((t^{\mathbb{N}^0}))$  isomorphic as plain fields?

Their support-ring pairs are isomorphic exactly when  $F \cong F'$ , but a plain field isomorphism need not preserve the outer valuation. The natural valuation can combine coefficient and exponent scales. The Archimedean residue argument therefore does not settle this question.

*Question 29.7* (Definability of the standard real form). Which additional set-sized predicates or parameters select the standard conjugacy type inside  $(\mathbf{No}[i], \mathbf{Oz}[i])$ ? Can the set-cut obstruction be expressed in a natural expanded language without naming the whole standard real axis?

The multiplicity theorem rules out uniqueness from the existing algebraic, valuation, and strong-summation conditions alone. A successful definition must distinguish  $\mathbb{R}$  from its proper Archimedean real closed subfields, not merely detect discreteness or the existence of floors. *Related:* `odg:def:q:realform` (Section 28.4).

*Question 29.8* (Effective inversion of the norm conjugator). Can  $\mathcal{H}^{-1}$  be described by a support-controlled transfinite algorithm with explicit complexity bounds on natural finitely presented inputs?

Maximality proves existence of a preimage but is not an algorithm. The rank-two example suggests coefficientwise inversion when the action is triangular in a specified scale decomposition. An effective theorem should specify a representation of supports and of the original automorphisms. *Status* (*batch 34*; [merge]). *Partly addressed.* Source 03 gives an explicit formula,  $\mathcal{H}^{-1} = (\sum_{r \geq 0} (-\mathcal{T})^r) D_{\mathcal{H}}^{-1}$  with  $\mathcal{T} = D_{\mathcal{H}}^{-1} \mathcal{H} - \text{id}$  (33.2), resting on the imported formal inversion of [6]; it gives no algorithm on a specified representation and no complexity bound, and its own Question 39.5 on certified support bounds is open.

*Question 29.9* (Formal verification). Can the full set-sized norm theorem be formalized using a Hahn-series library, followed by a separate universe-indexed or class-local transfer?

The best first targets are the finite-product diagonal summability lemma, the invariant section identities, the constant-term pairing criterion, and the immediate-extension surjectivity lemma. These are clean interfaces. Cardinality and set-cut results can be formalized after the core conjugacy theorem rather than being mixed into its implementation. [merge] Source 03 asks the same and proposes its own order of interfaces, with formal inversion in place of maximality (Section 39).

## 30 Conclusion, claim audit, and non-claims of source 02

### 30.1 Conclusion

Finite symmetries of a full Hahn field preserving its coefficient field and negative-support part admit a concrete normalization: their action on monomials can be absorbed into a single strong change of monomial section. The

invariant product  $\prod_g g(t^{\gamma/|G|})$  is the key construction; fullness and a separate inverse-summability argument turn it into a genuine automorphism.

For Gaussian omnific integers, this gives a complete classification in the strong valued category ([merge]: strongness is automatic there, so this is the valued category). The invariant is not an additional surreal scale, but the real closed coefficient field fixed by the coefficient involution. Archimedeanity of that field is exactly the condition for the fixed ring to be an integer part. Being the full real field is exactly the condition for the fixed class field to be a copy of **No**.

These distinctions produce the central negative conclusion: even all the stated normal-form and integer-part requirements leave  $2^c$  mutually nonisomorphic real forms. The unresolved next step is not to repeat the finite norm calculation, but to understand which hypotheses on arbitrary Gaussian-ring automorphisms force the summation and valuation structure needed for it. [merge] Summation is now known to follow from the valuation (opa:as:thm:gaussian); the valuation is the open part (opa:as:q:gaussian).

### 30.2 Proof dependency and claim audit

| Claim                                               | Main dependencies                                                                                                                                  | Status in the source                                            |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Hahn support arithmetic<br>Pairing criterion        | Well-ordered support sum and finite convolution<br>Row-finite Baire lemma; monomial tests;<br>descending-support witness                           | Classical input, cited<br>Complete supporting proof             |
| Set Hahn maximality                                 | Compatible coefficients in a nest; improvement in<br>an immediate extension                                                                        | Classical fact, reproved                                        |
| Class-local closure<br>Finite norm descent          | Set replacement; countable iteration; strongness<br>Divisible exponents; diagonal product; maximality;<br>support decomposition; pairing criterion | Complete written proof<br>Proposed main contribution            |
| Coefficient faithfulness<br>Gaussian reconstruction | Intertwining and surjectivity<br>Set-bounded denominator shift; integer divisibility;<br>multiplier units                                          | Consequence of descent<br>Prior support material, re-<br>proved |
| Involution classification                           | Norm descent; reconstruction; set-field Artin–<br>Schreier                                                                                         | Proposed main application                                       |
| Integer-part criterion                              | Leading-coefficient order; Archimedean coeffi-<br>cients                                                                                           | Complete written proof                                          |
| Standard-form criterion<br>$2^c$ multiplicity       | Explicit omitted set cuts; Conway interpolation<br>Transcendence-basis subfields; rational cuts; natu-<br>ral residues                             | Complete written proof<br>Complete written proof                |
| Finite example tests                                | Exact arithmetic in $\mathbb{Q}(i)[u]/(u^N)$                                                                                                       | Executed regression checks<br>only                              |
| Automatic strongness<br>([merge])                   | Three support regions; Corollary 18.3; opa:as:<br>thm:ctiso                                                                                        | Observation of the merge                                        |

**High-priority review points.** The finite-product family is indexed by  $S^G$  before its diagonal is taken; this is why no unjustified union of arbitrary supports enters the construction. The least exponent of each  $m_\gamma$  is exactly  $\gamma$ , so the leading coefficient of the least input term cannot cancel with later terms. Surjectivity is proved locally in full set-sized Hahn fields, not by manipulating a proper-class chain of balls.

The inclusion  $\mathcal{H}(\mathcal{I}) \subseteq \mathcal{I}$  is insufficient by itself to preserve the support ring in both directions. Equality uses surjectivity and the three-region direct sum. Constant-term preservation then follows, and only then does the pairing criterion give inverse strongness.

For the large-family theorem, the coefficient fields are Archimedean, so the outer Hahn valuation really is the natural valuation. No such identification is used for non-Archimedean coefficients. The exact upper bound is stated only in the strong valued Gaussian-preserving category, that is, for  $\mathcal{G}$ .

### 30.3 A formalization-oriented interface

A set-sized formalization can be organized around six interfaces, none of which requires a global surreal implementation at the outset.

**Summation interface.** A Hahn family carries proofs of well-ordered union support and point-finiteness. Required operations are subfamilies, coefficient rescaling, finite products, and regrouping. The diagonal lemma should consume this interface directly, rather than a topological convergence API.

**Section interface.** A multiplicative map  $m : \Gamma \rightarrow K^\times$  has value  $\gamma$  at  $\gamma$ , and maps every well-ordered monomial family to a summable family. Its strong  $k$ -linear extension is a field embedding. This lemma is reusable independently of group actions.

**Maximality interface.** A full Hahn field has no proper immediate extension. A self-embedding preserving values and coefficient residues therefore has a surjective underlying function. It is preferable to prove this once rather than repeat transfinite inversion arguments in every application.

**Support interface.** The strictly negative, zero, and strictly positive support parts form a direct sum. A bijection fixing constants and preserving values which maps the negative part into itself maps it onto itself and preserves the zero coefficient.

**Inverse-summation interface.** A  $k$ -linear field automorphism preserving the zero coefficient preserves Hahn summability in both directions. The pairing criterion supplies this interface; its only analytic-looking ingredient is the elementary Baire argument on a countable product of discrete fields.

**Finite-action interface.** A finite group acts by strong valued maps, stabilizing constants and negative support. Ordered finite actions on the value group are trivial; the product section is invariant; the preceding interfaces give the conjugacy theorem. The class-field transfer should be a separate theorem using invariant set-sized exponent subgroups, rather than a hidden universe coercion in this interface.

This is an implementation plan, not a claim that any of these interfaces has been compiled in Lean for the delivery. [merge] The collection's generic Hahn-series library is `Surreal/HahnSeries/`; no `saut:fs:` label has an implementation mapping.

### 30.4 What source 02 does not claim

The source states its limitations where they arise, and each is kept in place. The list collects them, with the place where each stands in Sections 16–30 (or in the delivered README and audit). Items 1–22 are the source's; items 23–27 were added in the merge.

1. Not refereed: complete written proofs are supplied relative to classical inputs; they were reviewed during preparation but not checked by an independent referee or a proof assistant (source title page and audit).

2. No Lean formalization is delivered; the interface of the previous subsection is a plan (Section 28, Question 29.9).
3. Publication priority and correctness are not independently certified; the literature and repository searches were focused, not exhaustive; the norm argument has familiar algebraic antecedents and may have formulations in a broader valuation-theoretic language; the word “theorem” implies no certification (Section 28).
4. No named unrestricted published conjecture or automorphism question is claimed settled (Section 16.4).
5. It is not proved that an arbitrary automorphism of  $\mathbf{Oz}[i]$  preserves the valuation or all Hahn sums (Section 16.4, Remark 22.2). The summation half is now settled for valued automorphisms by the collection (item 23); the valuation half is open.
6. No classification of all real forms of the plain class field  $\mathbf{No}[i]$ ; the  $2^c$  upper bound concerns  $\mathcal{G}$  only, not all involutions of the plain field (Section 16.4, Remark 25.3).
7. Nothing is claimed about exponential compatibility, a derivation, simplicity or a distinguished monomial section (Section 16.4, Question 29.5).
8. No classification of infinite or profinite group actions, no conjugacy of the entire automorphism group to a coefficient group, and no vanishing of cohomology for infinite groups (Section 21).
9. The outer Hahn valuation is identified with the natural valuation of the ordered field only for Archimedean coefficients; Proposition 24.6 is not stated in general (Section 24).
10. Artin–Schreier and Artin’s fixed-field theorem are applied only in the set field  $\mathbb{C}$ ; no class Galois theory is assumed (Section 23).
11. The reconstruction of Proposition 22.1 is prior supporting material, reproved, not a novelty claim; the omnific Diophantine report’s material was identified through guides and not independently audited by the source (Sections 22 and 28).
12. No priority is attached to the pairing criterion (Section 18); the order-two restriction is classical (Section 23).
13. The finite verifier checks exact identities in  $\mathbb{Q}(i)[u]/(u^{12})$  and a finite window of the lexicographic support example only; it does not verify infinite Hahn sums, the Baire lemma, maximality, class theory or cardinalities, and its success is not a machine verification of the main theorems (Section 26.3).
14. The repository comparison was selective: selected guides and report descriptions at the pin, read through a connector (a local clone failed), with some responses truncated; a failed keyword search does not establish absence; the repository was neither modified nor built (Section 28, audit).
15. Divisibility of the exponent group is sufficient, not shown necessary for each individual action (Section 27).
16. The fullness example  $k(t)$  is not claimed to arise from a finite norm action or to preserve the negative-support ring (Section 27).
17.  $\mathcal{G}$  is second-order shorthand; the count asserts a set-indexed uniform family, not that class automorphisms form sets or a quotient group; no class-sized transcendence basis, class Zorn lemma or large cardinal is used (Sections 17 and 28).
18. The norm conjugator is not claimed unique; the formula selects one (Section 26).
19. The research questions are proposed, not longstanding published problems (Section 29).

20. Kaplan–Krapp–Serra (arXiv v3) was checked as context only; no theorem is presented as solving its exponential-automorphism questions. The overview [16] is orientation, not a proof reference (Section 28, audit).
21. The delivered build helper overwrites the generated PDF and verification files of its own package; PDF metadata and compression may differ between TeX installations; no third-party papers or font files are included (delivered README).
22. The fixed-ring reconstruction and the phase twists are not claimed as new examples; nonstandard plain real forms are prior (Sections 26 and 28).
23. (*Merge.*) The redundancy of strongness (Proposition 18.5, Remark 22.3) and the answer to Question 29.1 rest on `opa:as:thm:ctiso` and `opa:as:thm:gaussian`; they are not results of the source.
24. (*Merge.*) No `saut:fs:` label has a Lean implementation mapping.
25. (*Merge.*) The consequences drawn for `odg:def:q:realform`, `univ:q:compatible`, `opa:as:q:otherrings` and the report `surcomplex/first-kappa-coefficients` are negative or partial information; none of those questions is answered.
26. (*Merge.*) The source’s citations of Kuhlmann–Serra (Definition 4.0.5), Poonen (Lemma 1, Theorem 1, Corollary 4) and Conrad (Theorem 3.1, Section 4) were not rechecked for the merge.
27. (*Merge.*) Of the verifier’s 3,751 assertions, 3,047 are order comparisons of exponent pairs; the identity checks number 704 (Section 26.3).

## ADDITION FROM A THIRD SOURCE

**Gaussian-omnific real forms and geometric symmetrization**

Sections 31–40: manuscript 03 of batch 34, pinned to repository commit 110efe9 (Appendix C.4)

**31 Gaussian-omnific real forms: conventions and statement map**

Sections 31–40 print the third source of this report, *Gaussian-Omnific Real Forms and Geometric Symmetrization* (Appendix C.4). The source is an *independent re-derivation* of the main results of source 02 (Sections 16–30): its pin 110efe9 contains the placement of source 02 (commit 7d04483) but not the text of source 02, which was written into this report afterwards (commit b8bbfee), and the source read only source 01’s article and README. Its results that source 02 already proves are printed once, in source 02’s sections, and credited below (Section 31.2); what is new or proved by a different route is printed here, with the source’s proofs. Text written for the merge is marked [merge]; the statement map, the conventions table and the relations of Section 40.2 are the merge’s throughout.

Source 03 asks whether the Gaussian omnific ring  $\mathbf{Oz}[i]$ , together with the Hahn valuation, selects a unique real form of  $\mathbb{K}$ . Its answer is the one of Theorems 16.2–16.4: no, but there is a precise classification. The nonlinear transformations of infinite series are removed by an arithmetic-preserving conjugation, and the coefficient real form is the residual invariant; even the integer-part property leaves  $2^c$  inequivalent possibilities. The word “geometric” in its title refers to the multiplicative symmetrization of the monomial section. There is no analytic limit and no assertion about a surcomplex exponential; the half-exponent is taken in the divisible ordered group  $\mathbf{No}$ .

**31.1 Conventions for the added sections**

The conventions of Sections 1.4 and 16.1 stay in force, with one exception: in Sections 32 and 33 the coefficient field  $k$  is an *arbitrary* field, of any characteristic, whereas in Sections 16–30 it has characteristic zero. The source’s symbols were renamed as follows so that each symbol keeps the meaning it already has in this report; no normalization changed.

| Source 03                                              | Here                                                               | Reason                                                                                                                |
|--------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| $\mathbf{No}(i), K; \mathbf{Og}, J; J^j$               | $\mathbb{K}; \mathbf{Oz}[i]; \mathbf{Oz}[i]^j$                     | this report’s names; $J$ is an index set in Section 17                                                                |
| $P, P_k$ (negative support)                            | $\mathcal{I}, \mathcal{I}_k$                                       | $P_\theta$ and $\mathcal{P}$ are the phase twists and the Puiseux field                                               |
| $o, R_o = o \oplus P_k$                                | $\mathfrak{o}, A_\mathfrak{o}$                                     | as in Section 17; $R_F$ is the descended ring                                                                         |
| $\langle x, y \rangle_0$                               | $\text{ct}(xy)$                                                    | written out, as in Theorem 18.2                                                                                       |
| $F : K_k \rightarrow K_l, \alpha, l$ (Proposition 3.3) | $\phi, \rho, k'$                                                   | $F$ is a coefficient field; $\alpha$ denotes automorphisms of $\mathbb{K}$ ; $\ell$ is the exponent coefficient (4.5) |
| $\alpha = j _{\mathbb{C}}, \alpha_g = g _k$            | $\rho, \rho_g$                                                     | the coefficient automorphism, as in Section 16.1                                                                      |
| $\widehat{\alpha}, \widehat{\alpha}_g$                 | $M_\rho, M_{\rho_g}$                                               | coefficientwise lifts, as there                                                                                       |
| $\beta \in \text{Aut}(\mathbb{C})$ (a conjugator)      | $\kappa$                                                           | as in Proposition 23.1                                                                                                |
| $H_G, H_j$                                             | $\mathcal{H}, \mathcal{H}_j$                                       | one symbol for the norm conjugator (the same map)                                                                     |
| $h_G(\gamma); m =  G $                                 | $m_\gamma; n$                                                      | the norm section and group order of Section 20                                                                        |
| $b, b_j$ (leading characters), $M_b$                   | $\chi_{\mathcal{H}}, \chi_{\mathcal{H}_j}, D_{\chi_{\mathcal{H}}}$ | diagonal maps are $D_\chi$ as in Theorem 6.1; $b$ is an exponent coordinate below                                     |
| $T, T_j$ (inversion remainder)                         | $\mathcal{T}, \mathcal{T}_j$                                       | $T$ is the transcendental of Theorem 10.2 and $T_s$ the flows                                                         |
| $S_n, \Omega$ (exponent sets)                          | $\Omega_r, \Omega$                                                 | $S_a$ are the dilations (4.4)                                                                                         |
| $N$ (kernel group); $u$ (cocycle)                      | $\mathcal{N}; \xi$                                                 | as in Section 21                                                                                                      |

| Source 03                                     | Here                                     | Reason                                                               |
|-----------------------------------------------|------------------------------------------|----------------------------------------------------------------------|
| $E, F_E, P_E, I_E$                            | $F, \mathbf{No}_F, \mathcal{I}_F, R_F$   | $E$ is reserved for an exponential; the others are source 02's names |
| $E_1, E_2; \phi: E_1 \rightarrow E_2$         | $F, F'; \psi$                            | as in Section 25                                                     |
| $A \subseteq B_1, E_A, \phi_A, \alpha_A, j_A$ | $Y, F_Y, \psi_Y, \rho_Y, j_Y$            | as there; $A$ is used for support rings                              |
| $\beta$ (a positive surreal bound)            | $\delta$                                 | as in Proposition 22.1                                               |
| $\mathcal{D}(F_E)$ (realized rational cuts)   | $\text{Cut}_{\mathbb{Q}}(\mathbf{No}_F)$ | $D$ is the displacement and $\mathcal{D}$ a derivation               |
| $q, s \in \mathbb{Q}$ (cut bounds)            | $q, q'$                                  | $s$ is a flow parameter                                              |
| $e_0, e_1; x = t^{e_0}, s = t^{e_1}$          | $(1, 0), (0, 1); z, u$                   | the two monomials of $K_{\text{lex}}$ (Section 26)                   |
| $D$ (a derivation)                            | $\mathcal{D}$                            | as in Theorem 5.2                                                    |
| $U_{\lambda}; \lambda, \mu$ (flow parameters) | $T_s; s, r$                              | the flows of Theorem 5.2; $\mathcal{U}$ is the leading-term kernel   |

The letters  $c$  (conjugation),  $\mathfrak{c} = 2^{\aleph_0}$ ,  $\mathcal{G}$ ,  $\tau$  (a value action),  $\chi_j$  (a leading character),  $B, B_0, B_1$  (a transcendence basis) and  $\mathbb{K}^j = \text{Fix}(j)$  keep their meanings. The following readings are tempting and false.

- “The orbit-product theorem is characteristic-free” does not make the Gaussian application characteristic-free: recovering  $\mathcal{I}$  as  $\bigcap_n n\mathbf{Oz}[i]$  uses  $\mathbb{Z} \subseteq k$  and  $\bigcap_n n\mathbb{Z} = 0$ .
- $\mathcal{T}$  is an operator on series; it is neither the transcendental  $T$  nor a flow  $T_s$ . The flows  $T_s$  of Section 38 are 1-automorphisms;  $\mathcal{H}$  in general is not.
- $\chi_j(\gamma) = \text{lc}(j(t^\gamma))$  is the leading character of the involution;  $\chi_{\mathcal{H}_j}(\gamma) = \text{lc}(m_\gamma)$  is that of its normalizer. They are related by a half exponent (Proposition 34.1), not equal.
- “Archimedean” does not mean “isomorphic to  $\mathbb{R}$ ” (Remark 35.5).
- Equal finite quotients and equal solvability of equations over  $\mathbb{Z}$  do not mean elementary equivalence (Section 37).
- $\text{Cut}_{\mathbb{Q}}(\mathbf{No}_F)$  is an external subset of  $\mathbb{R}$ ; no first-order definition of it is asserted.
- “ $H^1(C_2, \mathcal{N})$  is a singleton” is shorthand for the cocycle identity (34.4); no class of all class cocycles is formed.

### 31.2 Main statements and where they are printed

[merge] The source's two overview theorems are Theorems 16.2, 16.3 and 16.4: every nonidentity involution  $j \in \mathcal{G}$  is conjugate to  $M_\rho$ ,  $\rho = j|_{\mathbb{C}}$ , by the strongly  $\mathbb{C}$ -linear, value-fixing,  $\mathbf{Oz}[i]$ -preserving automorphism  $\mathcal{H}_j(\sum_\gamma a_\gamma t^\gamma) = \sum_\gamma a_\gamma t^{\gamma/2} j(t^{\gamma/2})$  (the case  $n = 2$  of (16.1), printed as (23.1)); conjugacy classes correspond to real closed fields  $F$  of transcendence degree  $\mathfrak{c}$ ; the fixed pair is  $(\mathbf{No}_F, R_F)$ ;  $R_F$  is an integer part exactly for Archimedean  $F$ ; there are exactly  $2^{\mathfrak{c}}$  classes, already among integer-part real forms, with pairwise nonisomorphic fixed fields and fixed rings. The table maps every numbered statement of source 03.

| Source 03        | Printed at          | Relation                                             |
|------------------|---------------------|------------------------------------------------------|
| Thm 1.1, Thm 6.1 | Thm 16.2, (23.1)    | same formula; re-derived                             |
| Thm 1.2          | Thms 16.2–16.4      | same statements                                      |
| Lem 3.1          | Prop 22.1; Rem 32.1 | 03's multiplier step works for every $\Gamma \neq 0$ |

| Source 03          | Printed at                                       | Relation                                                         |
|--------------------|--------------------------------------------------|------------------------------------------------------------------|
| Lem 3.2            | Thm 18.2; route in Section 32.2                  | opa:as:thm:detect; 03's proof is the binary-witness route        |
| Prop 3.3           | Prop 32.2                                        | opa:as:thm:ctiso; semilinear, between two fields                 |
| Thm 3.4, Rem 3.5   | Prop 18.5, Rem 22.3; Rem 32.3                    | opa:as:thm:gaussian, credited by 03                              |
| Thm 4.1, Cor 4.2   | Thm 32.4, Cor 32.5                               | new here (imported inversion)                                    |
| Thm 5.1            | Thm 33.1                                         | Thm 16.1, other hypotheses and route (Rem 33.2)                  |
| Cor 5.2            | Cor 33.3                                         | Cor 21.2 without $I$ -stability                                  |
| Rem 5.3            | Rem 33.4                                         |                                                                  |
| Cor 5.4            | Cor 33.5                                         | Cor 20.4, field part under 03's hypotheses                       |
| Eqs (6.1)–(6.4)    | Prop 34.1, (34.2), (34.3)                        | new here                                                         |
| Cor 6.2            | paragraph after Prop 23.1                        | same statement                                                   |
| Eq (6.5), $H^1$    | end of Section 21; (34.4)                        | the case $\Theta = C_2$                                          |
| Lem 7.1            | Prop 23.2, Prop 23.1(ii) $\Leftrightarrow$ (iii) | same proofs                                                      |
| Rem 7.2            | Rem 35.1                                         |                                                                  |
| Thm 7.3            | Props 23.1, 23.2                                 | same statement                                                   |
| Cor 7.4            | Cor 35.2                                         | the “only if” is added to Section 23                             |
| Thm 8.1            | (16.2); Prop 35.3                                | real closedness by a second route; the Gaussian splitting is new |
| Def 8.2, Lem 8.3   | Def 24.1, Lem 24.2                               | same                                                             |
| Thm 8.4            | Thm 24.3; Cor 35.4                               | same floor formula; adds a third condition and the transport     |
| Rem 8.5            | Rem 35.5                                         | cf. Section 27                                                   |
| Prop 8.6, Cor 8.7  | Prop 22.1, Cor 22.4                              | same (with $\mathfrak{o} = \mathbb{Z}$ )                         |
| Lem 9.1, 9.2       | proof of Lem 25.1; that lemma                    | identical fields $F_Y$                                           |
| Thm 9.3            | Thms 25.2, 16.4, Section 25                      | same count and bound                                             |
| Prop 9.4           | Thm 24.6; Rem 36.1                               | rational-cut form kept                                           |
| Cor 9.5            | Thm 25.2, Thm 24.4                               | same                                                             |
| Prop 9.6           | second consequence after Prop 23.1               | same proof                                                       |
| Thm 10.1, Rem 10.2 | Cor 36.2, Rem 36.3                               | weaker than Thm 24.4, Cor 24.5                                   |
| Props 11.1, 11.2   | Props 37.1, 37.2                                 | new here; the case $F = \mathbb{R}$ is prior                     |
| Prop 12.1          | Prop 38.1                                        | new example; cf. (26.2)                                          |
| Questions 1–12     | Section 39                                       | six merged, one answered, five new                               |

### 31.3 What is contributed, what is imported, and the scope

The source proposes as its contribution the orbit-product normalization with its arithmetic inverse, followed by the real-form classification, the exact integer-part criterion, the sharp count, and the intrinsic recognition of the ordinary surreal form. Classical field classification and finite-quotient consequences of a constant-term retraction are not new principles. It uses two kinds of prior work: the collection's recovery of coefficient fields from omnific arithmetic and its automatic strongness when the Gaussian integer part and the valuation ring are both preserved (opa:as:thm:detect, opa:as:thm:ctiso, opa:as:thm:gaussian), which it reproves with attribution so as not to assume the collection's claims; and the formal inversion of strongly contracting operators of Bagayoko, Krapp, Kuhlmann, Panazzolo and Serra [6], isolated in Theorem 32.4. Kuhlmann–Serra [3] and Kaplan–Krapp–Serra [5] supply the theory of Hahn automorphisms and the surreal class setting. Artin–Schreier theory is used only for the ordinary field  $\mathbb{C}$ , with a direct finite-polynomial argument for the class-sized fixed field (Proposition 35.3). [merge] In the current collection the classification package was

already proved by source 02; what is new is listed in Section 40.2.

The valuation hypothesis is essential to the scope of the proof. The source does *not* show that every automorphism of  $\mathbf{Oz}[i]$ , extended to its fraction field, preserves  $\mathcal{O}$ , nor does it show this for arbitrary involutions of  $\mathbf{Oz}[i]$ ; the collection leaves that compatibility open (opa:as:q:gaussian). The classification is complete within the valuation-compatible Gaussian sector, not within the full abstract Gaussian automorphism collection. The normalization is canonical relative to the selected Hahn monomials; it is not a parameter-free definition of those monomials from the ring. It does not identify all real closed fields of transcendence degree  $\mathfrak{c}$  by a finite list of parameters; it proves that no additional infinite-support twisting invariant survives once the coefficient real field is specified.

## 32 Summation detection and the formal-inversion input

### 32.1 Conventions recorded by source 03

The source's Hahn conventions are those of Section 17, for an arbitrary field  $k$  and ordered abelian group  $\Gamma$ : set-sized well-ordered supports (only these when  $\Gamma$  is a proper class), set-indexed summable families, strongly additive and strongly  $k$ -linear maps, and the additive decomposition  $K = \mathcal{I}_k \oplus k \oplus \mathfrak{m}_k$ , on which  $\text{ct}$  is linear but not multiplicative ( $\text{ct}(t^\gamma t^{-\gamma}) = 1$ );  $\text{ct}$  is multiplicative on  $A_k$  and on every  $A_{\mathfrak{o}}$ . The support facts for Hahn multiplication are Lemma 3.2; the source notes that they make products of summable families summable, including their subfamilies, and that they are stronger than convergence in a valuation topology and must not be replaced by topological language. For  $k = \mathbb{R}$ ,  $\mathfrak{o} = \mathbb{Z}$  and  $\Gamma = \mathbf{No}$  one obtains  $\mathbf{Oz}$ ; for  $k = \mathbb{C}$ ,  $\mathfrak{o} = \mathbb{Z}[i]$  one obtains  $\mathbf{Oz}[i]$ .

A valued automorphism  $\phi$  induces the ordered group automorphism  $\tau$  with  $v(\phi f) = \tau(v f)$ . The source defines it by  $\tau(\gamma) = v(\phi(t^\gamma))$ : independence of the representative holds because a ratio of two elements of equal value is a unit of  $\mathcal{O}_k$ , and the same construction for  $\phi^{-1}$  gives bijectivity. In the class case this avoids forming a quotient with proper-class equivalence classes. The Hahn valuation of  $F((t^\Gamma))$  agrees with the natural valuation of the ordered field when  $F$  is Archimedean, and need not otherwise; the name *Hahn valuation* is kept throughout. Set-sized Hahn fields are handled in ZFC; for  $\mathbf{No}$  and  $\mathbb{K}$  the source works in NBG with global choice, as in [5]. Every family summed is set indexed, every image of a set under the specified class maps is a set, and no class-indexed sum is used. A formula such as “conjugacy classes in  $\mathcal{G}$  are classified by a set” means that there is a set of coefficient data, each datum specifies a representative class map, and every specified automorphism is conjugate to a representative with the stated uniqueness property; the lower bounds are realized by actual set-indexed families.

### 32.2 Detection and covariance: source 03's routes

Source 03 reproves the part of the collection's argument it needs. Its reconstruction lemma is Proposition 22.1 (for  $K = \mathbb{C}((t^\Gamma))$  and  $\mathfrak{o} = \mathbb{Z}[i]$ ), with one difference of route.

*Remark 32.1* (The multiplier step for any exponent group). In  $K = k((t^\Gamma))$  with  $\Gamma \neq 0$  arbitrary,  $(\mathcal{I}_k : \mathcal{I}_k) = A_k$  and  $A_k^\times = k^\times$ . If  $x$  has a nonzero coefficient at some  $\delta > 0$ , then  $t^{-\delta} \in \mathcal{I}_k$  and  $\text{ct}(xt^{-\delta}) \neq 0$ , so  $xt^{-\delta} \notin \mathcal{I}_k$  and  $x$  is not a multiplier. No intermediate exponent is needed, unlike the proof of (22.3), which chooses  $0 < \delta < \epsilon$ . Together with  $\mathcal{I}_{\mathbb{C}} = \bigcap_n n(\mathbb{Z}[i] + \mathcal{I}_{\mathbb{C}})$ , every field automorphism of a full complex Hahn field with nonzero exponent group that stabilizes  $\mathbb{Z}[i] + \mathcal{I}_{\mathbb{C}}$  stabilizes  $\mathcal{I}_{\mathbb{C}}$  and  $\mathbb{C}$ .

**A second route to the pairing criterion.** The source proves Theorem 18.2, with the identity  $\text{ct}((\sum_i x_i)y) = \sum_i \text{ct}(x_i y)$  (a finite sum) and uniqueness of the sum, by the binary-witness route of opa:as:thm:detect, not

by the Baire argument of Lemma 18.1. If  $S, T$  are well ordered,  $S \cap (-T)$  is finite: an infinite subset would give an increasing sequence in  $S$  whose negatives decrease in  $T$ ; with  $S = \bigcup_i \text{supp } x_i$  and  $T = \text{supp } y$  this proves the forward direction and the displayed identity. Conversely, the tests  $y = t^{-\gamma}$  give point-finiteness. If the union of supports is not well ordered, take a strictly decreasing sequence in it; each support meets it finitely. Inductively choose a subsequence  $\gamma_0 > \gamma_1 > \dots$  and distinct indices  $i_r$  with  $[t^{\gamma_r}]x_{i_r} \neq 0$  and  $[t^{\gamma_q}]x_{i_r} = 0$  for  $q > r$ , discarding at each finite stage the finitely many places belonging to previously chosen supports. Choose  $b_r \in \{0, 1\}$  successively with  $\sum_{q \leq r} b_q [t^{\gamma_q}]x_{i_r} \neq 0$ ; at most one of the two choices is forbidden, since the new coefficient is nonzero. Then  $y = \sum_r b_r t^{-\gamma_r}$  is a Hahn series with  $\text{ct}(x_{i_r}, y) \neq 0$  for every  $r$ , a contradiction. Monomial tests prove uniqueness of the sum. The witness is countably supported, also in the class case.

**Proposition 32.2** (Constant-term covariance). *Let  $\phi : k((t^\Gamma)) \rightarrow k'((t^{\Gamma'}))$  be a field isomorphism between full Hahn fields (exponent groups sets, or **No** with set supports) and  $\rho : k \rightarrow k'$  a field isomorphism with  $\text{ct}(\phi x) = \rho(\text{ct } x)$  for every  $x$ . Then  $\phi|_k = \rho$ , and  $\phi$  and  $\phi^{-1}$  are strongly additive.*

*Proof.* For  $a \in k$  and every  $x$ ,  $\text{ct}((\phi(a) - \rho(a))\phi(x)) = 0$ . Surjectivity and nondegeneracy of  $(x, y) \mapsto \text{ct}(xy)$  give  $\phi(a) = \rho(a)$ . If  $(x_i)$  is summable and  $y = \phi(w)$ , then  $\text{ct}(\phi(x_i)y) = \rho(\text{ct}(x_i w))$ , nonzero for only finitely many  $i$ ; so the image family is summable by Theorem 18.2, and the pairing identity with nondegeneracy gives  $\phi(\sum_i x_i) = \sum_i \phi(x_i)$ . The same argument applies to  $\phi^{-1}$  with  $\rho^{-1}$ .  $\square$

This is `opa:as:thm:ctiso`. The source's Theorem 3.4 (a field automorphism of  $\mathbb{C}((t^\Gamma))$ ,  $\Gamma \neq 0$ , preserving  $\mathbb{Z}[i] + \mathcal{I}_{\mathbb{C}}$  and  $\mathcal{O}$  preserves  $\mathbb{C}$ , is  $\rho$ -semilinear with  $\text{ct} \circ \phi = \rho \circ \text{ct}$ , and is strong) is `opa:as:thm:gaussian`, printed here as Proposition 18.5 and Remark 22.3; its proof is the three-region argument given there.

*Remark 32.3* (Two circular arguments avoided). The proof does not infer strongness from continuity, and it does not assume that  $\text{ct}$  is multiplicative on  $K$ . Multiplication appears only through the pairing  $\text{ct}(xy)$ , while coefficient recovery uses the smaller nonpositive-support ring.

### 32.3 Formal inversion (imported)

The surjectivity of the normalizer does not follow from its leading terms alone. Source 03 imports the following theorem.

**Theorem 32.4** (Formal inversion; established input). *Let  $\mathcal{T}$  be a strongly  $k$ -linear map on a full Hahn series space such that*

$$\text{supp } \mathcal{T}(t^\gamma) \subseteq \{\delta : \delta > \gamma\} \quad \text{for every } \gamma.$$

*Then the geometric operator series*

$$(\text{id} + \mathcal{T})^{-1} = \sum_{r \geq 0} (-\mathcal{T})^r \tag{32.1}$$

*is well defined and strongly linear. In particular, for every summable family  $(f_i)$  the doubly indexed family  $(\mathcal{T}^r f_i)_{r,i}$  is summable.*

For set-sized exponent sets this is, according to the source, the one-operator specialization of [6, Theorem 3.11 and Proposition 1.45]: the support condition is their contraction condition in a totally ordered exponent set, no uniform lower bound on the positive shifts is required, and characteristic zero is not needed. The source imports that summability theorem rather than claiming that strictly increasing values alone make an arbitrary geometric operator series meaningful. The exact inverse is two-sided by the formal identity  $(1 + X) \sum_{r \geq 0} (-X)^r = 1$  in the operator-evaluation algebra.

For  $\Gamma = \mathbf{No}$  the source localizes. Given the union  $\Omega_0$  of the supports of a set-indexed input family, put

$$\Omega_{r+1} = \Omega_r \cup \bigcup_{\gamma \in \Omega_r} \text{supp } \mathcal{T}(t^\gamma), \quad \Omega = \bigcup_{r < \omega} \Omega_r.$$

The set  $\Omega$  need not be well ordered. The space of all well-ordered-support series with exponents in  $\Omega$  is  $\mathcal{T}$ -stable by strongness; the set-sized theorem applies there and gives (32.1) for all the specified inputs at once, independently of enlarging  $\Omega$ , since the same operator series is evaluated. This defines the inverse on the class field and proves its strongness for every set-indexed family. (Compare the invariant closure of Lemma 19.3.)

**Corollary 32.5** (An arithmetic inverse). *Suppose a strongly  $k$ -linear map  $H$  on  $K = k((t^\Gamma))$  has the form*

$$H = D_\chi(\text{id} + \mathcal{T}), \quad D_\chi(t^\gamma) = \chi(\gamma)t^\gamma,$$

where  $\chi : \Gamma \rightarrow k^\times$  is a character,  $\mathcal{T}$  is as in Theorem 32.4, and  $\mathcal{T}(\mathcal{I}_k) \subseteq \mathcal{I}_k$ . Then  $H$  is a linear bijection with strong inverse  $H^{-1} = (\sum_r (-\mathcal{T})^r) D_\chi^{-1}$ , and  $H^{\pm 1}(\mathcal{I}_k) \subseteq \mathcal{I}_k$ . If  $H$  is also a unital ring homomorphism and fixes  $k$ , it is a field automorphism preserving every  $A_0$  setwise.

*Proof.*  $D_\chi$  has the coefficientwise inverse  $D_{\chi^{-1}}$  and preserves supports. Every term  $\mathcal{T}^r f$  has negative support when  $f \in \mathcal{I}_k$ , and so does the summable sum; thus (32.1) preserves  $\mathcal{I}_k$ , and so does  $H$  by its factorization. A bijective unital field homomorphism is a field automorphism; fixing  $k$  preserves any chosen constant subring  $\mathfrak{o}$ .  $\square$

**Remark 32.6** (Relation to the leading-term kernel; [merge]). If  $H$  is a value-fixing field automorphism fixing  $k$  and  $\chi(\gamma) = \text{lc } H(t^\gamma)$ , then  $D_\chi^{-1}H = \text{id} + \mathcal{T}$  fixes every leading term; for  $\mathbb{K}$  this is the case  $\rho = \text{id}$ ,  $\tau = \text{id}$  of the four-layer factorization (6.1), with the 1-automorphism written to the right of  $D_\chi$  instead of to the left. Section 5 notes that an arbitrary homomorphism  $g \mapsto u_g \in 1 + \mathfrak{m}$  does not by itself give a summable or surjective map. Theorem 32.4 supplies the surjectivity half, with a strong inverse, once the forward map  $x \mapsto \sum_g x_g t^g u_g$  is known to be strongly  $\mathbb{C}$ -linear; forward summability and multiplicativity still have to be checked.

### 33 Orbit-product descent in any characteristic

The involution formula is part of a simultaneous normalization theorem, useful independently of surreal numbers. In this section  $k$  is an arbitrary field.

**Theorem 33.1** (Arithmetic orbit-product descent). *Let  $k$  be a field, let  $\Gamma$  be an ordered abelian group divisible by the positive integer  $n$  (a set, or **No** with set supports), and let  $G$  be a finite group of order  $n$  acting on  $K = k((t^\Gamma))$ . Assume each  $g \in G$  is valued, strongly additive and carries  $k$  onto itself; write  $\rho_g = g|_k$ . Put  $m_\gamma = \prod_{g \in G} g(t^{\gamma/n})$  and define  $\mathcal{H}$  by (16.1). Then  $\mathcal{H}$  is a strongly  $k$ -linear, value-fixing field automorphism with strong inverse, and*

$$g\mathcal{H} = \mathcal{H}M_{\rho_g} \quad (g \in G). \quad (33.1)$$

*If each  $g$  also preserves  $\mathcal{I}_k$ , then  $\mathcal{H}$  and  $\mathcal{H}^{-1}$  preserve  $\mathcal{I}_k$  and every  $A_0$  setwise.*

*Proof. Step 1: the value actions are trivial.* An increasing bijection of a totally ordered set has no nontrivial finite order: if  $\tau(x) > x$  its iterates strictly increase, and if  $\tau(x) < x$  they strictly decrease. So every  $g$  fixes values. Division by  $n$  is unique because an ordered group is torsion-free.

*Step 2: multiplicativity on monomials.* From  $(\gamma + \delta)/n = \gamma/n + \delta/n$ ,  $m_{\gamma+\delta} = m_\gamma m_\delta$  and  $m_0 = 1$ . Each of the  $n$  factors of  $m_\gamma$  has value  $\gamma/n$ , so  $v(m_\gamma) = \gamma$ . Put  $\chi_{\mathcal{H}}(\gamma) = \text{lc}(m_\gamma)$ ; multiplicativity of leading coefficients makes it a character.

*Step 3: the sum is strong.* Let  $(f_i)$  be summable and index all nonzero terms by pairs  $(i, \gamma)$ . The family  $(t^{\gamma/n})_{i, \gamma}$  is summable: division by  $n$  preserves order, and repetition of an exponent is finite. For every  $g$  the image family  $(g(t^{\gamma/n}))_{i, \gamma}$  is summable, the product family of these  $n$  families is summable by the Hahn product support lemma, and its diagonal subfamily is  $(m_\gamma)_{i, \gamma}$ . Multiplying terms by coefficients changes no supports; regrouping proves that (16.1) exists and that  $\mathcal{H}$  is strongly  $k$ -linear.

*Step 4: multiplicativity.* For  $f = \sum a_\gamma t^\gamma$  and  $q = \sum d_\delta t^\delta$ , both  $\mathcal{H}(fq)$  and  $\mathcal{H}(f)\mathcal{H}(q)$  are the sum of the summable family  $(a_\gamma d_\delta m_{\gamma+\delta})_{\gamma,\delta}$ . Constants are fixed.

*Step 5: the inverse.* Set  $\mathcal{T} = D_{\chi\mathcal{H}}^{-1}\mathcal{H} - \text{id}$ . On each monomial the leading coefficient of  $D_{\chi\mathcal{H}}^{-1}m_\gamma$  is 1 and all other exponents exceed  $\gamma$ , so  $\mathcal{T}$  satisfies Theorem 32.4, and

$$\mathcal{H}^{-1} = \left( \sum_{r \geq 0} (-\mathcal{T})^r \right) D_{\chi\mathcal{H}}^{-1} \quad (33.2)$$

is a strong inverse;  $\mathcal{H}$  is onto. It fixes values, as its leading terms show.

*Step 6: arithmetic.* Assume  $g(\mathcal{I}_k) = \mathcal{I}_k$  for all  $g$ . If  $\gamma < 0$ , then  $t^{\gamma/n} \in \mathcal{I}_k$ , every factor of  $m_\gamma$  is in  $\mathcal{I}_k$ , and so is their product; strongness gives  $\mathcal{H}(\mathcal{I}_k) \subseteq \mathcal{I}_k$ , hence  $\mathcal{T}(\mathcal{I}_k) \subseteq \mathcal{I}_k$ . By Corollary 32.5 the inverse preserves  $\mathcal{I}_k$ , and fixing  $k$  gives preservation of every  $A_\mathfrak{o}$ .

*Step 7: intertwining.* Left multiplication by  $h \in G$  permutes the factors of  $m_\gamma$ , so  $h(m_\gamma) = m_\gamma$ ; strongness and semilinearity give  $h\mathcal{H}(\sum a_\gamma t^\gamma) = \sum \rho_h(a_\gamma)m_\gamma$ , which is (33.1). All support families are sets; the class case uses only the localization after Theorem 32.4.  $\square$

*Remark 33.2* (Two routes to one normalizer; [merge]). The map  $\mathcal{H}$  is the norm conjugator of Theorem 16.1; Steps 1–4 and 7 are Lemmas 20.1, 20.2, 20.3 and the intertwining of Section 20, found independently. The hypotheses differ. Source 02 assumes characteristic zero and  $\mathcal{I}$ -stability, and strongness then follows (Proposition 18.5); source 03 assumes strongness and neither of the other two. The routes differ in Step 5: source 02 proves surjectivity by spherical completeness (Lemma 19.2) and local exponent closure (Lemma 19.3), and inverse strongness by the pairing criterion after proving  $\mathcal{H}(\mathcal{I}) = \mathcal{I}$  and  $\text{ct} \circ \mathcal{H} = \text{ct}$ ; source 03 obtains both at once from formal inversion, with the explicit inverse (33.2). Without  $\mathcal{I}$ -stability, source 02's surjectivity argument still applies (it uses only strongness, valuedness and  $k$ -stability), but its inverse-strongness argument does not, because  $\text{ct} \circ \mathcal{H} = \text{ct}$  is then unavailable; there Theorem 32.4 is the only route printed here. We found no use of characteristic zero in the proof of Section 20 either; the hypothesis there is needed for the Gaussian reconstruction, not for the descent.

**Corollary 33.3** (Faithfulness on coefficients). *Under Theorem 33.1, an element of  $G$  fixing  $k$  acts identically on  $K$ ; restriction to  $k$  is injective on  $G$ . A finite-order, valued, strongly  $k$ -linear automorphism of  $K$  is the identity whenever  $\Gamma$  is divisible by its order.*

*Proof.* Equation (33.1) and invertibility give  $g = \mathcal{H}M_{\rho_g}\mathcal{H}^{-1}$ . If  $\rho_g = \text{id}$ , then  $g = \text{id}$ .  $\square$

[merge] For  $K = \mathbb{K}$  and  $k = \mathbb{C}$  the last sentence is contained in Theorem 10.1, which needs no valuation or strongness. For general  $k$  and  $\Gamma$  it extends Corollary 21.2 to strong actions not assumed to stabilize  $\mathcal{I}_k$ , and to every characteristic.

*Remark 33.4* (An orbit product, not an additive average). The construction divides the exponent by  $|G|$  before taking a norm-like product. It does not average field automorphisms additively, which is generally not multiplicative, and it does not select roots of unrelated elements; coherence comes from the single monomial section and the homomorphism  $\gamma \mapsto \gamma/n$ . The orbit-product theorem is characteristic-free: the inverse uses the geometric series, not a logarithm with rational denominators. The Gaussian application takes place in characteristic zero.

**Corollary 33.5** (A finite-action descent principle). *Under the hypotheses of Theorem 33.1,  $K^G = \mathcal{H}(k^{\rho(G)}((t^\Gamma)))$ . If  $G$  preserves  $\mathcal{I}_k$  and  $\mathfrak{o}$  is a constant subring stable under the coefficient action, then  $(A_\mathfrak{o})^G = \mathcal{H}(\mathfrak{o}^{\rho(G)} + \mathcal{I}_{k^{\rho(G)}})$ .*

*Proof.* A series fixed by every coefficientwise map has all coefficients in  $k^{\rho(G)}$ . Transport by  $\mathcal{H}$  and use its arithmetic preservation.  $\square$

Under source 02's hypotheses this is Corollary 20.4.

### 34 Gaussian involutions: leading character and explicit inverse

Let  $j \in \mathcal{G}$  be an involution and  $\rho = j|_{\mathbb{C}}$ . Source 03 proves Theorem 16.2 by the route:  $j$  is strong, semilinear and  $\mathcal{I}$ -preserving (Proposition 18.5, from `opa:as:thm:gaussian`); its value action is an increasing involution, hence the identity; Theorem 33.1 with  $G = \{\text{id}, j\}$  gives the normalizer; if  $\rho = \text{id}$  the intertwining gives  $j = \text{id}$ ; and a nonidentity involution of  $\mathbb{C}$  has real closed fixed field and sends  $i$  to  $-i$  [8, Theorem 3.1]. For  $j = \text{id}$  the formula gives  $\mathcal{H}_j = \text{id}$ . The same arguments apply to a full set-sized complex Hahn field with a nonzero divisible exponent group, for automorphisms of the ambient field rather than merely of its support ring.

**Proposition 34.1** (The leading character and a coherent Hilbert 90 solution). *Let  $j \in \mathcal{G}$  be an involution,  $\rho = j|_{\mathbb{C}}$ , and  $\chi_j(\gamma) = \text{lc}(j(t^\gamma))$ . Then  $\chi_j : \mathbf{No} \rightarrow \mathbb{C}^\times$  is a character with  $\chi_j(\gamma)\rho(\chi_j(\gamma)) = 1$ . The normalizer  $\mathcal{H}_j$  has leading character*

$$\chi_{\mathcal{H}_j}(\gamma) = \chi_j(\gamma/2), \quad \frac{\chi_{\mathcal{H}_j}(\gamma)}{\rho(\chi_{\mathcal{H}_j}(\gamma))} = \chi_j(\gamma). \quad (34.1)$$

*Proof.* Since  $j$  fixes values, leading coefficients multiply, so  $\chi_j$  is a character. Write  $j(t^\gamma) = \chi_j(\gamma)t^\gamma + w$  with  $v(w) > \gamma$ . Applying the  $\rho$ -semilinear, value-fixing  $j$  gives  $t^\gamma = \rho(\chi_j(\gamma))j(t^\gamma) + j(w)$ , and comparing leading coefficients gives  $\rho(\chi_j(\gamma))\chi_j(\gamma) = 1$ . The leading coefficient of  $m_\gamma = t^{\gamma/2}j(t^{\gamma/2})$  is  $\chi_j(\gamma/2)$ , and  $\rho(\chi_j(\gamma/2)) = \chi_j(\gamma/2)^{-1}$  gives the quotient  $\chi_j(\gamma/2)^2 = \chi_j(\gamma)$ .  $\square$

Thus the multiplicative Hilbert 90 equation for characters has a coherent half-exponent solution, without choosing square roots separately. [merge] In the four-layer factorization (6.1),  $\chi_j$  is the character layer of  $j$  and  $\chi_{\mathcal{H}_j}$  that of  $\mathcal{H}_j$ ; (34.1) says that the character layer of  $j$  is the  $\rho$ -coboundary of that of its normalizer. The phase-character calculation of Section 26 is the case in which  $j = D_\chi c$  has no 1-automorphism layer.

The full inverse is not only the diagonal correction. Define

$$\mathcal{T}_j = D_{\chi_{\mathcal{H}_j}}^{-1} \mathcal{H}_j - \text{id}. \quad (34.2)$$

Then, by (33.2),

$$\mathcal{H}_j^{-1} = \left( \sum_{r \geq 0} (-\mathcal{T}_j)^r \right) D_{\chi_{\mathcal{H}_j}}^{-1}, \quad (34.3)$$

and all iterates  $\mathcal{T}_j^r$  preserve  $\mathcal{I}$ . This is where the proof keeps the Gaussian omnific arithmetic in the inverse, rather than only producing an embedding into the desired real form. [merge] Source 02 obtains  $\mathcal{H}^{-1}$  from maximality, which proves existence of a preimage without a formula; (33.2) and (34.3) are explicit formulas, but they come with no complexity bound, so Question 29.8 is only partly addressed (status note there).

**No further twisting at fixed coefficient action.** If  $j_1, j_2 \in \mathcal{G}$  are nonidentity involutions with the same restriction to  $\mathbb{C}$ , then  $\mathcal{H}_{j_2} \mathcal{H}_{j_1}^{-1}$  conjugates  $j_1$  to  $j_2$ , fixes  $\mathbb{C}$  and values pointwise and preserves  $\mathbf{Oz}[i]$ : both are conjugate to the same coefficientwise map. This is the paragraph after Proposition 23.1.

**A nonabelian descent reading.** Let  $\mathcal{N}$  be as in Section 21 with  $k = \mathbb{C}$ , and let a coefficient involution  $M_\rho$  act on it by conjugation. An element  $\xi \in \mathcal{N}$  with  $\xi M_\rho \xi M_\rho^{-1} = \text{id}$  gives an involution  $j = \xi M_\rho \in \mathcal{G}$ , and the normalization supplies  $\mathcal{H} \in \mathcal{N}$  with

$$\xi = \mathcal{H}(M_\rho \mathcal{H} M_\rho^{-1})^{-1}. \quad (34.4)$$

In customary notation  $H^1(C_2, \mathcal{N})$  is a singleton; in the class case this means the concrete identity (34.4). It is a Hilbert 90-type reading of the normalizer, not a claim to originate nonabelian descent or its notation; the general finite case is at the end of Section 21.

### 35 Classification, fixed pairs and integer parts: source 03's route

Source 03's Lemma 7.1 and Theorem 7.3 are Propositions 23.2 and 23.1, with the same proofs: nonidentity involutions of  $\mathbb{C}$  up to conjugacy correspond to real closed fields of transcendence degree  $c$ ; within  $\mathcal{G}$ , conjugacy is conjugacy of the coefficient involutions, and every such field occurs, so there is no additional character or infinite-support invariant. Its standard inputs are the existence and uniqueness of algebraic closures and transcendence bases [17, 18].

*Remark 35.1* (Cardinality is not substituted for transcendence degree). For an algebraically closed field of characteristic zero, uncountable cardinality equals transcendence degree, since an algebraic extension of  $\mathbb{Q}(X)$  has cardinality  $\max(\aleph_0, |X|)$ ; in particular  $\text{trdeg}_{\mathbb{Q}} \mathbb{C} = c$ . Equal cardinalities alone would not classify arbitrary fields; algebraic closedness is indispensable.

**Corollary 35.2** (Classification with coefficients fixed). *Under conjugation by elements of  $\mathcal{G}$  fixing  $\mathbb{C}$  pointwise, two nonidentity involutions in  $\mathcal{G}$  are conjugate exactly when their restrictions to  $\mathbb{C}$  are identical. No change of value-group coordinates is needed for sufficiency.*

*Proof.* Necessity follows by restricting the conjugacy identity to  $\mathbb{C}$ ; sufficiency is the paragraph after Proposition 23.1.  $\square$

**Why a real-axis hypothesis would hide the answer.** If  $j|_{\mathbb{C}}$  is ordinary conjugation,  $j = \mathcal{H}_j c \mathcal{H}_j^{-1}$  and its real form is  $\mathcal{H}_j(\text{No})$ . These include real forms moved by nonlinear phase shears, but all lie in the standard coefficient class. A nonstandard involution of the ordinary field  $\mathbb{C}$ , by contrast, survives as an invariant that no conjugation in  $\mathcal{G}$  removes. The main distinction is therefore not between visible and infinitesimal motion of monomials, but between different ordered coefficient fields.

**Proposition 35.3** (Real closedness directly, and the Gaussian splitting). *Let  $j \in \mathcal{G}$  be a nonidentity involution. Then  $\text{Fix}(j)$  is real closed,  $\mathbb{K} = \text{Fix}(j)(i)$ , and*

$$\mathbf{Oz}[i] = \mathbf{Oz}[i]^j \oplus i \mathbf{Oz}[i]^j, \quad \mathbf{Oz}[i]^j = \mathbf{Oz}[i] \cap \text{Fix}(j). \quad (35.1)$$

*Proof.* The fixed-pair identification (16.2) comes from transporting the coefficientwise fixed pair by  $\mathcal{H}_j$ . Real closedness can be proved directly, without applying a set-field theorem to a proper-class extension. Since  $j(i) = -i$ , every  $z \in \mathbb{K}$  decomposes uniquely as  $z = a + ib$  with  $a = (z + jz)/2$  and  $b = (z - jz)/(2i)$  in  $\text{Fix}(j)$ ; so  $i \notin \text{Fix}(j)$  and  $\mathbb{K} = \text{Fix}(j)(i)$ . The field  $\mathbb{K}$  is algebraically closed. For  $a, b \in \text{Fix}(j)$  take a square root  $c + id$  of  $a + ib$  with  $c, d \in \text{Fix}(j)$ ; then  $a^2 + b^2 = (c^2 + d^2)^2$ , so sums of squares in  $\text{Fix}(j)$  are squares. A nonzero  $a \in \text{Fix}(j)$  has a square root  $z \in \mathbb{K}$ , and  $jz = \pm z$ ; so  $a$  or  $-a$  is a square in  $\text{Fix}(j)$ , and not both, since  $i \notin \text{Fix}(j)$ . Thus the nonzero squares define an order. An odd-degree polynomial over  $\text{Fix}(j)$  splits in  $\mathbb{K}$ , and  $j$  pairs its nonfixed roots with their multiplicities, so it has a fixed root. These are the usual criteria for real closedness; each assertion involves only finitely many coefficients or roots.

If  $z \in \mathbf{Oz}[i]$ , its Gaussian integer constant term splits by the same formulas into ordinary integers, since  $\rho(i) = -i$ ; its purely infinite part splits into two members of  $\mathcal{I}$ , because  $\mathcal{I}$  is a  $\mathbb{C}$ -vector space and is  $j$ -stable. Hence  $a, b \in \mathbf{Oz}[i]^j$ , which proves (35.1).  $\square$

[merge] The first two assertions also follow from Theorem 10.1, for every involution of  $\mathbb{K}$ ; the splitting (35.1) of the ring is not stated elsewhere in the report.

**Corollary 35.4** (Exact integer-part criterion, three forms). *For a set-sized real closed field  $F$  the following are equivalent: (i)  $R_F$  is an integer part of  $\mathbf{No}_F$ ; (ii)  $\mathbb{Z}$  is an integer part of  $F$ ; (iii)  $F$  is Archimedean. The floor is then (24.1). For a nonidentity involution  $j \in \mathcal{G}$  with Archimedean  $F = \mathbb{C}^{j|c}$ , the floor in  $\mathbf{Oz}[i]^j$  is obtained by transporting (24.1) through the isomorphism of pairs induced by  $\mathcal{H}_j$ .*

*Proof.* Assume (i) and apply the floor property to a coefficient  $a \in F$ . If a proposed floor  $p + n$  has  $p \neq 0$ , then  $a - (p + n)$  has a nonzero negative-exponent part and cannot lie in  $[0, 1)$ ; hence  $p = 0$ , which proves (ii). Condition (ii) bounds every coefficient between two ordinary integers and implies (iii). Conversely, in an Archimedean ordered field each element lies in some  $[-N, N]$ , and among finitely many candidate integers there is a unique  $n$  with  $n \leq a < n + 1$ ; so (iii) implies (ii). For (ii) $\Rightarrow$ (i): if  $a > n$ , the positive constants  $a - n$  and  $n + 1 - a$  dominate  $\varepsilon$ , so  $0 < a - n + \varepsilon < 1$ ; if  $a = n$ , the sign of  $\varepsilon$  decides whether to subtract one. Discreteness is Lemma 24.2, and uniqueness holds because two distinct elements of  $R_F$  differ by at least one in absolute value. An isomorphism of real closed fields preserves their unique orders, so it transports the floor inequalities.  $\square$

**Remark 35.5** (Two distinctions). A non-Archimedean  $F$  does not make  $R_F$  nondiscrete: an infinite coefficient lies above every ordinary integer and below every positive purely infinite element, leaving a gap in the available floors. Conversely, an Archimedean real closed field need not be  $\mathbb{R}$ : proper real closed subfields of  $\mathbb{R}$  can have transcendence degree  $\mathfrak{c}$  and cardinality  $\mathfrak{c}$ . Replacing “Archimedean” by “isomorphic to  $\mathbb{R}$ ” in Corollary 35.4 would be false and would erase the main nonuniqueness phenomenon.

Source 03’s Proposition 8.6 and Corollary 8.7 are Proposition 22.1 with  $\mathfrak{o} = \mathbb{Z}$ ,  $k = F$  and Corollary 22.4:  $\text{Frac}(R_F) = \mathbf{No}_F$ ,  $\mathcal{I}_F = \bigcap_n nR_F$ ,  $(\mathcal{I}_F : \mathcal{I}_F) = F + \mathcal{I}_F$  in  $\text{Frac}(R_F)$  with units  $F^\times$ , and  $R_F \cong R_{F'}$  exactly when  $F \cong F'$ . So the coefficient invariant is recoverable from the abstract ring  $\mathbf{Oz}[i]^j$ , without an a priori valuation or monomial section. The fraction argument uses the proper-class upper-bound property of  $\mathbf{No}$  and must not be applied silently to a fixed set-sized Hahn field, whose supports can be unbounded in its exponent group and whose negative-support ring can have a smaller fraction field (Section 17).

## 36 Counting, rational cuts and the standard form

Source 03’s counting argument (its Lemmas 9.1, 9.2 and Theorem 9.3) is that of Section 25, with the same fields  $F_Y$ , the same coefficientwise involutions  $j_Y = M_{\rho_Y}$  and the same bound  $|\mathbb{C}|^{|\mathbb{C}|} = 2^{\mathfrak{c}}$ , without the continuum hypothesis; ordered subfields of  $\mathbb{R}$  are rigid because they fix  $\mathbb{Q}$  and reals are separated by rationals, and a field isomorphism of real closed fields is an ordered one. The isomorphism  $\psi_Y : F_Y(i) \rightarrow \mathbb{C}$  is an abstract field isomorphism, not the inclusion into the ordinary complex plane, which is not onto when  $F_Y \neq \mathbb{R}$ ; equal transcendence degree supplies it, and no continuity in the complex topology is required or asserted.

For an Archimedean  $F \subseteq \mathbb{R}$ , source 03 separates the fixed fields by the external set of realized rational cuts,

$$\text{Cut}_{\mathbb{Q}}(\mathbf{No}_F) = \{r \in \mathbb{R} : \text{some } x \in \mathbf{No}_F \text{ has } q < x < q' \text{ for all } q, q' \in \mathbb{Q} \text{ with } q < r < q'\}. \quad (36.1)$$

**Remark 36.1** (Recovered rational cuts). For every Archimedean real closed  $F \subseteq \mathbb{R}$ ,  $\text{Cut}_{\mathbb{Q}}(\mathbf{No}_F) = F$ . An element realizing a cut is bounded by ordinary integers, so it has no negative-exponent part and is  $a + \varepsilon$  with  $a \in F$ ,  $\varepsilon \in \mathfrak{m}_F$ ; the infinitesimal is below every positive rational, so the cut is that of  $a$ , forcing  $r = a$ ; conversely  $x = r$  realizes the cut of  $r \in F$ . This is the argument of the Archimedean case of Theorem 24.4, and its valuation form is Proposition 24.6. Any field isomorphism of real closed fields preserves order and fixes  $\mathbb{Q}$ , hence preserves  $\text{Cut}_{\mathbb{Q}}$ ; so the fields  $\text{Fix}(j_Y)$  are pairwise nonisomorphic, the  $j_Y$  are pairwise nonconjugate under arbitrary field automorphisms of  $\mathbb{K}$ , and none except the class with  $F_Y \cong \mathbb{R}$  has fixed field isomorphic to  $\mathbf{No}$ , whose realized cuts are all of  $\mathbb{R}$  (Theorem 25.2). No first-order definability of this external invariant is asserted.

Every set of elements of  $\mathbf{No}_F$ , for any real closed  $F$ , has an upper bound  $t^{-\delta}$ : choose a positive surreal  $\delta$  with  $-\delta < v(x)$  for all nonzero  $x$  in the set. So every fixed field in the valued Gaussian sector has no set-sized cofinal subset, and the countably cofinal real form of Theorem 10.2 has no involution preserving both  $\mathbf{Oz}[i]$  and  $\mathcal{O}$  (second consequence after Proposition 23.1). The nonstandard Archimedean examples thus differ from  $\mathbf{No}$  by missing bounded rational cuts, not by a small cofinal sequence. None of this bounds the number of conjugacy classes of all involutions of the plain field: the exact upper bound belongs to the valued Gaussian sector (Remark 25.3).

**Corollary 36.2** (Standard arithmetic form criterion). *Let  $j \in \mathcal{G}$  be a nonidentity involution and  $F = \mathbb{C}^{j|c}$ . The following are equivalent:*

- (i)  $j$  is conjugate to  $c$  by an element of  $\mathcal{G}$ ;
- (ii)  $F \cong \mathbb{R}$  as a field;
- (iii)  $\mathbf{Oz}[i]^j \cong \mathbf{Oz}$  as a ring;
- (iv)  $(\text{Fix}(j), \mathbf{Oz}[i]^j) \cong (\mathbf{No}, \mathbf{Oz})$  as a field-subring pair;
- (v)  $\text{Fix}(j) \cong \mathbf{No}$  as a plain field.

*If  $\mathbf{Oz}[i]^j$  is an integer part of  $\text{Fix}(j)$ , they are also equivalent to (vi):  $\text{Fix}(j)$  realizes every rational cut arising from an ordinary real number.*

*Proof.* (i) $\Leftrightarrow$ (ii) is Proposition 23.1. Since the fixed ring is  $\cong R_F$  and  $\mathbf{Oz} = R_{\mathbb{R}}$ , (ii) $\Leftrightarrow$ (iii) is Corollary 22.4. (i) implies (iv) by restricting a conjugator, and (iv) implies (iii); alternatively (ii) gives (iv) by a coefficientwise isomorphism followed by  $\mathcal{H}_j$ . The source proves (v) and (vi) under the integer-part hypothesis: then  $F$  is Archimedean, embed it in  $\mathbb{R}$  by its rational-cut embedding; by Remark 36.1  $\text{Fix}(j)$  realizes exactly the cuts of that copy of  $F$ , all of  $\mathbb{R}$  exactly when the copy is  $\mathbb{R}$ ; a field isomorphism  $\text{Fix}(j) \cong \mathbf{No}$  forces equality of realized-cut sets, and the converse follows from (iv). [merge] Without that hypothesis, (ii) $\Leftrightarrow$ (v) is Theorem 24.4 (Corollary 24.5), so (v) needs no hypothesis. Clause (vi) does need it: if  $F$  is the real closure of  $\mathbb{R}(X)$  ordered with  $X$  above every real, then  $F$  is non-Archimedean of transcendence degree  $\mathfrak{c}$ ,  $\mathbf{No}_F \supseteq \mathbb{R}$  realizes every real rational cut, and  $\mathbf{No}_F \not\cong \mathbf{No}$ .  $\square$

**Remark 36.3** (The source's caution, and its status). The source notes that for non-Archimedean  $F$  the Hahn residue field and the natural ordered residue field need not coincide, and so does not use the plain-field criterion  $\text{Fix}(j) \cong \mathbf{No}$  without the integer-part hypothesis; the ring criterion (iii) holds for every  $j$  in the sector. [merge] The caution is correct as a statement about its own proof. The plain-field criterion was already proved for every set-sized real closed  $F$  by explicit omitted set cuts (Theorem 24.4); the cut  $\mathbb{N} < F_{\infty}$  of its proof avoids the residue-field comparison.

The proof also explains why agreement with a familiar collection of ordinary congruences is insufficient to recognize  $\mathbf{Oz}$ : those congruences do not see the recovered coefficient field.

## 37 What finite arithmetic cannot distinguish

The following are direct retraction arguments. For the ordinary omnific ring their counterparts are already in the collection; the point is that they hold uniformly for the mutually nonisomorphic descended rings of Theorem 25.2.

**Proposition 37.1** (Uniform finite quotients). *For every real closed field  $F$  and every positive ordinary integer  $n$ ,*

$$nR_F = n\mathbb{Z} \oplus I_F, \quad R_F/nR_F \cong \mathbb{Z}/n\mathbb{Z}.$$

Every unital ring homomorphism from  $R_F$  to a finite ring factors through  $\text{ct} : R_F \rightarrow \mathbb{Z}$ . Consequently all nonzero finite quotients are the ordinary  $\mathbb{Z}/n\mathbb{Z}$ , and the inverse system of ordinary congruence quotients is independent of  $F$ .

*Proof.* The additive group  $I_F$  is divisible by every positive ordinary integer, so the first identities follow by taking constant terms. If a finite target ring has additive exponent  $m$ , then for  $p \in I_F$  write  $p = mq$  with  $q \in I_F$ ; every homomorphism kills  $p$ , so it factors through  $R_F/I_F = \mathbb{Z}$ . A nonzero finite quotient is a nonzero quotient of  $\mathbb{Z}$  of finite cardinality.  $\square$

**Proposition 37.2** (Ordinary polynomial solvability is unchanged). *A finite system of polynomial equations with coefficients in  $\mathbb{Z}$  has a solution in  $R_F$  if and only if it has a solution in  $\mathbb{Z}$ .*

*Proof.*  $\mathbb{Z}$  embeds in  $R_F$ . Conversely, apply the ring retraction  $\text{ct} : R_F \rightarrow \mathbb{Z}$  to every coordinate of a solution; polynomial evaluation commutes with ring homomorphisms.  $\square$

Thus  $2^c$  pairwise nonisomorphic integer-part rings have identical finite quotient systems and identical truth values for existential polynomial equation systems over ordinary integer coefficients. This is not an assertion of elementary equivalence, and it does not cover disequations, order constraints, or coefficients from the nonordinary parts of the rings. Constant extraction can kill a nonzero purely infinite element, so a disequation cannot be transported by the same argument.

[merge] For  $F = \mathbb{R}$  these are `odg:thm:finitequotients` and `odg:thm:transfer` of the omnific Diophantine report (the latter also for positive existential formulas at ordinary integer parameters, and the same retraction proves that extension here), and for the definable rings  $I_A$  they are `dsn:thm:finite-quotients` of the definable-surreals report. The factorization clause is the finite-target case of `osq:thm:universal(ii)` of the set-sized-quotient report, applied with the set field  $F$  and the subring  $\mathbb{Z}$  (that report's  $D$ ): every unital homomorphism from  $R_F$  to any set-sized ring factors through  $\text{ct}$ . What is new here is the uniformity over the  $2^c$  nonisomorphic rings, which gives negative information on `odg:def:q:realform` beyond Section 28.4: congruences and integer equations do not distinguish the fixed rings.

## 38 A rank-two flow example and the finite checks of source 03

The classification should not be read as saying that all involutions act coefficientwise before conjugation. Work in  $K_{\text{lex}} = \mathbb{C}((t^{\mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}}))$  with  $z = t^{(1,0)}$  and  $u = t^{(0,1)}$  as in Section 26, and define on monomials

$$\mathcal{D}(z^a u^b) = a z^a u^{b+1} \quad (a, b \in \mathbb{Q}). \quad (38.1)$$

The coefficient field is  $\mathbb{C}$ , on which  $\mathcal{D}$  acts trivially; additivity of the first exponent coordinate gives the derivation rule, translation by  $(0, 1)$  gives strongness, and  $\mathcal{D}^r(z^a u^b) = a^r z^a u^{b+r}$ . The sums

$$T_s(z^a u^b) = z^a u^b \exp(sau) = \sum_{r \geq 0} \frac{(sa)^r}{r!} z^a u^{b+r} \quad (s \in \mathbb{C}) \quad (38.2)$$

define strong automorphisms with  $T_s T_{s'} = T_{s+s'}$ . Here  $\exp$  is a formal power series at the positive-value element  $sau$ , not a global surcomplex exponential. For a summable input support  $S$ , all output supports lie in  $S + \mathbb{N}(0, 1)$ , and the support lemma gives the required summability. [merge] These are the flows of Theorem 5.2 in the set-sized workspace  $\mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}$ , with the additive map  $(a, b) \mapsto a$  and shift  $\delta = (0, 1)$ ; that theorem also shows they are 1-automorphisms.

**Proposition 38.1** (A phase-twisted real form). *Let  $c$  be coefficientwise conjugation on  $K_{\text{lex}}$  and  $j = T_i c$ . Then  $j$  is an involution preserving the valuation ring and  $\mathbb{Z}[i] + \mathcal{I}$ , it does not preserve the real-coefficient Hahn field  $\mathbb{R}((t^{\mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}}))$  setwise, and its normalizer is  $\mathcal{H}_j = T_{i/2}$ .*

*Proof.* Since  $cT_i c = T_{-i}$ ,  $j^2 = \text{id}$ . For a negative exponent  $(a, b)$ : if  $a < 0$ , every  $(a, b + r)$  is negative; if  $a = 0$  and  $b < 0$ , the factor  $a^r$  makes the monomial fixed. Strongness gives preservation of  $\mathcal{I}$ , and the inverse has the same property; constants in  $\mathbb{Z}[i]$  are fixed by  $T_i$  and preserved by  $c$ . The image  $j(z) = z \exp(iu)$  has a nonzero imaginary coefficient at the next exponent. Finally, for  $\gamma = (a, b)$ ,

$$t^{\gamma/2} j(t^{\gamma/2}) = z^a u^b \exp(iau/2) = T_{i/2}(t^\gamma).$$

□

The fixed field is  $T_{i/2}(\mathbb{R}((t^{\mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}})))$ , and the fixed Gaussian part is the transported real support ring. The example illustrates the theorem; phase shears themselves are not claimed new, since related shears and support-cut criteria already occur in the collection (opa:thm:realaxis, Theorem 4.2). [merge] It belongs to the same family as the example (26.2): for  $\varphi \in 1 + u\mathbb{C}[[u]]$  with  $\bar{\varphi} = \varphi^{-1}$ , the map  $z^a u^b \mapsto z^a u^b \varphi(u)^a$  composed with  $c$  is an involution in the same class, with normalizer  $z^a u^b \mapsto z^a u^b \varphi(u)^{a/2}$ ; source 02 takes  $\varphi = W = (1 + iu)/(1 - iu)$  and source 03 takes  $\varphi = \exp(iu)$ .

### 38.1 What source 03's verification program checks

The program code/03-gaussian-real-forms-verify.py uses exact arithmetic in  $\mathbb{Q}(i)$  and finite truncations in the inner monomial  $u$ : its test algebra is  $\mathbb{Q}(i)[z^{\mathbb{Q}}][u]/(u^{N+1})$  with finite outer support, at order  $N = 8$  and seed 20260923. It tests the formal identities  $T_s T_r = T_{s+r}$ ,  $j^2 = \text{id}$ ,  $j\mathcal{H}_j = \mathcal{H}_j c$  and  $\mathcal{H}_j(fq) = \mathcal{H}_j(f)\mathcal{H}_j(q)$ , finite geometric inverses, and the sign of sampled lexicographic support shifts. These are finite regression checks of signs, coefficients and normalization formulas; they do not establish universal Hahn summability, class-sized assertions, existence of transcendence bases or cardinality claims, which rest on the proofs and the stated inputs. No Lean source is supplied.

[merge] The recorded run passes 457 assertions: 9 each for the orbit-product formula, the monomial involution and the monomial intertwining (the monomials  $z^a$ ,  $a \in \{-2, -\frac{3}{2}, \dots, 2\}$ ), 24 each for eleven identities on random series (264), and 166 in the family `negative_support_preservation`. Those 166 are comparisons  $(a, b + r) < (0, 0)$  with  $a < 0$ , or  $a = 0$ ,  $r = 0$ ,  $b < 0$ , which hold by the lexicographic order itself. The normalizer is implemented as the flow  $T_{i/2}$ ; the orbit-product formula  $t^{\gamma/2} j(t^{\gamma/2})$  is compared with it only on the nine monomials. In the truncated algebra the remainder  $\mathcal{T}$  is nilpotent, so the geometric inverse is a finite sum there (the check `nilpotent_remainder`); the program does not test the summability of an infinite operator series. Rerunning it for this report reproduced the record (README).

## 39 Questions of source 03

The source proposes twelve questions, in increasing distance from its proof; they are research questions, not assertions that each is a previously published open problem. [merge] Six coincide with questions already in the collection and are recorded as paragraphs that cite those questions' labels; one is printed with its answer, which was already in the report; five are new.

**Its Questions 1 and 2: automatic valuation preservation.** “Does every unital automorphism of  $\mathbf{Oz}[i]$ , extended to  $\text{Frac}(\mathbf{Oz}[i]) = \mathbb{K}$ , preserve the Hahn valuation ring?” and, even if not, “must an automorphism

of order two of  $\mathbf{Oz}[i]$  have a valuation-compatible extension?” These are `opa:as:q:gaussian` and the involution clause of source 02’s second question (Section 29). The source’s comments: an affirmative answer would turn the classification into one of all abstract Gaussian omnific involutions; the multiplier construction recovers  $\mathbb{C}$  and  $\mathcal{I}$ , and what is missing is recovery of the complementary positive-support part without an order. Finite order might rule out behaviour available to arbitrary maps, but the orbit-product formula cannot be used before strongness is proved, since its diagonal summability would be unavailable; a useful intermediate goal is to recover constant extraction for finite-order maps. *Status*: open. [merge] With the valuation, strongness is automatic (Remark 22.3), so for involutions the missing step is exactly the valuation.

*Question 39.1* (Nonvalued arithmetic real forms). Can a real form occur whose involution preserves  $\mathbf{Oz}[i]$  but not its Hahn valuation? If so, what invariant replaces the coefficient-field classification, and can there be more than  $2^c$  conjugacy classes?

The bound of Theorem 16.4 holds in the valued sector because every action there has a coefficientwise representative; it gives no bound once that hypothesis is removed. A counterexample would have to explain exactly how the intrinsic negative-support and coefficient substructures fail to determine the valuation. [merge] A negative answer to the first clause is the involution case of `opa:as:q:gaussian`. The proper-class family of `fk:sb:thm:involutions` (Appendix B) lies outside  $\mathcal{G}$ ; whether any of its members preserves  $\mathbf{Oz}[i]$  is not known.

**Its Question 4: removing exponent divisibility.** “For an ordered abelian group not divisible by the order of a finite action, what is the obstruction to orbit-product normalization?” This is Question 29.3, with the same example  $t \mapsto -t$  on  $\mathbb{C}((t^{\mathbb{Z}}))$  (Example 27.1). The source adds: the half-exponent cannot simply be written down; the example shows that divisibility is a real boundary, not an artifact of notation; a classification should separate character obstructions from genuinely nonlinear ones. *Status*: open.

**Its Question 5: general arithmetic coefficient rings.** “For a characteristic-zero field  $k$  and a unital subring  $\mathfrak{o}$ , which hypotheses on  $\mathfrak{o}$  make preservation of  $A_{\mathfrak{o}}$  recover  $k$  and constant extraction?” This is `opa:as:q:otherrings`. The source adds that the orbit-product theorem already handles the descent once strongness and coefficient preservation are available, that a systematic reconstruction theorem would extend the Gaussian case to other arithmetic coefficient rings and their finite Galois actions, and that the intersection  $\bigcap_n n\mathfrak{o}$  must be controlled: it is zero for  $\mathbb{Z}$  and  $\mathbb{Z}[i]$  but not for every coefficient ring. *Status*: open.

*Question 39.2* (Positive characteristic). Which arithmetic predicates force coefficient reconstruction and automatic strongness in positive characteristic, especially for finite actions whose orders are divisible by the characteristic?

The monomial orbit-product theorem already works in every characteristic (Theorem 33.1): it divides in the exponent group, not in the coefficient field. Automatic strongness from integer divisibility has no direct analogue, and wild finite-order automorphisms can appear when exponent divisibility fails. A precise theorem must specify both the summability structure and the arithmetic subring.

*Question 39.3* (Intrinsic theories of descended integer parts). For Archimedean real closed fields  $F, F'$  of transcendence degree  $c$ , when are  $R_F$  and  $R_{F'}$  elementarily equivalent as rings?

They have identical ordinary equation solvability and finite quotients (Section 37) but need not be isomorphic. Coefficient recovery uses an intersection over all positive integers, an external construction that is not automatically a first-order interpretation. Locating the first-order information about the omitted real cuts is a separate model-theoretic problem. [merge] Compare `fk:sb:q:omnific` of `surcomplex/first-kappa-coefficients`, which asks whether the rings of short omnific integers are elementarily equivalent.

**Question 39.4** (Which fixed fields are abstractly surreal?). For non-Archimedean  $F$  of transcendence degree  $\mathfrak{c}$ , characterize when  $F((t^{\mathbf{No}}))$  is isomorphic as an ordered field to  $\mathbf{No}$ .

The source comments that the integer-part property fails for  $R_F$ , but the plain field may have a natural valuation different from the Hahn valuation, so the Archimedean cut invariant no longer suffices; one should compare natural residue fields, value groups and realization of set-sized cuts, keeping the arithmetic pair separate from the plain field. *Status* ([merge]): *answered* before the source was merged, and negatively: by Theorem 24.4,  $F((t^{\mathbf{No}})) \cong \mathbf{No}$  (even as a plain field) only if  $F \cong \mathbb{R}$ , so never for non-Archimedean  $F$ ; the omitted set cut is  $\mathbb{N} < F_\infty$ . The finer question of when two such fields are isomorphic to each other is Question 29.6, open.

**Question 39.5** (Effective normalizers). For computable finite-rank or finitely described Hahn actions, can one compute specified coefficients of  $\mathcal{H}$  and  $\mathcal{H}^{-1}$  with certified bounds on the amount of input support required?

The inverse (33.2) is a genuine strong operator series, but that does not by itself give a termination bound in an arbitrary representation. A useful first domain is rational lexicographic rank with finite shift rules; the program of Section 38.1 covers only a homogeneous rank-two test case, not an algorithm for arbitrary surreal inputs. [merge] Related: Question 29.8, which (33.2) addresses in part.

**Its Question 10: compatibility with extra analytic structure.** “Which normalized real forms are compatible with a specified derivation, exponential, or selected class of Hahn-analytic functions?” This is Question 29.5. The source adds: the conjugators preserve ring arithmetic and valuation, not the Gonshor exponential or the Berarducci–Mantova derivation; additional structure may shrink the coefficient possibilities dramatically; a rigidity claim must identify the additional operations and check that the normalizer transports them rather than treating them as automatic field-theoretic data. *Status*: open.

**Question 39.6** (Hermitian and norm descent). How do integral Hermitian forms, unitary groups and norm equations over  $\mathbf{Oz}[i]$  change as its real structure ranges over the classes of Theorem 16.4?

The splitting (35.1),  $\mathbf{Oz}[i] = \mathbf{Oz}[i]^j[i]$ , and the transported positive norm are a starting point. Ordinary finite quotient information is unchanged, so distinctions must use coefficients, order or support geometry. This connects with the collection’s work on omnific groups and lattices (surreal/omnific-groups-and-lattices); no classification of those objects is claimed.

**Its Question 12: formal verification and foundation minimization.** “Can the orbit-product normalization be formalized for set-sized Hahn fields and then transported to a universe-controlled surreal implementation with explicit smallness assumptions?” This is Question 29.9. The source proposes the order: the scalar detector, strong semilinear maps, diagonal product summability, the operator-inverse input, arithmetic preservation of the inverse, and the conjugacy identity; the counting argument and the real closed coefficient classification can be formalized separately in ordinary set mathematics, and a final stage should express class-map assertions without a type of all class maps. *Status*: open; no `saut : gr : label` has a Lean mapping.

## 40 Position, audit and non-claims of source 03

### 40.1 The source’s comparison at its pin, and a stale claim

The source inspected the collection at 110efe9: the root README, the catalogue, the omnific-preserving report with its source audit, and this report’s `article.tex` and README. It records that the omnific-preserving report proves the valuation-compatible Gaussian automatic-strongness theorem, and that this report discusses general real forms, finite subgroups, phase twists and nonconjugate real forms without a Gaussian classification.

It then states that, at that snapshot, the inspected sources do not supply the orbit-product normalizer with its arithmetic inverse, the classification, or the integer-part and cardinality package, and that its comparison is targeted, not exhaustive over files, earlier commits, unintegrated manuscripts or unpublished work.

[merge] That statement is stale and is not printed as current. At the pin this report's `article.tex` was source 01 alone, as the source says; but the directory already held source 02's delivered audit (placed in 7d04483, an ancestor of the pin), which names the section  $m_\gamma = \prod_g g(t^{\gamma/n})$ , the classification of valued Gaussian-preserving involutions, the Archimedean integer-part criterion and the exact  $2^e$  count; the source did not read that file. Source 02's text was written into this report in b8bbfee, after the pin (Sections 16–30). In the current collection the normalizer, the classification and the integer-part and cardinality package are source 02's, and source 03 is an independent re-derivation of them. Its inverse formula is new (Section 40.2).

On novelty the source says: the proposed package rests on established Hahn support theory, formal operator calculus, Artin–Schreier theory and classical transcendence-basis arguments; a norm-like construction and a Hilbert 90-type vanishing statement naturally have classical analogues, and novelty of the exact Hahn and Gaussian formulations requires expert comparison and cannot be certified by a web search. It announces no solution of a named published open problem; it resolves a stated classification problem within the valued Gaussian sector, retains the broader compatibility question as open, and treats priority and independent correctness certification as separate questions. Of Kaplan–Krapp–Serra [5] (arXiv v3), no unresolved question is claimed solved by the classification.

## 40.2 Relations checked against the current collection

[merge]

- *Printed once, credited.* The classification package: Theorems 1.1, 1.2, 6.1, 7.3, 8.1, 8.4, 9.3, Corollaries 6.2, 8.7, 9.5, Lemmas 7.1, 8.3, 9.1, 9.2, Propositions 8.6, 9.4, 9.6 and the  $H^1$  statement are source 02's results (Section 31.2); the reconstruction, detection and automatic strongness of its Section 3 are `opa:lem:divisible`, `odg:def:thm:multiplicator`, `opa:as:thm:detect`, `opa:as:thm:ctiso` and `opa:as:thm:gaussian`, as the source says.
- *Genuinely different routes, kept.* The binary-witness proof of the pairing criterion (that of `opa:as:thm:detect`); the multiplier step for every  $\Gamma \neq 0$  (Remark 32.1); the direct real-closedness proof (Proposition 35.3); surjectivity and inverse strongness by formal inversion (Theorem 33.1, Step 5).
- *New relative to the collection.* (a) The explicit strong inverse (33.2), (34.3) and the leading-character identity (34.1); (b) descent for strong finite actions that need not stabilize  $\mathcal{I}_k$ , in every characteristic (Theorem 33.1, Corollaries 33.3, 33.5), resting on the imported formal inversion; (c) the uniform arithmetic of the descended rings (Propositions 37.1, 37.2; the case  $F = \mathbb{R}$  and the factorization are prior, as credited there); (d) the Gaussian splitting (35.1), the three-form integer-part criterion and Corollary 35.2; (e) the flow example of Proposition 38.1; (f) Questions 39.1, 39.2, 39.3, 39.5 and 39.6. A search of the collection found no other report proving (a)–(e); priority beyond the collection is not certified.
- *Weaker than already printed.* The source's plain-field clause needs the integer-part hypothesis; Theorem 24.4 does not (Corollary 36.2, Remark 36.3); its question on abstractly surreal fixed fields is thereby answered (Question 39.4).
- *Partial progress.* Question 29.8 (explicit inverse, no complexity bound); the obstacle noted in Section 5 (Remark 32.6).
- *Other reports.* `opa:as:q:gaussian` and `opa:as:q:otherrings` stay open (Section 39). The factorization of Proposition 37.1 is a case of `osq:thm:universal`; Propositions 37.1 and 37.2 extend `odg:thm:finitequotients`, `odg:thm:transfer` and `dsn:thm:finite-quotients` to all  $R_F$  and

add negative information on `odg: def: q: real form`; Question 39.3 is a companion of `fk: sb: q: omnific`. None of these reports is edited here.

### 40.3 Conclusion of source 03

Gaussian omnific arithmetic together with the Hahn valuation does not select one real form. It makes finite symmetries regular enough for a multiplicative normalization of the monomial section; all support-level twisting can then be removed, leaving the coefficient real field as the complete conjugacy invariant. The integer-part property detects Archimedeaness of that field, not Dedekind completeness, which is enough to produce  $2^c$  mutually inequivalent real forms with integer parts, including examples not isomorphic to **No** even as plain ordered fields, while their ordinary finite quotients and polynomial-equation solvability over  $\mathbb{Z}$  coincide. The strongest next question is whether Gaussian omnific arithmetic alone forces valuation compatibility, at least for involutions; the answer would decide whether the classification covers every arithmetic real structure or only its valued part.

### 40.4 Proof dependency and hypothesis audit

The main proof has the dependency order: **Oz**[ $i$ ],  $\mathcal{O} \Rightarrow \mathcal{I}, \mathbb{C}, \mathfrak{m} \Rightarrow \text{ct} \Rightarrow \text{strong semilinearity}; \text{finite action} \Rightarrow \text{trivial value action}; \text{orbit product and strongness} \Rightarrow \text{strong multiplicative } \mathcal{H}; \text{contracting remainder and formal inversion} \Rightarrow \mathcal{H}^{-1} \text{ preserves } \mathcal{I}; \text{conjugacy normal form} \Rightarrow \text{coefficient invariant} \Rightarrow \text{integer-part criterion and counting. No arrow uses the conclusion of a later one.}$

| Hypothesis or distinction                       | Where it is used                                                                                                 |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Full Hahn field                                 | The scalar detector needs every countably supported test series; formal inversion acts on full support spaces.   |
| Surjectivity of the original maps               | Constant-term tests in the target must have preimages; preservation of support pieces must be onto.              |
| Preservation of $\mathcal{O}$                   | Gives preservation of $\mathfrak{m}$ , hence full constant-term covariance after arithmetic reconstruction.      |
| Divisibility by $ G $                           | Makes $t^{\gamma/ G }$ a coherent monomial family; it cannot be replaced by isolated root choices.               |
| Strongness                                      | Makes the diagonal orbit-product family summable and transports the conjugacy identity from monomials to series. |
| Contracting-operator summability                | Proves that the normalizer is onto; leading terms alone give only injectivity.                                   |
| Preservation of $\mathcal{I}$                   | Ensures that the normalizer and its geometric inverse preserve the negative-support ring.                        |
| Archimedean coefficient field                   | Exactly the floor condition for constants; identifies the Hahn valuation with the natural one.                   |
| Proper-class exponent group <b>No</b>           | Every set support has an upper bound, needed for $\text{Frac}(R_F) = \mathbf{No}_F$ .                            |
| Fixed coefficient field vs. fixed residue field | They coincide under the reconstruction, not for arbitrary valued surcomplex automorphisms.                       |
| Class maps vs. a class of maps                  | Only individual maps and set-indexed representatives are constructed.                                            |

The inverse is checked on both sides by the formal identity  $(1 + X) \sum_r (-X)^r = 1$ ; it is not only a pointwise solution of the intertwining equation. Preservation of  $\mathcal{I}$  by each summand gives preservation by the sum, since all sums are Hahn sums with coefficientwise finite contributions.

## 40.5 What source 03 does not claim

The source states its limitations where they arise, and each is kept in place. The list collects them, with the place where each stands (in Sections 31–40, or in the shipped 03-gaussian-real-forms-SOURCES.md, the verification record, and the delivery README, which is not shipped). Items 1–25 are the source’s; items 26–31 were added in the merge.

1. An AI-assisted, unrefereed research draft; the proofs are intended to establish the claims under their assumptions, but independent correctness certification is not claimed (title page, README).
2. Novelty and priority are not certified; the literature and repository searches were targeted, and failing to find the package is not proof of priority; independent expert review is needed before any publication claim (Section 40.1, SOURCES.md).
3. Classical facts (real closed coefficient fields, Galois descent, Hilbert 90 for characters, constant-term retractions) are not presented as new; the finite-quotient and equation propositions explain what fails to detect the classified rings and are not a new general retraction theorem (Sections 31.3 and 37).
4. The collection’s reconstruction and automatic-strongness arguments are reproduced with attribution, not as novelty (Section 32.2).
5. Valuation preservation is an assumption of the Gaussian application; whether every automorphism of  $\mathbf{Oz}[i]$  preserves the Hahn valuation is not resolved (Section 31.3, Section 39).
6. Nor is it shown for arbitrary involutions of  $\mathbf{Oz}[i]$ ; the classification is complete only in the valuation-compatible Gaussian sector (Section 31.3).
7. The normalizer is canonical only relative to the chosen Hahn monomial section; the ring is not shown to define that section (Section 31.3).
8. Real closed fields of transcendence degree  $\mathfrak{c}$  are not identified by a finite list of parameters (Section 31.3).
9. “Geometric” refers to a multiplicative symmetrization; there is no analytic limit and no assertion about a surcomplex exponential; exp in the example is a formal power series (Sections 31 and 38).
10. The formal-inversion theorem of [6] is an imported input; strictly increasing values alone are not claimed to make an arbitrary geometric operator series meaningful, and the inverse is not justified by a topological convergence slogan (Theorem 32.4).
11. Strongness is not inferred from continuity, and  $\mathfrak{c}$  is not assumed multiplicative on  $K$  (Remark 32.3).
12. Archimedean real closed fields need not be Dedekind complete or isomorphic to  $\mathbb{R}$  (Remark 35.5).
13.  $\text{Frac}(R_F) = \mathbf{No}_F$  uses the boundedness of every set of surreal exponents; it is not asserted for set-sized Hahn support rings (Section 35).
14. Exactly  $2^{\mathfrak{c}}$  classes is a theorem about the valued Gaussian sector; the constructed examples stay distinct under arbitrary conjugators, but no upper bound for all pure-field involutions is claimed (Section 36).
15. Identical finite quotients and integer-coefficient equation solvability do not imply elementary equivalence; disequations, order constraints and nonordinary coefficients are not covered (Section 37).
16. No first-order definability of the rational-cut invariant is asserted (Remark 36.1).
17. The plain-field criterion is not used without the integer-part hypothesis (Remark 36.3; see item 29).
18. The  $H^1$  statement is a Hilbert 90-type reading of the normalizer, not a claim to originate nonabelian descent; in the class case it is the concrete cocycle identity (Section 34).

19.  $\mathcal{G}$  for class maps is shorthand; no class of all class maps is formed, and counting uses set-indexed representatives (Section 32).
20. Phase shears are not claimed new (Section 38).
21. The finite checks do not verify infinite Hahn summability, class-sized constructions, transcendence bases, cardinalities or all possible automorphisms; the program covers a homogeneous rank-two test case and is not an algorithm (Section 38.1, Question 39.5, the record's limitations).
22. No Lean verification was performed, the repository's Lean build was not run, and no repository files were modified (README, Section 38.1).
23. The research questions are proposed, not published open problems; no named published open problem is announced as solved, and no unresolved question of [5] is claimed solved (Sections 39 and 40.1).
24. The source comparison was targeted, not exhaustive over files, earlier commits, unintegrated manuscripts or unpublished work (Section 40.1).
25. The Wikipedia page supplied with the task was used for orientation, not as an input to a technical claim (SOURCES.md).
26. (*Merge.*) The source's statement that the collection did not supply the orbit-product normalizer, the classification or the integer-part and cardinality package is stale (Section 40.1); those results are source 02's, and source 03 is credited as an independent re-derivation.
27. (*Merge.*) The citations of [6] (Theorem 3.11, Proposition 1.45), including their characteristic-free reading and their application to series on the ordered exponent set  $\Omega$  of the class localization, and of Conrad and the Stacks Project, were not rechecked for the merge.
28. (*Merge.*) The observation that source 02's descent proof uses no characteristic zero, the comparison of routes (Remark 33.2) and Remark 32.6 are the merge's.
29. (*Merge.*) Clause (v) of Corollary 36.2 without the integer-part hypothesis rests on Theorem 24.4 (source 02), and the example showing that clause (vi) needs it is the merge's; the credits to `osq:thm:universal`, `odg:thm:transfer`, `odg:thm:finitequotients` and `dsn:thm:finite-quotients` are the merge's.
30. (*Merge.*) Of the program's 457 assertions, 166 are lexicographic comparisons that hold by the order; the orbit-product formula is compared with the flow on nine monomials only; the geometric inverse is tested only where the remainder is nilpotent (Section 38.1).
31. (*Merge.*) No `saut:gr:` label has a Lean implementation mapping.

## A A quick reference of explicit examples

In the table, “values fixed” means  $v(\alpha z) = v(z)$  for every nonzero  $z$ , not merely preservation up to a value-group automorphism. The term “constants” means the embedded ordinary complex field.

| Map                               | Preserved data                                                          | Nontrivial feature                                                                              |
|-----------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| $c$                               | Real axis pointwise; modulus; values; strong sums                       | $i \mapsto -i$ ; not complex fine-differentiable                                                |
| $S_2$                             | Constants; real axis; conjugation; strong sums                          | $v \mapsto 2v$ ; $\omega \mapsto \omega^2$ ; derivative zero                                    |
| $D_\lambda, \lambda \neq 0$       | Constants; real axis; conjugation; values; strong sums                  | Rescales leading coefficients by $e^{\lambda(g)}$                                               |
| $P_{\pi/2}$                       | Constants; values; strong sums; fine topology                           | $t \mapsto it$ ; moves the real axis and changes modulus                                        |
| $\Phi_d, d(b) = 1$                | Monomials; leading terms; strong sums; real axis when $d$ is real       | $b \mapsto b + t^\delta$ ; ordinary constants need not be fixed                                 |
| $T_1, \lambda = \ell, \delta = 1$ | Constants; real axis; conjugation; leading terms; strong sums           | $t \mapsto t/(1-t)$ ; not a total-real-exponential automorphism                                 |
| Ordered extension of Section 7    | Constants; real axis; conjugation; fine topology                        | Fixes the summands of $a$ but sends $a \mapsto a + t^\omega$                                    |
| Pure-field back-and-forth         | Any prescribed set of parameters, with ordinary $\mathbb{C}$ also fixed | Can send a positive infinite $x$ to $x^{-1}$ or $ix$ ; no canonical normal-form action supplied |

## B A conjugacy criterion for the standard real form

The cofinality obstruction in [theorem 10.2](#) has a useful positive counterpart. Say that an ordered class field has the *set-cut interpolation property* if for every pair of sets  $\mathcal{L}, \mathcal{R}$  with  $\mathcal{L} < \mathcal{R}$ , there exists  $x$  with  $\mathcal{L} < x < \mathcal{R}$ . Empty sides are permitted.

**Proposition B.1.** *An involution  $j$  of  $\mathbb{K}$  is conjugate to  $c$  if and only if its fixed real field has the set-cut interpolation property.*

*Proof.* The property holds in **No** by Conway cuts and is preserved by ordered isomorphisms. Conversely let  $F = \text{Fix}(j)$  have the property. It is a proper class, since  $\mathbb{K} = F(i)$ , and it is real closed by [theorem 10.1](#). Run the proof of [theorem 9.2](#) between  $F$  and **No** rather than within **No**. At every stage, the only special existence condition used to extend the partial ordered-field isomorphism is interpolation of a set-sized cut. It holds on both sides, so back-and-forth gives  $F \simeq \mathbf{No}$ . The real-form conjugacy criterion in [Section 10](#) then gives  $j \sim c$ .  $\square$

This is a structural characterization, not an effective decision procedure for a given class involution. It singles out precisely the conjugacy class of the usual surreal real form among all real forms of the pure field.

*Related (batch 32; [merge]).* The report `surcomplex/first-kappa-coefficients` constructs, under Global Choice, one involution of  $\mathbb{K}$  for each uncountable cardinal, agreeing with  $c$  on  $\mathbb{C}$ , whose fixed field is isomorphic to the field of surreals with fewer terms than that cardinal (`fkcsb:thm:involutions`). Those fields have set-presented gaps, so they fail the set-cut interpolation property, and by this criterion none of these involutions is conjugate to  $c$ ; that report records this route (`fkcsb:rem:univ`). Two of them attached to cardinals of different cofinalities are not conjugate, which gives a proper class of pairwise nonconjugate real forms. They are pure-field involutions. By [Proposition 23.1](#) none of them lies in the group  $\mathcal{G}$  of

valued automorphisms preserving  $\mathbf{Oz}[i]$ , whose involutions Theorem 16.2 classifies; the two families are complementary.

## C Provenance and the pinned comparison

### C.1 Provenance of this report

This report was built from *one* manuscript (source 01): the archive `surcomplex_automorphisms`, manuscript 07 of batch 20 of the collection, dated 22 September 2026 and pinned to repository commit `dcf86662b574988e75d08d515d3a2a5ba7bbc0d7`. It was placed in commit `5fe7f8d` as a new report rather than merged into `surreal/exponential-automorphism-rigidity`: it continues that report's section on  $\mathbf{No}(i)$ , but with plain field automorphisms instead of exponential ones, and answers no question that report names. At placement there was no second source: nothing was merged, and nothing was selected away. Every result, proof, example, question and limitation of the manuscript is printed here. [merge] A second source was added in batch 32 (Appendix C.3), and a third in batch 34 (Appendix C.4); this subsection and Appendix C.2 concern source 01 only.

The delivery consisted of the article source, its PDF and a README; it contained no code and no data. The source was renamed `article.tex` and edited as listed below; the README was rewritten for the collection; the delivered PDF is not shipped and is replaced by a build of this text.

The editorial changes are these.

- (1) Every label carries the prefix `saut:.`. The source's 79 labels are kept, unchanged after the prefix, and seven were added (`saut:sec:conventions`, `saut:sec:position`, `saut:eq:Lchain`, `saut:sec:nonclaims`, `saut:app:conjugacy`, `saut:app:provenance`, `saut:app:pinned`).
- (2) The title page gives the report's status and provenance, and the abstract gains one sentence placing  $\text{Aut}_L(\mathbb{K})$ .
- (3) Sections 1.4 and 1.5 and the consolidated non-claims of Section 15.3 are new. Short pointers were added after Theorem 6.1, at the start of Section 13, and in the first question of Section 14.
- (4) Six symbols were renamed so that each has one meaning (Section 1.4):  $E_{\rho,\tau} \rightarrow M_{\rho,\tau}$ ,  $E_d \rightarrow \Phi_d$ , the derivation  $D \rightarrow \mathcal{D}$ , the Puiseux field  $P \rightarrow \mathcal{P}$ , the automorphism  $\tau \rightarrow \psi$  of the non-density argument, and cut sides  $(L, R) \rightarrow (\mathcal{L}, \mathcal{R})$ ; the source's  $\text{Aut}(\mathbb{K}, L)$  is written  $\text{Aut}_L(\mathbb{K})$ .

The placement edit preserved the source's mathematical statements. It retained the results reproduced from the rigidity report (its Lemma 3.1, the faithfulness of Corollary 4.2, the content of Theorem 5.1 and Corollary 5.2, Theorem 8.4, and Theorem 9.2 with its kernel clause). They are printed here once, with the source's proofs and attribution, rather than replaced by pointers, because they are part of one manuscript's argument; Section 1.5 names each counterpart. The letter  $E$  was freed for the exponential, following the rigidity report, rather than renaming that report's symbol. The title is the source's; the directory is named `surcomplex-field-automorphisms` to say that the automorphisms are field automorphisms.

The later proof review clarifies rational cuts and strong coefficient action. It expands the class and support arguments and excludes every derivative value in  $\mathbb{K}$ , including infinite values. The finite-observation non-density proof is extended to all surcomplex field automorphisms. Current generic Lean coverage is separated from the pinned inspection. See `REVIEW.md` for the changes and validation. The added arguments are not attributed to the original delivery.

## C.2 Pinned repository scope

Section 13 is the source’s comparison at its pin `dcf8666`, an ancestor of the placement commit `5fe7f8d` (sixteen commits earlier). Checked against the repository for this report:

- The descriptions of `Surreal/Algebra/Complexify.lean`, `Surreal/Algebra/FineDerivative.lean`, the normal-form bridge note, and the rigidity report with its appended audit correction are accurate at the pin. None of these files, nor the root `README.md`, changed between the pin and the placement; the report catalogue `docs/README.md` changed only by listing reports of another batch and extending other entries.
- The bibliography cites the rigidity report by its article title, *Growth-Scale Rigidity for Surreal and Surcomplex Automorphisms*; the collection’s `README` for the same directory is headed “Cofinal displacement and the rigidity of exponential automorphisms”. Both name `surreal/exponential-automorphism-rigidity`.
- The source did not relate its results to `found:thm:discrete` and its Lean proof, to the collection’s zero-derivative witnesses, to the character twists of `spec:lem:finitindex`, to the Cayley parametrization of the trigonometry report, or to the Lean modules `Surreal/Surcomplex/Basic.lean` and `Surreal/HahnSeries/ComplexNumbers.lean`, all present at the pin. The hidden-negative-directions and bounded-support reports and `Surreal/Algebra/SigmaDerivation.lean` were added after the pin and could not have been seen. Section 1.5 supplies these relations.
- The `README` of the rigidity report states that nothing there proves a theorem about the automorphism group of **No** that any other report also proves. That was accurate before this report was placed; this report reprints several of its results (Section 1.5).

## C.3 Source 02: finite symmetry descent

Sections 16–30 are built from a second manuscript: the archive `surreal_finite_symmetry_descent`, manuscript 02 of batch 32 (delivered in commit `aa268a4`), *Finite Symmetry Descent and Real Forms of Gaussian Omnific Integers*, research article prepared with ChatGPT for Vladimir Reshetnikov, dated 23 September 2026, 27 pages, pinned to repository commit `343dc2c471212bb9b53ff4623bace2e1943f255b`. It was placed in commit `7d04483` as an addition with local number 02 (source 01 counting as 01), because it answers the valued case of this report’s open real-form classification (Section 14). The pin is an ancestor of the placement commit, and this report’s text did not change between them; the parts of `surreal/omnific-preserving-automorphisms` on summation detection and automatic strongness were written in parallel and were not in the pinned tree.

**Files.** The manuscript source, its PDF and its delivery `README` are not shipped; its text is Sections 16–30. Shipped byte-identical, with the prefix `02-finite-symmetry-descent-`: the source and proof audit (report root), the build helper and the finite verifier (`code/`), and the verification record, its console copy and the build validation record (`data/`). Those files still use the delivery names (`code/verify_finite.py`, `data/verification.json`, `finite_symmetry_descent.tex`); the `README` of this directory maps them.

### What it contributes, and where the merge chose.

- (1) Placement: the source’s Sections 1–14 are Sections 16–29; its Section 15 and Appendices A–B are Section 30. They follow Section 15 and precede the appendices, so that no number of source 01 changes.

- (2) Labels: the source's labels are kept after the prefix `saut:fs:`, except `eq:surreal-hahn` (the identification (3.1), printed once) and the section and appendix labels of its closing matter; unlabelled questions, remarks, the integer-part definition and several sections received `saut:fs:` labels.
- (3) Printed once, with credit: the pairing criterion and its corollary (`opa:as:thm:detect`, `opa:as:thm:ctiso`; the Baire proof kept as a further route), the reconstruction (`opa:lem:divisible`, `odg:def:thm:multiplicator`, `odg:thm:fractions`), Corollary 23.3 (Theorem 10.1; the source's route kept), and the definition of strong maps (Section 3, restated for general  $k$  and  $\Gamma$ ).
- (4) Stale against the current tree, corrected in place and marked [merge]: the source's strongness hypotheses (Proposition 18.5 and Remark 22.3); its first question, now answered; its second question, which is `opa:as:q:gaussian` and is not repeated as a new question; and the second half of Remark 22.2. The source's statements are kept beside the corrections.
- (5) Notation: the renamings of Section 16.1.
- (6) Bibliography: the entries [1, 2, 3, 5] and the Artin–Schreier notes [8] are shared by both sources, and [14, 15, 16] were added. The source's four repository entries (this report and the omnific-preserving, omnific Diophantine and large-cardinal reports) are cited as collection paths.
- (7) Added in the merge: Proposition 18.5, Remark 22.3, the consequences listed after Proposition 23.1 and Corollary 24.5, the identification  $\mathcal{H}_{j_\theta} = P_\theta$ , the assertion count of Section 26.3, Section 28.4, the status notes, and items 23–27 of Section 30.4. Within source 01, the title page, the abstract, Sections 1.5, 10, 14 and 15.3 and Appendix C received pointers or status notes (marked [merge] outside the front matter); no mathematical statement of source 01 was changed.

The source's own comparison with the collection (its Section 13.2) was read against the pinned tree. This report's nonconjugate real form and conjugacy criterion, the phase twists and the omnific reconstruction are present there, as it says. The automatic-strongness results were not in its tree.

## C.4 Source 03: Gaussian-omnific real forms

Sections 31–40 are built from a third manuscript: the archive `gaussian_omnific_real_forms`, manuscript 03 of batch 34 (delivered in commit `a4611ad`), *Gaussian-Omnific Real Forms and Geometric Symmetrization: finite-action descent, coefficient-field classification, and  $2^c$  inequivalent integer-part real forms*, research manuscript prepared with ChatGPT for Vladimir Reshetnikov, dated 23 September 2026, 24 pages, pinned to repository commit `110efe9dea646a092a640838515af4ad2d7ee2e0`. It was placed in commit `a7a435f` as an addition with local number 03, because it re-derives source 02's normalization and count and adds an explicit inverse, a characteristic-free descent and the arithmetic of the fixed rings. The pin contains the placement of source 02 (`7d04483`) but not its write (`b8bbfee`); the source's comparison read this report's article and README, which were then source 01 alone (Section 40.1).

**Files.** The manuscript source, its PDF, its delivery README and the checksum manifest `SHA256SUMS.txt` of the delivery (seven entries, verified at placement) are not shipped; its text is Sections 31–40. Shipped byte-identical, with the prefix `03-gaussian-real-forms-`: the source and claim audit `SOURCES.md` (report root), the build helper `build.sh` and the verification program `verify.py` (code/), and the verification record `verification.json` (data/). Those files still use the delivery names; the README of this directory maps them.

**What it contributes, and where the merge chose.**

- (1) Placement: the source's Sections 1–13 and Appendices A–B are Sections 31–40, after source 02's closing section and before the appendices, so that no number of sources 01 and 02 changes. Its Section 1 is Section 31; Sections 2–4 are Section 32; Section 5 is Section 33; Section 6 is Section 34; Sections 7–8 are Section 35; Sections 9–10 are Section 36; Section 11 is Section 37; Section 12 is Section 38; Section 13 is Section 39; Section 14 and Appendices A–B are Section 40.
- (2) Printed once, with credit: every statement that source 02 or the collection already proves (the map of Section 31.2); the source's genuinely different proofs are kept as second routes (Section 40.2).
- (3) Labels: the source's labels are not carried (its results are either mapped to existing labels or printed under new ones); every new label has the sub-prefix `saut:gr:`.
- (4) Stale against the current tree: its statement that the collection lacked the normalizer and the classification (Section 40.1); its question on abstractly surreal fixed fields, already answered (Question 39.4); its plain-field clause, weaker than Theorem 24.4 (Corollary 36.2).
- (5) Notation: the renamings of Section 31.1.
- (6) Bibliography: [6, 3, 5, 8] are shared with the earlier sources; [17, 18] were added. The source's two repository entries (the omnific-preserving report and this report) are cited as collection paths.
- (7) Added in the merge: the statement map, the conventions table, Remarks 32.6 and 33.2, the notes after Corollary 33.3, Proposition 34.1, Proposition 35.3 and Proposition 37.2, clause (v) and the example in the proof of Corollary 36.2, the status notes of Section 39, the assertion count of Section 38.1, Section 40.2 and items 26–31 of Section 40.5. In the earlier text: the title page and abstract, Sections 1.5 and 5, Theorem 16.1, Questions 29.3, 29.5, 29.8 and 29.9, and Appendix C received pointers or status notes, marked [merge] outside the front matter; no mathematical statement of sources 01 and 02 was changed.

The source's own comparison with the collection (its Appendix B.1) was read against the pinned tree: this report's article and README at 110efe9 are source 01 alone, as it says, and the omnific-preserving report contains `opa:as:thm:detect`, `opa:as:thm:ctiso` and `opa:as:thm:gaussian`, which it credits.

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